V7 and V9 comparison

After our discussion, I compared the additional V9 file you shared with V7, the draft I sent. I also used ChatGPT to organize the comparison, then checked the passages against both files. This comparison replaces the V8-specific findings where V9 differs.

V9 retains the baseline-image explanation, detailed baking and assembly procedures, print-preparation setting and coating-height explanation. Its Methods contains 2,194 words versus 2,226 in V7. The title, all five legends, supplementary text, ten tables and seven image contents are unchanged.

The Table S2 citations and reference list are still absent, and the mixed citation styles remain. V9 adds useful Figure 3/4B links and pin-height statistics in Results. The newer count dataset, calibrated images and saved counting settings can be integrated alongside the remaining citation, layout, ethics and archive fixes.

I kept the fabrication method central in the suggestions, with neuronal culture as one tested use. I also marked added warnings about tests the paper does not claim, and places where the rewrite repeats existing results or limits.

The links let you compare the passages and return to the rows without searching through the files. In Word for Mac, right-click a handle and choose Open Hyperlink; use the R links to return to the table, or Contents in the footer.

The main part to review is the revision table (pp. 3–16). You can leave comments in its rows.

The remaining 86 pages are linked supporting material: the saved ChatGPT discussion, cited passages, sources and images.

Reading guide

V7 P8 = version 7, paragraph 8. V9 P50 = version 9, paragraph 50. R = table row; L = line in the saved ChatGPT discussion.

Source-handle colors compare wording; comment colors show whether the point is covered.

The discussion below is the one shared with V8; its comments now check the corresponding points against V7 and V9.

Contents

Comparison (details)	pp. 2–3 (2 pages)
Revision table (leave comments here) ←	**→ pp. 3–16 (14 pages)**
ChatGPT discussion, with comments (mostly accessed via links)	pp. 16–40 (25 pages)
Cited V7 and V9 passages	pp. 40–85 (46 pages)
Additional sources and images	pp. 85–99 (15 pages)
Published Lab on a Chip comparisons	pp. 99–102 (4 pages)

Comparison

Already covered in V7

Table S2 was cited twice before the rewrite removed its links

Supplementary Table S2 is not cited anywhere in the main manuscript.

Feedback L415

Laser removal of material (ablation), direct printing, hot embossing, hard-plastic fabrication, and modular inserts combine these scales using different equipment and added operations (Supplementary Table S2).

V7 P8

V7 P7 and V7 P8 both cite Table S2; V9 P15 retains a narrower fabrication comparison without either citation. R06 and R07 propose keeping the comparison and its links; Feedback L488 repeats the request.

The baseline explanation is retained in V9

I would clarify that these are baseline measurements from an experiment conducted before axotomy

Feedback L543

These were baseline images collected before axotomy; Supplementary Protocol S1 describes the dataset.

V9 P50

V7 P44 and V9 P50 both give this explanation; R23 keeps it with the imaging procedure.

Procedures and outcomes already occupy different sections

V7 P23 and V9 P33 give the EPDM/clamping procedure in Methods; V7 P52 and V9 P64 report the height spread and seating result in Results. V7 P33 and V9 P41 describe cast inspection, while V7 P54 and V9 P68 report cast output and elapsed use. R14, R19, R31 and R33 cover those examples.

Figure 1 already shows the two-well geometry; Figure 2 shows the pin and EPDM backing; the five legends already have titles and Figure 5 has three application rows. R62 covers the presentation changes.

Useful V9 edits

V9 P67 and V9 P78 add direct Figure 3 and Figure 4B links (R33; R38); V9 P64 brings Table S1's mean and SD into main Results beside the existing height spread (R31). R27 covers moving generated layouts beside design, and R25 covers four versus six Results groups.

Work still needed

The direct Figure S1/S2 links are still missing (R29); the mixed citation styles and absent bibliography remain (R52). V9 P13 contains main-text PMIDs, while V9 P108 retains author–year citations in the unchanged Figure 5 legend; some Methods references also remain author–year.

R53–R55 cover integrating the newer count dataset, calibrated CTβ views and saved counting configuration. R59 covers the later width estimates. R56 covers animal approval details, R49 the study archive, and R50 the conflicts/data-availability order.

V9's rendered Table S1 now breaks 5.567 into 5.56 and 7, despite unchanged values; its cell text no longer has V7's explicit 10-point size and inherits the 12-point default. R57 shows this formatting change and covers widening the comparison tables.

Removed explanations and added warnings

Removed explanations

V9 P72 omits V7 P60's reason for choosing Opera Phenix, and V9 P85 omits V7 P68's explicit 17-fold margin ratio while retaining the underlying dimensions. V9 P13 narrows V7 P6's application breadth. R36, R42 and R04 propose retaining these explanations.

Added warnings and repetition

V9 P70 adds an unperformed joint-strength test to the observed adhesive result (R34); V9 P87 adds unmeasured labor/time/cost comparisons to the one-time-mold workflow (R44). V7 reports intact joints and avoided processing steps, not measured strength or cost savings.

The one-preparation limit is explicit in Methods, repeated at the end of neuronal Results and repeated in Discussion (V9 P46; V9 P51; V9 P77; V9 P82; V9 P88). The tracer limit appears in Methods, Results and Discussion (V9 P45; V9 P76; V9 P88). R41 and R37 cover reducing this repetition without dropping the limits.

V9 P11 omits V7 P4's measured reuse totals and edge-growth limitation. It calls alignment "precise", while V9 P85 says actual accuracy was not directly quantified; it also describes compartmentalized axonal growth more strongly than the indirect-labeling account in V9 P78. R03 keeps the fabrication framing, reuse observations and measured culture limit.

Tim offered to read the draft. I think his input could be valuable given that he has published in Lab on a Chip twice. I would value his feedback on V7, with your revised version alongside it if you would like. Paper 9 Paper 10

Revision table

The rows cover the V7/V9 differences and the remaining reporting and presentation work; unchanged items are retained where they answer a point in the saved discussion.

ID	Change or work	My suggestion	What the sources show
R01	Title	Keep the shared title	V9 retains the V7 title, including PDMS, SU-8 and the hybrid wafer mold. V7 P1 V9 P1
R02	Authors and affiliations	V9 layout	V9 adds numbered affiliations and corresponding-author notation; its first affiliation now includes the department and institution. V7 P2 V9 P3 V9 P6 V9 P7 V9 P8

ID	Change or work	My suggestion	What the sources show
R03	Abstract	V7 framing with a concise device count	V9 begins with neuronal applications and drops V7's measured reuse totals (>50 casts; >8 months) and edge-growth limitation. Its "precise alignment" and "compartmentalized axonal growth" wording is stronger than the nominal registration and indirect-labeling accounts in Results/Discussion; keep the fabrication workflow, reuse observations and culture limit, using the reconciled device count from R53. V7 P4 V9 P11 Related feedback: Feedback L82 Feedback L136
R04	Introduction: applications	V7 account of the applications	V7 connects the method to neuroscience, stem-cell biology, cancer, immunology and developmental biology; V9 replaces this with neuronal connectivity and perturbations, understating the generalizability of the fabrication platform. V7 P6 V9 P13
R05	Introduction: prior plate systems	Platform context; fabrication methods remain the main comparison	V9 adds high-capacity, modular and fluid-walled neuronal systems; check each new reference and the claim it supports, then use them as application context alongside the comparison of fabrication operations and equipment in V7 and Table S2. V7 P7 V7 P8 V9 P14
R06	Introduction: access to fabrication	V7 equipment-access comparison and Table S2 link	V7 compares obtaining a mold from a shared facility with acquiring molding or machining equipment and cites Table S2; V9 omits this Introduction comparison and citation, while retaining a shorter shared-facility account in Discussion. V7 P7 V9 P15 V9 P87 Related feedback: Feedback L415 Feedback L488
R07	Introduction:	V7 fabrication	V9 narrows the comparison to hybrid micro–

ID	Change or work	My suggestion	What the sources show
	combining channels and wells	comparison with its Table S2 link	macro molding and additive neuronal devices and removes the second Table S2 citation; retain the comparison of operations each fabrication route adds. V7 P8 V9 P15 Sources Related feedback: Feedback L415 Feedback L488 Feedback L498
R08	Introduction: insert-based prior work	Earlier prior-work comparison with V7's array-transfer explanation	V9 introduces Ristola's individually placed metal inserts earlier and retains the single-operation transfer of 96 inserts in Discussion; keep that distinction, including its applicability to other insert arrangements. V7 P8 V7 P9 V7 P68 V9 P16 V9 P85
R09	Introduction: what this study tests	Concise study overview; retain the untested-layout qualification	Both describe the same fabrication and neuronal demonstration; V9 retains the computational-only status in Methods and Results but expands the overview and repeats dimensions. V7 P10 V7 P16 V7 P66 V9 P17 V9 P27 V9 P60 Related feedback: Feedback L84 Feedback L327
R10	Methods: one-time fabrication and repeated casting	Restore the short Protocol S1/S2 workflow explanation	V9 removes the opening Methods sentence distinguishing routine production in Protocol S1 from one-time mold fabrication in Protocol S2; the two protocols are unchanged, and Discussion retains the underlying distinction. V7 P12 V9 P24 V9 P87 Related feedback: Feedback L539
R11	Methods: design and generated files	Keep separate design, tested-geometry and generated-layout paragraphs	V9 preserves the software/export description, shared coordinates, device dimensions and untested layouts in three paragraphs; its leading heading is incorporated into the first paragraph rather than standing separately. V7 P13 V7 P14 V7 P15 V7 P16 V9 P25 V9 P26 V9 P27

ID	Change or work	My suggestion	What the sources show
			Related feedback: Feedback L19 Feedback L62
R12	Methods: SU-8 and SUEX layers	Keep the layer and bake details	V9 retains the base, microchannel and well layers, developer replacement, heating/cooling conditions and direct oven placement without controlled ramps; the procedural losses identified in V8 are absent here. V7 P17 V7 P18 V7 P19 V7 P20 V7 P21 V9 P28 V9 P29 V9 P30 V9 P31 Related feedback: Feedback L39 Feedback L96
R13	Methods: resin printing	Keep design scaling separate from print-preparation compensation	V9 retains the 2.26% design enlargement and explicitly says no shrinkage or tolerance compensation was applied in the print-preparation software. V7 P15 V7 P22 V9 P26 V9 P32
R14	Methods: array transfer and bonding	Keep the procedure in Methods	V9 retains the rubber position, thickness and hardness, clamps, epoxy, washes, cure and release sequence; the corresponding height variation and seating outcome remain in Results. V7 P23 V9 P33 Related feedback: Feedback L11 Feedback L39
R15	Methods: parylene coating	Keep the separate coating paragraph	V9 preserves the coating thickness, facility, equipment and pre-coating design dimensions; its added PMID should be verified and formatted with the other references. V7 P24 V9 P34 Related feedback: Feedback L96
R16	Methods: frame printing	Keep the supplied printing-file details	V9 retains HIPS printing conditions, Table S6 and the supplied Creality Print 6.0.4 G-code file. V7 P26 V9 P35 Related feedback: Feedback L562

ID	Change or work	My suggestion	What the sources show
R17	Methods: casting, demolding and glass bonding	Keep the detailed casting and bonding sequence	V9 retains the rack, parallel processing, wafer support, peeling direction, guillotine, glass dimensions and bonding parameters; it adds prior-work references to the tapered-sidewall paragraph. V7 P25 V7 P27 V7 P28 V7 P29 V9 P36 V9 P37 V9 P38 Related feedback: Feedback L21 Feedback L96
R18	Methods: frame assembly and surface treatment	Keep the assembly sequence and distinct plasma steps	V9 retains frame/glass orientation, adhesive volume and cure, trimming, and the separate second plasma treatment; its explanatory final sentence repeats a distinction already made by the sequence. V7 P30 V7 P31 V9 P39 V9 P40 Related feedback: Feedback L96
R19	Methods: mold and device assessment	Keep the inspection criteria and unframed control	V9 retains individual pin measurements, inspection criteria, usable-cast definition, adhesive culture screening and the single unframed control; it also retains the statement that toxicity was not quantified. V7 P32 V7 P33 V7 P34 V7 P35 V9 P41 V9 P42 V9 P43 Related feedback: Feedback L46 Feedback L528
R20	Methods: pipetting and microscopy	Keep the experiment-specific procedure	Both identify eight fitted tips and imaging on two unmodified high-content systems. V7 P36 V9 P44 Related feedback: Feedback L74
R21	Methods: fluorescent tracers	Keep the tracer procedure and its stated limit	V9 retains the secondary antibodies, volumes, dilution, incubation, objective, binning, exposure and visual assessment; it also retains the unmeasured flow-rate and duration limit. V7 P37 V7 P38 V9 P45 Related feedback: Feedback L78 Feedback L432

ID	Change or work	My suggestion	What the sources show
R22	Methods: neuronal culture and CTβ	Keep V7's terminology; reconcile the exclusions in R53	V9 retains the one-embryo preparation, coating, plating, medium, CTβ timing and exclusion rule; it moves the technical-replicate statement into the culture paragraph and explains soma/axon terminology. V7 P39 V7 P40 V7 P41 V9 P46 V9 P48 Related feedback: Feedback L51
R23	Methods: imaging and counting	Keep the imaging and baseline explanation	V9 retains the pixel scale, exposures, stitching, common counting area, reason for Plate 3's larger grid and the pre-axotomy baseline sentence; the baseline clarification requested in the saved discussion is already present in both V7 and V9. V7 P42 V7 P43 V7 P44 V9 P49 V9 P50 Related feedback: Feedback L25 Feedback L543 Feedback L608
R24	Methods: statistics	Keep the descriptive summaries; update them with R53	V9 retains the within-plate SD and explicit technical-replicate explanation without V8's extra warning about inferring other preparations. V7 P45 V7 P46 V9 P51 Related feedback: Feedback L26 Feedback L432
R25	Results: four groups become six	Discuss four versus six Results groups	V9 separates design, mold development, casting, compatibility, tracer transport and neuronal testing; V7 covers these topics in four groups, and both already separate procedures from findings. V7 P48 V7 P55 V7 P62 V7 P65 V9 P58 V9 P61 V9 P67 V9 P71 V9 P74 V9 P77 Related feedback: Feedback L58
R26	Results: coordinated design and plate geometry	V7's concise account under the proposed design heading	V9 explains the same coordinated components and dimensions over two paragraphs; these dimensions are already specified in Methods, so the Results can concentrate on what the coordinated design enables.

ID	Change or work	My suggestion	What the sources show
			V7 P49 V9 P58 V9 P59 Related feedback: Feedback L62 Feedback L136
R27	Results: alternative layouts moved earlier	Earlier placement with concise generated-layout wording	V9 places alternative fabrication files beside the design result and retains the untested status; V7 explains the same coordinated changes in its final Results group. V7 P65 V7 P66 V9 P60 Related feedback: Feedback L84 Feedback L327
R28	Results: general mold-development statements	Begin with the failures and solutions; omit the recap	V9 adds an opening requirement statement and a closing summary around the existing filter, adhesion and clamping observations; the intervening paragraphs already show how the mold was developed. V7 P49 V7 P50 V7 P51 V9 P61 V9 P66
R29	Results: filter and base-layer observations	Existing observations with direct Figure S1/S2 links	Both report the raised edges, detachment and successful process changes in Results; V9 still cites only Notes S1/S2 rather than the figures themselves. V7 P50 V7 P51 V9 P62 V9 P63 Related feedback: Feedback L33 Feedback L34 Feedback L467 Feedback L471
R30	Results: additional SU-8 bake	Keep the batch observations without calling the process optimized	V9 keeps the eight-wafer and subsequent two-wafer observations; “optimization” and “optimized” also appear in its mold-development account, although the sequential batches do not isolate variables or establish an optimum. V7 P51 V9 P63
R31	Results: pin heights and clamping	Keep the added mean and SD with the existing clamping result	V9 brings the mean 5.551 mm and SD 26.7 µm from Table S1 into main Results alongside V7’s existing 116 µm range, uneven rigid clamping and successful EPDM seating. V7 P52 V9 P64

ID	Change or work	My suggestion	What the sources show
			Sources Related feedback: Feedback L11 Feedback L127
R32	Results: open channels and mold release	Report the observation; integrate the width check in R59	Both report open cast channels and unmeasured channel dimensions; V9 retains the limit beside that result, while its later Discussion repeats it. V7 P53 V9 P65 Related feedback: Feedback L136 Feedback L149
R33	Results: repeated casting and mold use	Keep the reuse observations and V9's Figure 3 link	V9 retains five assembled molds, three used routinely, more than 50 casts, both losses and more than eight months of use; it adds a direct Figure 3 callout and does not retain V8's standardized-durability warning. V7 P54 V7 P56 V9 P67 V9 P68 Sources Related feedback: Feedback L48 Feedback L138 Feedback L522
R34	Results: frame materials and adhesives	V7's direct observations without a new strength warning	V9 retains lower visible HIPS warping and the surviving SI 5140 cultures but changes the failed-adhesive wording and adds an unperformed mechanical-strength test; Methods already states that toxicity was not quantified. V7 P57 V7 P58 V9 P69 V9 P70
R35	Results: multichannel loading	Keep the eight-well loading observation	V9 reports the manual eight-well operation without V8's robotic-dispensing warning; its closing compatibility paragraph repeats the preceding loading and imaging observations. V7 P55 V7 P56 V9 P71 V9 P73 Related feedback: Feedback L74
R36	Results: microscope use	Restore the reason for choosing Opera Phenix	Both report use of two unmodified systems; V9 omits V7's explanation that Opera Phenix was faster for these experiments. V7 P60 V9 P72

ID	Change or work	My suggestion	What the sources show
R37	Results: tracer movement	V7's concise result and limit	V9 spreads loading, redistribution and the existing flow/isolation limit over three paragraphs; V7 already reports the two-time-point result and its limits in one Results paragraph. V7 P61 V9 P74 V9 P75 V9 P76 Related feedback: Feedback L78 Feedback L432
R38	Results: neuronal labeling	Concise neuronal result with the Figure 4B link	V9 adds the direct Figure 4B link and retains the inference of uptake by axons and retrograde transport; its abstract uses stronger "compartmentalized axonal growth" wording than this result supports directly. V7 P62 V7 P63 V9 P77 V9 P78 Related feedback: Feedback L430
R39	Results: assessed and counted devices	Integrate the updated DMSO counts and plot in R53	Both report 72 interior devices, three exclusions and 69 analyzed devices, with plate means and within-plate SD; the later DMSO dataset needs a reconciled exclusion account before replacing those totals. V7 P64 V9 P79 V9 P80 Related feedback: Feedback L53 Feedback L551
R40	Results: unframed control and edge growth	Keep the control observation with its limits	V9 moves the unframed control into neuronal Results and explicitly lists the unresolved frame, adhesive, sealing and culture-environment contributions; V7 already reports the control and unresolved cause. V7 P59 V7 P64 V7 P73 V9 P81 Related feedback: Feedback L55
R41	Results: repeated neuronal-summary paragraph	Keep the result; state the independent-preparation limit once	V9 adds a closing neuronal-validation summary and future independent-preparation sentence, then repeats the same limit in Discussion; the one-preparation design is already explicit in Methods. V7 P75 V9 P82 V9 P88

ID	Change or work	My suggestion	What the sources show
R42	Discussion: contribution and prior inserts	V7's process-centered account with the earlier prior-work citation	V9 retains simultaneous positioning and transfer across other array layouts but expands the contribution and adds an alignment-accuracy warning; its introductory statement remains centered on the fabrication strategy. V7 P67 V7 P68 V9 P84 V9 P85
R43	Discussion: printer resolution and parylene	Keep printed parts confined to the mold; vet the added references	V9 retains the distinction between resin in the mold and PDMS/glass in the finished device, adds printed-mold/coating references, and appends a future configuration-compatibility qualification. V7 P69 V9 P86 Related feedback: Feedback L260 Feedback L500
R44	Discussion: repeated use and bench production	Keep the one-time mold/repeated-casting explanation	V9 retains the eight-month use observation and later bench production, then adds unmeasured time, labor and cost comparisons; V7 describes avoided processing steps rather than claiming measured savings. V7 P70 V7 P71 V9 P87 Related feedback: Feedback L140 Feedback L432
R45	Discussion: coating and working volume	Keep the coating explanation and integrate R59	Unlike V8, V9 retains the relative-height explanation: coating raises both ridge and surrounding surface; it also retains the expected width increase and 50 µL working-volume constraint, while adding dimensional-fidelity and scaling-refinement proposals. V7 P72 V9 P89
R46	Discussion: edge growth	V7's concise wording with the unresolved frame factors	Both report the poorer peripheral growth and unframed control; V9 expands the same unresolved mechanism into a separate Discussion paragraph after describing it in Results. V7 P73 V9 P90 Related feedback: Feedback L55

ID	Change or work	My suggestion	What the sources show
R47	Discussion: adapting other layouts	Keep the coordinated-design explanation and tested/generated distinction	Both explain changes to pins, holes, adhesive space and layout; V9 retains the shared-file logic but adds a list of future alignment, casting, transport and biological checks for the untested layouts. V7 P74 V9 P91 Related feedback: Feedback L84 Feedback L327
R48	Conclusion	V7 conclusion with concise observed reuse and assay results	V9 repeats the contribution and adds future quantitative reproducibility and peripheral-culture work; its “48 independent” devices should mean separate device units rather than independent biological replicates. V7 P76 V7 P77 V9 P92 V9 P93 Related feedback: Feedback L44 Feedback L140 Feedback L432
R49	Code, files and data statement	Accurate file list and dataset availability, with a completed study archive	V9 compresses the file list and states that experimental data are provided in the supplement or repository; the embedded supplement is unchanged and lacks individual counts, so name the actual supplied dataset and reconcile it through R53, while completing the promised versioned archive. V7 P78 V7 P79 V7 P80 V7 P81 V9 P94 V9 P95 Sources Related feedback: Feedback L551 Feedback L560 Feedback L562 Feedback L564
R50	Conflict-of-interest statement	Keep the declaration and put it before Data availability	V9 adds the conflicts declaration after Data availability; the required order is conflicts, data availability, acknowledgements. V7 P78 V7 P82 V9 P94 V9 P96 V9 P97 Sources Related feedback: Feedback L581 Feedback L583
R51	Acknowledgements and funding	V7 acknowledgements plus accurate funding details	V9 adds CIHR and MSCanada funding but removes V7’s separate thanks; verify the funding and retain the existing acknowledgements as appropriate. V7 P82 V7 P83 V9 P98

ID	Change or work	My suggestion	What the sources show
R52	References and citation style	V7 bibliography with vetted new entries and consistent citations	V7 has a 24-entry bibliography and author–year citations; V9 omits that list and adds 20 PMID/PMIDs labels while retaining author–year text in the legends, supplement and some Methods paragraphs; include the complete references, vet the additions, and apply the final numbered journal style. V7 P84 V7 P85 V7 P108 V9 P13 V9 P16 V9 P27 V9 P34 V9 P108 Sources Related feedback: Feedback L436 Feedback L569 Feedback L571
R53	Updated DMSO counts and Figure 4 text	Integrate the recent DMSO plot and match the text	The later DMSO package has 72 rows and five counting-artifact exclusions, giving 22, 21 and 24 devices (67 total); both drafts still report three exclusions and 69 devices, so reconcile the flags with the exclusion rule, then update Methods, Protocol S1, Results, Figure 4 and its legend. V7 P41 V7 P64 V7 P81 V7 P116 V7 P245 V7 P247 V9 P48 V9 P79 V9 P80 V9 P95 V9 P106 V9 P235 V9 P237 Sources Related feedback: Feedback L51 Feedback L291 Feedback L551 Feedback L607
R54	Figure 4 scale bars	Use the recent calibrated CTβ views; add tracer bars	The newer supplementary-biology views supply calibrated 500 µm CTβ bars and crop dimensions; integrate these views into Figure 4, while adding the acquisition scale and bar to the separate tracer panel. V7 P43 V7 P116 V9 P49 V9 P106 Sources Related feedback: Feedback L301
R55	Counting-method settings	Integrate the saved supplementary-biology counting method	The newer analysis supplies software, count tables, overlays and saved settings; identify the configuration matching the chosen dataset and integrate it into Methods and Protocol S1, with the reconciled exclusions from R53. V7 P241 V9 P231 Sources

ID	Change or work	My suggestion	What the sources show
			Related feedback: Feedback L51
R56	Animal approval details	Add the approving body, guidelines and approval number	Both protocols mention an approved procedure but omit the approving body, guidelines and approval number. V7 P226 V9 P216 Sources Related feedback: Feedback L576
R57	Supplementary table layout	Restore compact S1 numbers; widen the S2/S3 columns	The data are unchanged, but V9 removes S1's explicit 10-point cell text and inherits a 12-point default; in its rendered page 21, 5.567 splits into 5.56 and 7, and the other heights wrap similarly. The earlier V7/V8 render kept these values on one line. V9 also splits long words in S2; restore readable numerical cells and widen the comparison columns. V7 P121 V7 P129 V7 P131 V9 P111 V9 P119 V9 P121 Sources Related feedback: Feedback L494 Feedback L507 Feedback L606
R58	Separate supplement and height CSV	Use the existing separate supplement and height CSV	The 96-height CSV matches Table S1 and the existing separate supplement matches V7's supplementary text and ten tables; V9 retains those contents unchanged. V7 P119 V7 P121 V9 P109 V9 P111 Sources Related feedback: Feedback L511 Feedback L513 Feedback L609
R59	Later channel-width measurements	Report the width estimates with their measurement method	The later check supplies nine brightfield ruler estimates from three devices, spanning 10.7–13.1 µm; moving each blurred-wall endpoint by one pixel changes a span by up to 2.37 µm, so report the pixel scale, endpoint rule and overlays with the estimates. V7 P53 V9 P65 V9 P89 Width data Related feedback: Feedback L136
R60	Individual mold-production	Keep the reuse observations; add per-mold records if	Both report more than 50 casts, more than eight months of use and both mold losses; existing per-mold records could add detail, but

ID	Change or work	My suggestion	What the sources show
	records	available	no new durability study is proposed. V7 P54 V9 P68 V9 P87 Related reuse report Related feedback: Feedback L138 Feedback L522
R61	Channel depth and between-cast comparisons	No new depth or cast-to-cast study proposed	The saved discussion proposes depth and between-cast measurements; integrate the available width estimates in R59 without adding those studies. V7 P53 V7 P72 V9 P65 V9 P89 Related feedback: Feedback L136 Feedback L346
R62	Figure layout and additional plots	Revisit image sizes and Figure 3/5 layout; integrate the supplied biology views	All five legend titles, device/alignment views and Figure 5's three application rows already exist; V9 removes the legends' explicit 11-point size and inherits 12-point text. Revisit image sizes and layout and integrate the recent device-dot plot and calibrated views; height and aggregate mold plots are possible uses of existing data, not proposed additions here. V7 P110 V7 P112 V7 P114 V7 P116 V7 P118 V9 P100 V9 P102 V9 P104 V9 P106 V9 P108 Sources Related feedback: Feedback L214 Feedback L238 Feedback L239 Feedback L262 Feedback L291 Feedback L313 Feedback L327 Feedback L398
R63	Small supplement edits	Add note citations and clarify the protocol inspection instruction	Add Figure S1/S2 links within the corresponding notes; Table S3 already separates tested and generated-only designs, and the protocol's open-channel result can become an inspection instruction; no raised-edge measurement experiment is proposed. V7 P152 V7 P153 V7 P155 V7 P157 V7 P158 V7 P159 V7 P161 V7 P332 V9 P142 V9 P143 V9 P145 V9 P147 V9 P148 V9 P149 V9 P151 V9 P322 Sources Related feedback: Feedback L450 Feedback L451 Feedback L475 Feedback L477 Feedback L541

Saved ChatGPT discussion, with V7/V9 comments

The pasted discussion follows. The comments link to table rows and quoted passages.

Source transcript

Feedback L1

Prompt: I'm confused about what is materials and methods vs what is results in this version of the manuscript

The confusion is understandable because your manuscript describes both the development of a fabrication method and the experimental validation of that method. As a result, several technical details could appear in either section.

For Lab on a Chip, I recommend a clear distinction:

Feedback L7

Materials and Methods should explain how the molds and devices were fabricated and how their performance was evaluated. Results should explain what was achieved, which technical problems were overcome, and how well the devices performed.

For example, your draft describes the use of a compressible EPDM rubber backing to align 96 resin inserts despite differences in their printed heights.

Feedback L11

The procedural details—rubber thickness, clamping arrangement, adhesive and curing conditions—belong in Materials and Methods. The observation that pin heights varied by 116 µm, that rigid clamping produced uneven contact, and that the compressible backing permitted simultaneous insert alignment belongs in Results.

Feedback L13

1. How I would reorganize your manuscript

Feedback L14

Your existing draft already has a reasonable Methods structure. I would retain most of its headings but reorganize the Results to tell a more compelling engineering story.

Materials and Methods	Corresponding Results
L19 OpenMFD design and fabrication-file generation	Parametric design of medium-throughput microfluidic devices
L20 Hybrid mold fabrication	Development and optimization of reusable hybrid molds
L21 PDMS casting and plate assembly	Repeated device fabrication and mold durability
L22 Mold and device characterization	Device integrity and compatibility with automated workflows
L23 Fluorescent tracer assay	Directional fluid transport between compartments

Materials and Methods	Corresponding Results
L24 Primary cortical neuron culture and CTβ retrograde axon tracing	Neuronal culture and compartment-specific axonal labeling
L25 Imaging and cell counting	Incorporated into neuronal validation Results
L26 Statistical analysis	Quantitative outcomes reported in the relevant Results sections

The important difference is that the Methods describe the experimental procedures, whereas the Results follow the engineering development and validation of the technology.

Feedback L29

2. Specific examples from your draft

Example A: Fabrication optimization

Feedback L31

Your current Results appropriately report several technical problems encountered during development:

- **Feedback L33**
  Raised SUEX edges interfered with insert bonding.
- **Feedback L34**
  SU-8 features detached when patterned directly onto silicon.
- **Feedback L35**
  Differences in printed insert heights prevented uniform bonding with rigid clamping.
- **Feedback L36**
  A compressible backing improved contact across the insert array.

These observations belong in Results because they demonstrate why particular design modifications were necessary.

Feedback L39

By contrast, exposure energies, baking temperatures, solvent treatments, adhesive application and clamping procedures belong in Methods.

I would retain these optimization experiments because they provide evidence that your method solves genuine fabrication problems.

Feedback L43

Example B: Mold durability

Feedback L44

The statement that three molds produced more than 50 usable casts is a Result.

Feedback L46
The procedure used to inspect molds and determine whether a cast was usable belongs in Methods.

Feedback L48
However, I would strengthen this section by distinguishing the number of successful casts from mold survival over time. These are related but different measurements of durability.

Feedback L50
Example C: Neuronal validation

Feedback L51
Your Methods should specify the neuronal preparation, plating density, culture conditions, timing and concentration of CTβ treatment, imaging procedure and analysis criteria.

Feedback L53
Your Results should report the observed axonal labeling, number of successfully analyzed devices, variability between plates and limitations in culture performance.

Feedback L55
Importantly, the poorer neuronal growth at the plate edges is a Result, not a methodological detail to omit.

Feedback L57
3. A recommended Results structure

Feedback L58
I suggest six principal Results subsections:

1. Parametric design of medium-throughput microfluidic devices

Feedback L62
Introduce OpenMFD, the coordinated generation of photomasks and insert arrays, the 48-device/96-well format, and the principal design features.

Feedback L64
2. Development of reusable hybrid photolithographic–3D-printed molds

Feedback L66
Describe the successful integration of SU-8/SUEX photolithography with printed inserts. This is where the optimization experiments involving SU-8 adhesion, SUEX exposure, insert alignment and compressible bonding should appear.

3. Reproducible fabrication and assembly of plate-format devices

Feedback L70
Present the finished PDMS devices, elimination of manual well punching, repeated casting, mold durability and integration with the supporting frame.

Feedback L72
4. Compatibility with automated liquid handling and high-content imaging

Feedback L74
Demonstrate the 9 mm well spacing, multichannel pipetting and imaging on standard high-content microscopy systems.

5. Characterization of fluid transport between compartments

Feedback L78
Present the fluorescent tracer experiments, including the observed directional redistribution and limitations of the measurements.

Feedback L80

6. Validation of compartmentalized primary neuronal culture

Feedback L82
Present neuronal survival and morphology, CTβ retrograde labeling, the 69 analyzed interior devices and the edge-related culture limitations.

Feedback L84
The alternative device designs generated by OpenMFD could be incorporated into Section 1 or presented as a short final Results subsection demonstrating design flexibility, provided it is clear that those configurations were generated computationally but not fabricated or biologically tested.

Feedback L86

4. One important editorial change

Feedback L87
I would move most of the highly detailed fabrication information into the supplementary protocols while retaining the essential technical specifications in the main Materials and Methods.

Feedback L89
Your current manuscript contains unusually detailed descriptions of printer settings, specific manufacturing steps and troubleshooting. That information is valuable for reproducibility, but presenting all of it in the main text risks obscuring the technological contribution.

Feedback L91
For example, the main Methods should describe the SU-8/SUEX layer architecture, the resin-insert fabrication and alignment procedure, bonding conditions and parylene coating. The supplementary protocol can provide every exposure setting, printer movement parameter, washing step and detailed assembly instruction.

I would not shorten the Methods at the expense of reproducibility. Rather, I would organize the information so that readers can understand the procedure from the main manuscript and reproduce it precisely using the supplement.

5. Recommended next step

Feedback L96
I recommend rewriting Materials and Methods first, using the existing draft as the technical source.

We should preserve the experimental details and numerical parameters, eliminate repetition, move observations and interpretations into Results, and identify any missing information needed for reproducibility.

Feedback L100
Once that is complete, we can rewrite the Results around the six sections above. This will allow

us to present the fabrication innovation much more effectively while ensuring that every experimental claim has a corresponding methodological description.

Feedback L102
I've also opened a Work-mode session for a more detailed section-by-section review of the manuscript, where we can reorganize the actual Word document without losing the original technical content.

Feedback L106

Figures

Feedback L108
One weakness is that Figures 2 and 3 primarily illustrate fabrication procedures rather than demonstrate fabrication performance.

Feedback L110
For a stronger engineering manuscript, I would consider adding a quantitative characterization figure documenting mold yield, repeated casting performance, and microchannel geometry. This could substantially strengthen the Results without requiring an additional biological experiment.

I'll also check the draft's supplementary tables to see which quantitative panels can be made from measured data, and which would need additional experiments.

The supplementary tables contain enough measured data for a useful preliminary figure: all 96 insert-pin heights, mold outcomes (five assembled, three used routinely), and the reported total of more than 50 usable casts. The limitation is that the draft does not give exact cast counts per mold or measured channel dimensions, so I won't imply those are available.

2. Sample quantitative characterization figure

Yes, you have enough data to produce a meaningful preliminary characterization figure, although additional measurements would strengthen it considerably.

I used data from Supplementary Tables S1 and S5 to generate three panels:

Panel	Measurement	Data available
L126 A. Spatial variation in insert height	Heatmap showing deviations in pin height across the 8 × 12 printed array	All 96 individual measurements
L127 B. Insert-height distribution	Histogram showing variability in 3D-printed insert heights	Mean 5.551 mm; SD 26.7 μm; range 116 μm
L128 C. Mold fabrication outcomes	Number of molds successfully used versus molds lost during fabrication or handling	3 routinely used; 2 lost; 5 total

The first two panels are particularly useful because they quantitatively establish a technical challenge that your fabrication method addresses: substantial variation in the heights of simultaneously printed inserts.

Feedback L131
The compressible EPDM backing is an important component of your innovation, and these measurements provide quantitative justification for its incorporation.

Feedback L133
3. What is missing for a stronger figure?

For a fabrication-focused Lab on a Chip manuscript, I would ideally add two additional measurements.

Feedback L136
Microchannel dimensional fidelity: Measure the width and height of several representative microchannels in finished PDMS devices. This would allow comparison with the nominal 10 μm width and demonstrate how accurately the photolithographic features are reproduced.

Feedback L138
Repeated casting performance: Report the number of successful PDMS casts from each of the three reusable molds. If records are available, these could establish manufacturing consistency and provide a more informative measure of mold durability.

Feedback L140
These experiments are more directly relevant to the central technological claims than additional biological assays.

4. My recommendation

I would develop the sample into a dedicated figure titled:

Feedback L145
Figure X. Fabrication reproducibility and durability of hybrid photolithographic–3D-printed molds.

The strongest final version would include spatial variation in resin-insert heights, finished microchannel dimensions, and repeated casting performance.

Feedback L149
Your current measurements are sufficient for an initial figure, but microchannel dimensional fidelity is the most important missing quantitative characterization.

That measurement would provide direct evidence that the hybrid mold retains the micrometer-scale precision needed for compartmentalized neuronal assays.

Sample_Quantitative_Device_Characterization.png

Feedback L163
Editorial Assessment

Feedback L165
One issue deserves particular attention before submission: The manuscript would benefit from direct measurements of finished microchannel dimensions. This would support the central claim that combining photolithography with 3D-printed inserts preserves the fine structures required for neuronal compartmentalization.

Figures

Recommendations for the figures in your Lab on a Chip manuscript

Feedback L174
I reviewed the five figures in your original manuscript against the revised Results and Discussion. My main recommendation is to make the figures less like illustrations of a fabrication protocol and more like evidence supporting the technological innovation and experimental validation.

The existing figures contain most of the visual material you need, but their presentation could be substantially improved.

I would retain the five-figure structure, consider adding one quantitative fabrication-characterization figure, and standardize the legends.

Feedback L180

1. Incorporating figure titles into legends

Feedback L181
For Lab on a Chip, I recommend using a short, descriptive title as the first sentence of each figure legend, followed immediately by the explanation of the individual panels.

For example:

Fig. 1. Parametric design of hybrid molds for medium-throughput microfluidic devices. (A) OpenMFD generates coordinated fabrication files from shared design parameters. (B) Single-device photomask layout. (C) Wafer-scale photomask layout. (D) Three-dimensional models of individual well-forming inserts and the complete insert array. (E) Supporting plate-frame model. (F) Assembled microfluidic culture plate containing 48 two-compartment devices and 96 wells.

This is preferable to placing a large descriptive title within the figure itself and then repeating that title in the legend.

I suggest three conventions throughout the manuscript:

- **Feedback L191**
  Within the figure: Use panel letters (A–F), concise labels, dimensions, and annotations necessary to interpret the images.

- **Feedback L192**
  First sentence of the legend: State the principal message of the figure in bold.

- **Feedback L193**
  Remaining legend: Explain the panels, relevant experimental conditions, sample sizes, and quantitative measurements.

The figure titles should communicate what each figure demonstrates, not simply describe its contents.

Feedback L196
For instance, Development of reusable hybrid molds for wafer-scale microfluidic fabrication is stronger than Insert transfer and wafer bonding for the hybrid mold.

2. Figure-by-figure recommendations

Feedback L199

Figure 1. Parametric design and device architecture

Feedback L200

Current figure: Page 13 of the original manuscript.

This figure shows the OpenMFD design interface, photomasks, individual resin inserts, insert arrays, supporting frame, and assembled device.

The essential information is present, but the figure is visually fragmented. The software outputs and photographs are displayed at different scales, and the relationships between the components are not immediately obvious.

Suggested improvements

Feedback L208

Make the central message that one parametric design generates every component necessary to manufacture the device.

I would retain the existing panels but reorganize them into a more continuous workflow:

Feedback L212

OpenMFD parameters → Photomasks and resin inserts → Reusable mold → Complete microfluidic plate

Feedback L214

I would particularly recommend adding a larger schematic of one two-compartment device showing the wells and connecting microchannels. The current figure does not sufficiently emphasize the neuronal compartmentalization geometry.

Feedback L216

The 48-device/96-well organization and 9 mm spacing should also be clearly indicated.

Proposed title and legend

Feedback L220

Fig. 1. Parametric design enables coordinated fabrication of medium-throughput microfluidic devices. (A) OpenMFD generates matched photolithographic masks, three-dimensional resin inserts, and supporting plate components from shared design parameters. (B) Schematic of an individual two-compartment device showing culture wells connected by parallel microchannels. (C) Wafer-scale photomask layout containing 48 two-compartment devices. (D) Models of individual resin inserts and the complete 96-insert array. (E) Three-dimensional model of the supporting plate frame. (F) Assembled PDMS microfluidic plate containing 96 wells arranged on a standard 9 mm grid.

Note: Panel B would require modifying the existing figure.

Feedback L224

Figure 2. Hybrid mold fabrication and alignment

Feedback L225

Current figure: Page 14.

This is potentially one of the most important figures because it illustrates your central engineering innovation.

However, the current arrangement gives similar visual emphasis to the clamping equipment, alignment geometry, and photographs of the bonded inserts.

Feedback L231
I would make the simultaneous alignment of 96 inserts the dominant visual element.

The figure should establish how the shared coordinate system, corresponding pins and holes, and compressible backing work together.

I recommend emphasizing the following features:

- **Feedback L237**
  Registration of the entire resin-insert array with the wafer.
- **Feedback L238**
  Enlargement of the square alignment pin and SUEX opening.
- **Feedback L239**
  Cross-sectional schematic showing the compressible EPDM backing.

- **Feedback L240**
  Photograph of the successfully assembled mold.

Feedback L241
The 1.85 mm pin, 2.00 mm opening, and 75 µm clearance per side should remain clearly labeled.

Proposed title and legend

Fig. 2. Simultaneous alignment and integration of 3D-printed inserts into reusable photolithographic molds. (A) Patterned silicon wafer and resin-insert array before assembly. (B) Schematic of the alignment interface showing the 1.85 × 1.85 mm resin pin positioned within a 2.0 × 2.0 mm SUEX opening, providing 75 µm nominal clearance per side. (C) Clamping assembly used to bond the inserts to the wafer. (D) Cross-sectional schematic illustrating the compressible EPDM backing used to accommodate variations in insert height. (E) Photograph of bonded resin inserts. (F) Micrograph of the patterned SUEX alignment openings.

Feedback L247
I would consider incorporating the measured 116 µm variation in insert height into this figure as a quantitative panel. That would directly connect the engineering problem with the solution shown.

Feedback L249
Figure 3. Repeated casting and plate assembly

Feedback L250
Current figure: Page 15.

This figure illustrates PDMS pouring, degassing, curing, demolding, trimming, bonding, and frame assembly.

Feedback L254
Although informative, it is currently closer to a supplementary fabrication protocol than a central Results figure.

I would simplify it considerably.

The central message should be:

Feedback L260
One reusable hybrid mold produces multiple complete microfluidic plates without additional well punching.

Feedback L262
I would retain a clear schematic of the principal stages and move some of the more detailed assembly photographs to the supplementary information.

Feedback L264
Ideally, this figure would also include a quantitative demonstration of mold reuse.

Proposed title and legend

Feedback L268
Fig. 3. Repeated fabrication of plate-format microfluidic devices using reusable hybrid molds. Schematic illustrating the production of complete PDMS microfluidic arrays from a reusable silicon-wafer mold. PDMS is cast over the hybrid mold, degassed, cured, and demolded to produce 48 two-compartment devices with 96 preformed culture wells. Each cast is trimmed, plasma-bonded to glass, and incorporated into a supporting plate frame. Because the resin inserts remain attached to the mold, subsequent casting cycles generate all culture wells without individual punching. Three molds collectively produced more than 50 usable casts.

For the last sentence, adding a quantitative panel would be particularly valuable.

Feedback L272
Figure 4. Functional validation of neuronal microfluidic devices

Feedback L273
Current figure: Page 16.

Feedback L275
This is the most important biological validation figure and, in my opinion, the one that would benefit most from substantial revision.

The current figure combines fluorescent tracer redistribution, CTβ-labeled neurons, and quantitative neuronal labeling. These are appropriate experiments, but the small images make it difficult to assess the results.

Suggested improvements

Feedback L281
I would reorganize Figure 4 into five panels:

Feedback L283
A. Experimental schematic: Show a two-compartment device indicating the direction of fluid movement and the site of CTβ application.

Feedback L285

B. Fluid transport: Retain the fluorescent tracer images at 0 and 72 hours, but enlarge representative devices so that redistribution between compartments is evident.

Feedback L287

C. Neuronal culture: Include a high-resolution image illustrating neuronal cell bodies, microchannels, and axonal extension. A representative image spanning both compartments would be particularly informative.

Feedback L289

D. Retrograde labeling: Include higher-magnification images showing CTβ-positive neuronal cell bodies following tracer administration to the axonal compartment.

Feedback L291

E. Quantitative analysis: Retain the CTβ-positive cell count data, but consider displaying individual device measurements rather than only the plate means and standard deviations.

Feedback L293

Displaying individual data points would demonstrate the range of responses across the 69 analyzed devices. It would also avoid creating the impression that the three plates represent independent biological replicates.

Feedback L295

Proposed title and legend

Feedback L297

Fig. 4. Functional validation of compartmentalized neuronal culture and retrograde axonal labeling. (A) Schematic illustrating the two-compartment microfluidic architecture, direction of fluid movement established by a hydrostatic volume difference, and selective application of CTβ to the axonal compartment. (B) Fluorescence images showing tracer distributions immediately after loading and following 72 h of incubation. Alexa Fluor 568-conjugated antibody (red) and Alexa Fluor 488-conjugated antibody (green) were introduced into opposite compartments at initial volumes of 25 and 50 µL, respectively (24 devices). (C) Representative image of primary cortical neurons cultured in the microfluidic devices at DIV11. (D) Representative fluorescence images showing retrogradely labeled neuronal cell bodies following application of CTβ-647 to the axonal compartment at DIV10. (E) Quantification of CTβ-positive neuronal cell bodies across three independently assembled plates. Data represent 69 analyzed interior devices derived from one embryonic neuronal preparation; devices and plates are technical replicates.

Panels A and C would require additional figure preparation. Panel E could be improved using the existing individual cell-count dataset.

Feedback L301

I would also ensure that all fluorescence micrographs have appropriate scale bars.

Feedback L303

Figure 5. Adaptability of the fabrication approach

Feedback L304

Current figure: Page 17.

This figure shows OpenMFD-generated designs for three alternative applications:

- Oligodendrocyte myelination.
- Axon guidance.
- Three-compartment neuronal culture.

The scientific purpose is appropriate, but the figure currently contains considerable redundancy because each design is represented by several very similar software outputs.

Feedback L313

I would simplify the figure into three rows, one for each application, with three columns:

Application	Device geometry	Corresponding resin-insert array
Myelination	Two cell populations connected by microchannels	Matched insert configuration
Axon guidance	Orthogonal compartment arrangement	Matched insert configuration
Three-compartment culture	Three interconnected compartments	Matched insert configuration

Feedback L321

The wafer-scale layout could be included as a small inset rather than a separate full-sized panel.

Feedback L323

Proposed title and legend

Fig. 5. Parametric generation of alternative compartmentalized microfluidic device configurations. OpenMFD-generated photomasks and corresponding resin-insert arrays for three literature-inspired microfluidic architectures. (A, B) Configuration for studying oligodendrocyte myelination and neuron–glia interactions. (C, D) Configuration incorporating orthogonal compartments for axon-guidance experiments. (E, F) Three-compartment neuronal culture configuration. For each design, changes in compartment geometry, well arrangement, and microchannel organization are incorporated into matched photolithographic and three-dimensional fabrication files. These configurations were generated computationally and have not been fabricated or experimentally validated.

Feedback L327

One important point: I would make it clear within the figure itself that these are computational designs, not experimentally validated devices.

Feedback L329

3. Should you add a quantitative characterization figure?

Yes. I believe this would be the single most valuable improvement to the manuscript.

The strongest publications describing new microfabrication technologies generally combine an explanation of the fabrication process with quantitative evidence of manufacturing performance.

Currently, your manuscript contains excellent illustrations of how the devices are made, but less quantitative evidence demonstrating how reliably they can be manufactured.

I recommend developing a new figure with the following structure.

Proposed Figure 6. Quantitative characterization of hybrid mold fabrication

L341 Panel	Content	Availability
A	Heatmap of measured insert-height variation across the 96-position array	Available
B	Distribution of individual insert heights	Available
C	Mold fabrication outcomes and repeated casting performance	Partially available
L345 D	Measured microchannel widths in finished PDMS devices	Additional measurements required
L346 E	Comparison of microchannel dimensions between successive casts or different molds	Additional measurements required

The existing measurements provide a mean pin height of 5.551 mm, standard deviation of 26.7 µm, and range of 116 µm. These results are directly relevant to the compressible alignment method.

Feedback L349
However, I would not present the three usable molds out of five assembled molds as a general manufacturing success rate without explaining the two failures and the limited sample size.

Feedback L351
The most important new measurement would be the dimensional accuracy of the microchannels in the finished PDMS devices.

Suggested title and legend

Fig. 6. Quantitative characterization of hybrid mold fabrication and device reproducibility. (A) Spatial distribution of measured resin-insert heights across a 96-position array, expressed as deviations from the mean. (B) Distribution of insert heights (mean, 5.551 mm; standard deviation, 26.7 µm; range, 116 µm). (C) Fabrication outcomes for five hybrid molds, three of which entered routine use and collectively produced more than 50 usable PDMS casts. (D, E) Proposed quantitative measurements of microchannel dimensional fidelity and variation between successive PDMS casts.

Feedback L357
Panels D and E should only be included once actual measurements are available.

Feedback L359
An alternative would be to combine this quantitative figure with the existing Figure 2, producing

a stronger composite figure that links fabrication challenges, engineering solutions, and manufacturing performance. This would avoid increasing the number of main figures.

Feedback L361

4. Additional recommendations for figure presentation

There are several changes I would apply consistently.

Feedback L364

Remove the large titles embedded within the figures. The current figures sometimes contain descriptive headings that duplicate information in the legends. These take up space and reduce the size available for scientific images. The short bold title at the beginning of the figure legend should carry that information.

Feedback L366

Use consistent labeling. Panel letters should be placed in the same position, using the same font and size, throughout all figures. I recommend capital letters (A, B, C) and a consistent sans-serif font for labels within figures. Your manuscript text and Word figure legends will continue to use Times New Roman, 12 pt, black, justified, and single-spaced.

Feedback L368

Emphasize quantitative findings. Where possible, integrate measurements directly into the relevant figure rather than placing all numerical information in supplementary tables.

Feedback L370

Make the microchannels more visible. The connecting microchannels are central to the biological application and technical demands of the platform. At least one figure should show them clearly at high magnification.

Feedback L372

Reduce unnecessary photographs of equipment. Photographs of clamps, racks, and assembly stages can be useful for reproducibility but are often better suited to supplementary protocols. The main figures should prioritize the fabricated structures and evidence of device performance.

5. Recommended final figure organization

Feedback L375

I would ultimately aim for five main figures rather than six:

L379 **Final figure**	**Principal message**	**Source**
L380 Figure 1	Parametric design generates coordinated fabrication components	Revised existing Fig. 1
Figure 2	Hybrid fabrication enables simultaneous insert alignment, with quantitative validation	Existing Fig. 2 + characterization data
L382 Figure 3	Reusable molds enable repeated fabrication of complete plate-format devices	Revised existing Fig. 3

L379 Final figure	Principal message	Source
Figure 4	Devices support directional fluid movement and compartmentalized neuronal assays	Expanded existing Fig. 4
Figure 5	The design framework accommodates alternative microfluidic architectures	Simplified existing Fig. 5

If you obtain quantitative measurements of microchannel fidelity and between-cast reproducibility, a dedicated sixth figure may become justified.

My recommendation

Feedback L388

I would prioritize revisions in this order:

Feedback L390

First, improve Figure 2 by integrating the existing insert-height measurements with the alignment and bonding schematics. This would strengthen the presentation of the central engineering innovation.

Feedback L392

Second, improve Figure 4 by adding larger representative neuronal images and displaying the individual CTβ-positive cell counts. This would make the biological validation more convincing.

Feedback L394

Third, simplify Figures 1, 3, and 5 to remove redundant information and establish a consistent visual style.

The central principle should be that each figure demonstrates one important finding from the Results, while the figure legend provides the experimental detail necessary to interpret it.

Feedback L398

I would also keep the legends reasonably concise—ideally approximately 60–120 words each, depending on the number and complexity of panels—rather than repeating extensive methodological descriptions already provided in Materials and Methods.

Feedback L400

This is a conceptual layout only, not a presentation of experimental results. The microscopy images and individual data points are illustrative and must be replaced with your actual data before publication.

PastedGraphic-1.png

Prompt: are all figures including supplemental figures explained and cited in the manuscript draft. Is the supplemental material appropriate and what modifications would you suggest

Feedback L406

Review of figures and Supplementary Information — Lab on a Chip

Overall, the supplementary material is appropriate for a methods-development manuscript, but I identified several issues that should be corrected before submission.

Feedback L409

I reviewed the latest expanded manuscript, including the five main figures, two supplementary figures, six supplementary tables, two supplementary notes, and two detailed fabrication and experimental protocols.

The main findings are:

- Feedback L413
  All five main figures are cited in the manuscript, and their content generally corresponds to the Results.
- Feedback L414
  Neither Supplementary Figure S1 nor Supplementary Figure S2 is explicitly cited in the main manuscript.Both are discussed indirectly through references to Supplementary Notes S1 and S2.
- Feedback L415
  Supplementary Table S2 is not cited anywhere in the main manuscript. It contains a useful comparison of existing microfluidic fabrication approaches.
- The remaining supplementary tables, notes, and protocols are referenced.
- Some supplementary information could be reorganized to improve readability and strengthen the evidence supporting the fabrication method.

I would retain most of the supplementary content. For a Lab on a Chip manuscript introducing a fabrication technology, detailed protocols, printing parameters, design specifications, and quantitative measurements are particularly valuable.

Feedback L420

1. Audit of the main figures

Figure	Content	Cited in main text?	Assessment
Figure 1	OpenMFD design, fabrication files, and assembled device	Yes — Methods and Results	Appropriate
Figure 2	Insert alignment and hybrid mold assembly	Yes — Methods and Results	Appropriate
L426 Figure 3	Repeated PDMS casting and plate assembly	Yes — Results	Appropriate

Figure	Content	Cited in main text?	Assessment
Figure 4	Directional tracer movement and neuronal validation	Yes — Results	Appropriate, with some qualifications
Figure 5	Alternative microfluidic configurations generated by OpenMFD	Yes — Results	Appropriate

Figure 4 warrants particular attention

Feedback L430
The manuscript appropriately cites Figure 4A for fluorescent tracer redistribution, Figure 4B for CTβ retrograde labeling, and Figure 4C for neuronal cell counts.

Feedback L432
The interpretation is also appropriately cautious: the tracer experiment demonstrates redistribution but does not establish sustained fluidic isolation, and the neuronal experiments involve three technical-replicate plates derived from one embryonic preparation.

I would retain those qualifications.

Feedback L436
One minor modification is needed in the Figure 5 legend. It still identifies the source papers using author–year citations, whereas you have established a preference for PMIDs throughout this manuscript. These should be converted to the corresponding verified PMIDs.

2. Audit of the supplementary material

Supplementary item	Cited in main manuscript?	Assessment and recommendation
L442 Figure S1 — Effect of LP360 filtering on SUEX	No direct citation	Retain. Add an explicit citation in the fabrication-optimization Results.
L443 Figure S2 — SU-8 delamination	No direct citation	Retain. Directly supports an important fabrication challenge. Add a Results citation.
L444 Table S1 — Resin-insert height measurements	Yes	Essential. Keep the 96 individual measurements. Improve table formatting.
L445 Table S2 — Comparison of fabrication approaches	No	Useful but needs revision. Cite in the Introduction, simplify the comparison, and standardize references.

Supplementary item	Cited in main manuscript?	Assessment and recommendation
L446 Table S3 — Design parameters and validation checks	Yes	Very valuable. Retain, but improve readability and distinguish validated parameters from design assumptions.
L447 Table S4 — Resin-printing parameters	Yes	Essential for reproducibility. Retain.
L448 Table S5 — Mold fabrication and casting outcomes	Yes	Important but insufficiently quantitative. Add individual mold-production records if available.
L449 Table S6 — HIPS frame-printing settings	Yes	Appropriate. Retain.
L450 Note S1 — LP360 exposure optimization	Yes	Appropriate, but explicitly reference Figure S1 within the note.
L451 Note S2 — SU-8 adhesion and thermal processing	Yes	Appropriate, but explicitly reference Figure S2 within the note.
L452 Protocol S1 — Device assembly and neuronal culture	Yes	Essential. Retain the detailed protocol.
L453 Protocol S2 — Hybrid mold fabrication	Yes	Essential. This is one of the most valuable components of the supplementary material.

Feedback L454

Supplementary Figures S1 and S2

These are scientifically relevant because they document technical problems encountered during fabrication.

Figure S1 provides a qualitative comparison of SUEX structures exposed with and without the LP360 filter. This supports your observation that the filter reduces raised-edge artifacts.

Figure S2 shows SU-8 delamination when microchannel features are patterned directly onto silicon without the continuous SU-8 base layer.

Both images provide useful evidence supporting the rationale for fabrication optimization.

I would add these references directly to the Results.

Feedback L465

For Figure S1, the relevant Results sentence could be revised to:

Feedback L467

Incorporating the LP360 long-pass filter substantially reduced the raised-edge artifacts observed following unfiltered SUEX exposure, producing visibly flatter surfaces suitable for resin-insert bonding (Supplementary Fig. S1).

For Figure S2:

Feedback L471

SU-8 microchannel structures patterned directly onto silicon exhibited delamination during subsequent processing (Supplementary Fig. S2), whereas incorporating a continuous, flood-exposed SU-8 base layer prevented this observed failure.

I would also add explicit references to the figures within Supplementary Notes S1 and S2.

Feedback L475

One potential improvement: Figure S2 currently documents the unsuccessful fabrication condition. A corresponding image showing intact SU-8 structures following incorporation of the base layer would provide a more convincing visual demonstration of the improvement.

Feedback L477

Similarly, Figure S1 is qualitative. Quantitative measurements of the raised-edge features would strengthen the comparison, although they are not essential to retain the figure.

Feedback L479

3. Supplementary Table S2 needs attention

Feedback L480

This is probably the supplementary table I would revise most substantially.

It compares approximately ten fabrication approaches, including conventional soft lithography, hot embossing, 3D printing, hybrid molding, and commercial devices.

This comparison is relevant because the novelty of your manuscript depends on clearly distinguishing your fabrication strategy from earlier multiscale approaches.

However, I identified three problems.

Feedback L488

First, Table S2 is not cited in the Introduction or Discussion. I would add a reference to it in the Introduction, where previous fabrication strategies are described.

A suitable sentence would be:

Feedback L492

Existing approaches for integrating micrometer-scale channels with millimeter-scale culture compartments differ in their requirements for specialized equipment, mold preparation, component alignment, and repeated device production (Supplementary Table S2).

Feedback L494

Second, the table is difficult to read. The six columns contain considerable text, and several rows extend across multiple pages.

I would shorten the entries substantially and consider using a landscape page orientation.

Feedback L498
Third, the representative references are currently given largely in author–year format. These should be converted to your established PMID convention, using DOIs only when a verified PMID is unavailable.

Feedback L500
I would also describe the table as a qualitative comparison of fabrication strategies, not a quantitative comparison of manufacturing efficiency. The manuscript does not contain experiments directly comparing fabrication time, costs, yields, or dimensional accuracy across the different approaches.

4. The supplementary tables need formatting improvements

I inspected the rendered supplementary pages, and this is a significant presentation issue.

Feedback L505
Supplementary Table S1: Some numerical values wrap across lines because the columns are too narrow. For measurements reported to three decimal places, this makes the data difficult to interpret and creates a risk of transcription errors.

Feedback L507
Supplementary Tables S2 and S3: These are particularly difficult to read. Their narrow columns produce substantial text wrapping, and individual tables extend across several pages.

Feedback L509
Supplementary Tables S4 and S6: These are easier to follow because they largely contain two-column lists of printing parameters.

Feedback L511
My recommendation is to prepare the Supplementary Information as a separate document, using landscape orientation for Tables S1–S3 and portrait orientation for the remaining material.

Feedback L513
I would also provide the individual height measurements from Table S1 as a CSV file.

This is consistent with RSC guidance, which encourages accessible, machine-readable supporting data and notes that supplementary files are published without editorial reformatting. Royal Society of Chemistry

Feedback L517

5. Supplementary Table S5 should provide more quantitative manufacturing information

Table S5 currently reports that five hybrid molds were assembled, three entered routine use, two were lost, and the three working molds collectively produced more than 50 usable casts.

These are useful findings, but the table would be substantially stronger if it contained mold-specific production data.

Feedback L522
Ideally, each mold would have an individual record of the number of casts attempted, number meeting the acceptance criteria, reason for failure or retirement, and duration of use.

This would allow readers to distinguish initial fabrication success from mold durability and casting reproducibility.

Feedback L526
If those records are unavailable, I would retain the existing summary but avoid implying that a precise manufacturing yield has been established.

Feedback L528
I would also ensure that the supplementary definition of a usable cast is presented clearly as an inspection-based criterion, rather than implying that channel dimensional accuracy was measured.

6. Are the supplementary protocols appropriate?

Yes. I would retain both protocols, including their detailed experimental parameters.

In fact, they are particularly valuable for a manuscript introducing a fabrication method intended for adoption by other laboratories.

Feedback L535
Supplementary Protocol S2 describes the novel hybrid-mold fabrication process and should remain comprehensive. Details such as photoresist processing, insert transfer, alignment, adhesive curing, parylene deposition, and demolding are central to reproducibility.

Feedback L537
Supplementary Protocol S1 provides the corresponding procedures for casting, device assembly, neuronal culture, tracer application, and imaging.

Feedback L539
The division between these protocols is logical: one describes the specialized fabrication of the reusable mold, while the other explains how the mold is used to produce and operate experimental devices.

Feedback L541
I would make only modest editorial changes, primarily eliminating duplicated explanations and ensuring consistency with the main Materials and Methods.

Feedback L543
One specific detail deserves clarification. Protocol S1 refers to neuronal images acquired before axotomy in a particular experiment. Since axotomy is not presented as an experimental endpoint in this manuscript, I would clarify that these are baseline measurements from an experiment conducted before axotomy, rather than imply that axotomy was part of the reported validation.

7. Additional items that should be addressed before submission

Feedback L546
A. Provide the underlying neuronal count data

Feedback L547
The manuscript reports CTβ-positive neuronal cell counts from 69 interior devices, but the individual measurements are not included in the supplementary tables.

The Data Availability statement indicates that these data will be deposited.

Feedback L551
I recommend supplying a CSV file containing the per-device counts, together with plate and device identifiers and an explicit record of the three excluded devices.

Feedback L553
This would allow reviewers to evaluate the variability represented in Figure 4C.

Feedback L555
B. Provide a complete set of fabrication files

An unusual advantage of your manuscript is that the complete fabrication design can be made available.

Lab on a Chip specifically encourages authors to submit chip-design masks as electronic supplementary information. Royal Society of Chemistry

Feedback L560
I recommend ensuring that the downloadable package contains the actual photomask DXF files, insert STL files, frame designs, and the version of OpenMFD used to generate the reported devices.

Feedback L562
The printing G-code and microscope XML files should also be included or linked to a stable repository.

Feedback L564
Providing a permanently archived software release with a DOI would be preferable to relying exclusively on a GitHub repository that may change.

Feedback L566
C. Standardize references throughout the supplement

Feedback L567
Several supplementary items still use author–year references, including Table S2, the Figure 5 legend, and portions of the protocols.

Feedback L569
These should be reconciled with the PMID-based citations used in the manuscript.

Feedback L571
For final submission, note that Lab on a Chip requires numbered superscript citations and a numbered reference list. The PMIDs can remain in your working draft and be converted during final reference formatting. Royal Society of Chemistry

Feedback L573
D. Add a complete animal ethics statement

The supplementary neuronal culture protocol states that the embryonic cortical neurons were prepared under an approved animal-use procedure.

Feedback L576
However, the main Materials and Methods does not identify the approving institution or protocol number.

I recommend adding this information to the main manuscript, using the actual approval details. RSC's author guidelines require appropriate ethical reporting for studies involving animals. Royal Society of Chemistry

Feedback L580

E. Correct the placement of the Data Availability statement

Feedback L581

The latest manuscript places Data availability before Conflict of Interest Statement.

Feedback L583

RSC specifies the reverse order: the Data Availability statement should follow the conflicts-of-interest statement and precede the Acknowledgements. Royal Society of Chemistry

This is a straightforward correction.

Feedback L587

8. Recommended final organization of the Supplementary Information

Feedback L588

I would organize it into three components.

Component	Contents
Supplementary Methods	**Protocol S1:** PDMS casting, assembly, neuronal culture and assays. Protocol S2: Hybrid mold fabrication.
Supplementary Characterization and Design	Figures S1–S2, Notes S1–S2, Tables S1–S3 and S5.
Supplementary Fabrication Parameters	Tables S4 and S6, with associated G-code, CAD files and microscopy settings available separately.

This would create a more logical distinction between the instructions needed to reproduce the device and the experimental evidence supporting the fabrication method.

Feedback L598

I would keep the source CAD and numerical datasets in machine-readable formats rather than relying solely on tables embedded in a Word document.

Feedback L600

9. Overall recommendation

Feedback L601

I would retain all five main figures, both supplementary figures, and most of the existing supplementary material.The content is scientifically relevant and broadly appropriate for Lab on a Chip.

My priorities, in order, would be:

- **Feedback L605**
  Correct the missing citations to Supplementary Figures S1–S2 and Table S2.

- **Feedback L606**
  Reformat Supplementary Tables S1–S3, especially the numerical measurements and wide comparison tables.

- **Feedback L607**
  Add the underlying CTβ cell-count dataset and more detailed mold-production records, if available.
- **Feedback L608**
  Standardize the references and clarify the baseline neuronal imaging experiment.
- **Feedback L609**
  Prepare a separate, well-organized Supplementary Information document and an archived package of the fabrication files.

The supplementary material is already one of the stronger aspects of the manuscript because it provides unusually detailed instructions for reproducing the fabrication process.

Feedback L612
The most significant remaining weakness is not the quantity of supporting information, but the limited quantitative characterization of mold performance and finished microchannel geometry.

Providing those measurements would strengthen both the main Results and the Supplementary Information.

Feedback L616
I would address the citation and formatting problems now, while treating additional microchannel measurements and mold-specific production records as the highest-value experimental improvements before submission.

Details and cited V7/V9 passages

R01. Title Table row R01

V9 retains the V7 title, including PDMS, SU-8 and the hybrid wafer mold.

My suggestion: Keep the shared title

V7 P1 Table row R01

> *Plate-format PDMS microfluidic culture devices from hybrid SU-8/3D-printed resin wafer molds*

V9 P1 Table row R01

> *Plate-format PDMS microfluidic culture devices from hybrid SU-8/3D-printed resin wafer molds*

R02. Authors and affiliations Table row R02

V9 adds numbered affiliations and corresponding-author notation; its first affiliation now includes the department and institution.

My suggestion: V9 layout

V7 P2 **Table row R02**

> *Authors: Tristan Simas, Yonatan Morocz, Jathav Puvirajan, David Juncker, and Alyson FournierAffiliations: McGill UniversityCorresponding author: Tristan Simas*

V9 P3 **Table row R02**

> *Tristan Simas1,#, Yonatan Morocz2, Jathav Puvirajan1, David Juncker2, and Alyson Fournier1*

V9 P6 **Table row R02**

> *1Department of Neurology and Neurosurgery, Montréal Neurological Institute, McGill University, Montréal, Québec, Canada*

V9 P7 **Table row R02**

> *2Victor Phillip Dahdaleh Institute of Genomic Medicine, McGill University, Montréal, QC, Canada.*

V9 P8 **Table row R02**

> *#Corresponding author*

R03. Abstract **Table row R03**

V9 begins with neuronal applications and drops V7's measured reuse totals (>50 casts; >8 months) and edge-growth limitation. Its "precise alignment" and "compartmentalized axonal growth" wording is stronger than the nominal registration and indirect-labeling accounts in Results/Discussion; keep the fabrication workflow, reuse observations and culture limit, using the reconciled device count from R53.

My suggestion: V7 framing with a concise device count

Related feedback: Feedback L82 Feedback L136

V7 P4 **Table row R03**

> *Plate-format devices made from polydimethylsiloxane (PDMS), a silicone rubber, can combine spatially controlled cell culture with multichannel liquid handling and automated imaging, but pipette-accessible wells are often punched individually after curing. We developed a design and fabrication method for reusable silicon-wafer molds that combine microfeatures patterned in SU-8/SUEX light-sensitive resins with 3D-printed resin inserts to form all wells during casting. OpenMFD generates the photomasks and insert array from shared device dimensions so that the printed pins match the alignment-hole positions on the wafer. The inserts are then transferred and bonded as one array. Three molds produced more than 50 usable casts, and the most recent molds remained in use for more than 8 months without visible degradation. The tested plate contained 48 two-compartment devices with 96 wells on the standard 9 mm grid, supporting multichannel pipetting and imaging on unmodified high-content systems. Assays demonstrated directional tracer movement, primary cortical neuron culture, and retrograde axon labeling. Culture validation was limited to interior devices because growth was poorer at the plate edges. Once the mold is made, subsequent batches use benchtop casting and assembly, with all wells formed in each cast.*

V9 P11 **Table row R03**

> *Microfluidic platforms enable spatially controlled neuronal culture and compartment-specific manipulation, but their adoption in medium-throughput assays is limited by fabrication*

> *methods that require labor-intensive processing and assembly. Here, we present an open-source, hybrid microfabrication strategy for producing reusable silicon-wafer molds that integrate photolithographically defined microchannels with 3D-printed structures for simultaneous formation of culture wells. A parametric design framework, OpenMFD, generates coordinated photomasks and three-dimensional insert arrays, enabling precise alignment and simultaneous bonding of millimeter-scale structures to SU-8/SUEX-patterned wafers. Using this approach, we fabricated polydimethylsiloxane (PDMS) culture plates containing 48 compartmentalized devices arranged on a standard microplate grid, eliminating individual well punching during device production. Devices supported multichannel liquid handling, automated high-content imaging, directional fluidic manipulation, and primary cortical neuron culture. Compartmentalized axonal growth was demonstrated by retrograde cholera toxin B labeling establishing compatibility with parallel neuronal assays. Furthermore, the parametric design framework generated matched fabrication files for additional neuronal culture configurations, illustrating the adaptability of the approach. By combining high-resolution photolithography with scalable three-dimensional molding, this method provides an accessible and reusable fabrication platform for medium-throughput microfluidic assays.*

R04. Introduction: applications Table row R04

V7 connects the method to neuroscience, stem-cell biology, cancer, immunology and developmental biology; V9 replaces this with neuronal connectivity and perturbations, understating the generalizability of the fabrication platform.

My suggestion: V7 account of the applications

V7 P6 Table row R04

> *Microfluidic devices give cell biologists spatial control over cell bodies, axons, local reagent exposure, gradients, migration, differentiation, and co-culture interactions. Established designs support applications across neuroscience, stem cell biology, cancer research, immunology, and developmental biology (Taylor et al., 2005; Taylor et al., 2015; Wang et al., 2018; Coquinco et al., 2014), and commercial devices are available from suppliers including Xona Microfluidics and ANANDA Devices. Plate-format devices extend these capabilities to experiments with many replicates using multichannel pipettes and automated imaging. For academic laboratories, custom designs and affordable local fabrication are important.*

V9 P13 Table row R04

> *Microfluidic compartmentalization has transformed the study of neuronal cell biology by enabling independent access to cell bodies, axons and their local extracellular environments. Microchannel-connected culture chambers permit spatially restricted treatments, isolation of axonal material and analysis of axonal injury, regeneration and transport (PMID: 16094385). Subsequent designs have enabled controlled axon-guidance gradients, localized neuron–glia interactions and more complex neuronal networks (PMIDs: 26000554; 22828584; 35466333). The geometry of connecting microchannels also influences device performance and the study of communication between neuronal populations (PMID: 36111641). Together, these developments have established compartmentalized microfluidics as a versatile approach for investigating neuronal connectivity and responses to spatially defined perturbations.*

R05. Introduction: prior plate systems Table row R05

V9 adds high-capacity, modular and fluid-walled neuronal systems; check each new reference and the claim it supports, then use them as application context alongside the comparison of fabrication operations and equipment in V7 and Table S2.

My suggestion: Platform context; fabrication methods remain the main comparison

V7 P7 **Table row R05**

> *Experiments with many biological replicates or treatment groups benefit from devices arranged for standard pipetting and long-term culture. Multiwell plates provide this format. Many academic laboratories make custom devices from polydimethylsiloxane (PDMS), a silicone rubber. They use light-based patterning (photolithography) to fabricate a reusable mold, then cast the devices, remove them from the mold (demolding), and bond them to glass. Commercial plates and hard-plastic methods based on hot embossing (pressing heated plastic against a mold), computer-controlled milling, or injection molding provide alternatives. For many academic biology groups, having a shared cleanroom fabricate a reusable mold is more accessible than acquiring precision molding or machining equipment. The laboratory can then cast and assemble devices on the bench. Hard-plastic methods require specialized equipment during each production run and may still require photolithography or another high-resolution process to define the micrometer-scale channels (Supplementary Table S2).*

V7 P8 **Table row R05**

> *The fabrication challenge is the difference in scale between micrometer-sized channels and millimeter-deep wells. Patterning the light-sensitive resin SU-8 reliably forms the channels, whereas pipette-accessible openings are commonly punched into the PDMS after curing. Laser removal of material (ablation), direct printing, hot embossing, hard-plastic fabrication, and modular inserts combine these scales using different equipment and added operations (Supplementary Table S2). Resin 3D printing can produce molds for PDMS casting as well as parts incorporated directly into the finished device. The choice of printer, resin, print orientation, and post-treatment affects printing accuracy and biological compatibility (Meynard et al., 2025; Venzac, 2025). Manually placed 3D-printed metal inserts have also been bonded to SU-8 molds to form open compartments around photolithographic microtunnels in neural co-culture devices (Ristola et al., 2019).*

V9 P14 **Table row R05**

> *Extending these experimental capabilities to parallel assays requires platforms that accommodate multiple treatment conditions, biological replicates and established liquid-handling and imaging workflows. A high-capacity microfluidic plate containing up to 96 independently addressable, three-compartment culture units has been used with primary rodent and human induced pluripotent stem cell-derived neurons, demonstrating compatibility with morphological and functional measurements (PMID: 38150531). Other approaches prioritizing experimental flexibility include modular microfluidic units that can be combined into customizable neuronal culture configurations, whereas fluid-walled chambers support human neuronal microcircuits and moderate-throughput axotomy assays (PMIDs: 35466333; 38841815). These studies establish that parallelization and adaptable neuronal compartmentalization are feasible. Nevertheless, the fabrication workflow remains an important design consideration when laboratories need to reproduce, modify or manufacture custom arrays of devices.*

R06. Introduction: access to fabrication Table row R06

V7 compares obtaining a mold from a shared facility with acquiring molding or machining equipment and cites Table S2; V9 omits this Introduction comparison and citation, while retaining a shorter shared-facility account in Discussion.

My suggestion: V7 equipment-access comparison and Table S2 link

Related feedback: Feedback L415 Feedback L488

V7 P7

V9 P15 **Table row R06**

> *A persistent fabrication challenge is the integration of micrometer-scale channels, which control neurite access and local fluid movement, with millimeter-scale reservoirs that permit straightforward cell seeding and pipetting. Conventional PDMS soft lithography can reproduce photolithographically defined microfeatures, but culture reservoirs are often opened individually after casting. This mismatch has previously been addressed with hybrid micro–macro molding combining milled and hot-embossed structures to replicate microscopic channels and macroscopic reservoirs without manual punching (PMID: 20049640). The approach was subsequently applied to multi-compartment neuron–glia cultures supporting localized treatments (PMID: 22828584). Other strategies use additive manufacturing to create compartmentalized neuronal devices without extensive manual post-processing (PMID: 32832365). Building on previous advances in multiscale molding and the elimination of manual reservoir punching, our approach introduces a customizable, parametric design strategy that enables the simultaneous alignment and assembly of large arrays of well-forming inserts onto reusable photolithographically patterned molds, facilitating reproducible, medium-throughput device fabrication.*

V9 P87 **Table row R06**

> *The fabrication results demonstrate the feasibility of repeated production using the hybrid molds. Three molds collectively produced more than 50 usable PDMS casts, and the most recently fabricated molds remained in active use for more than eight months without visible deterioration. Several processing modifications were important for achieving this performance, including the use of a continuous SU-8 base layer, optimized thermal processing, filtered SUEX exposure, compressible clamping, and a flexible build plate for releasing the bonded insert array. These modifications addressed distinct challenges involving photoresist adhesion, bonding-surface uniformity, printed insert-height variation, and mechanical stress during assembly. Once a mold was completed, subsequent production required only PDMS casting, demolding, glass bonding, and plate assembly, without repeating photolithography or individually punching culture wells. This separation of mold fabrication from repeated device production may improve accessibility for laboratories that can obtain molds through a shared microfabrication facility but do not have routine access to cleanroom equipment. Nevertheless, the present study did not directly quantify manufacturing time, labor requirements, or cost savings relative to established alternatives, and these potential advantages should be evaluated in future comparisons.*

R07. Introduction: combining channels and wells Table row R07

V9 narrows the comparison to hybrid micro–macro molding and additive neuronal devices and removes the second Table S2 citation; retain the comparison of operations each fabrication route adds.

My suggestion: V7 fabrication comparison with its Table S2 link

Related feedback: Feedback L415 Feedback L488 Feedback L498

V7 P8

V9 P15

R08. Introduction: insert-based prior work Table row R08

V9 introduces Ristola's individually placed metal inserts earlier and retains the single-operation transfer of 96 inserts in Discussion; keep that distinction, including its applicability to other insert arrangements.

My suggestion: Earlier prior-work comparison with V7's array-transfer explanation

V7 P8

V7 P9 Table row R08

> *Our approach uses shared design dimensions and transfers the entire insert array in one step to make a reusable mold. OpenMFD defines the microfeatures, insert pins, and matching SUEX alignment holes in the same coordinate system and applies the same correction for PDMS shrinkage to the masks and inserts. The matching positions allow the printed inserts to be aligned with the wafer as one array. A compressible backing accommodates pin-height differences during bonding. The bonded inserts remain in the mold and form every well in subsequent PDMS casts, so each batch follows the usual casting and assembly sequence.*

V7 P68 Table row R08

> *The method combines shared device coordinates with alignment and bonding of the entire insert array to form a reusable mold. Ristola et al. combined SU-8 features with three individually aligned, 3D-printed steel inserts for a neuron-oligodendrocyte co-culture device (Ristola et al., 2019). Here, matching pin and hole positions allow all 96 resin inserts to be aligned with a 6-inch wafer in a single operation. The square pins seat in matching SUEX holes, which limit sideways movement. Their nominal clearance is 75 µm per side, compared with a 1.275 mm chamber margin on each side of the channel array (Supplementary Table S3). This margin is 17 times the per-side clearance; final alignment also depends on printing accuracy and photolithographic layer alignment. For an array that fits on one wafer and print platform, the same alignment operation is used regardless of the number or arrangement of inserts. The compressible EPDM rubber sheet backing accommodates pin-height differences during bonding of the entire array. The resulting mold forms every well during casting.*

V9 P16 Table row R08

> *Photopolymer-based 3D printing offers accessible fabrication of larger structures and complex geometries, although printed resins can inhibit PDMS curing and may require additional treatments before use as casting molds (PMID: 40190179). Hybrid assemblies combining soft lithography with printed polymer components provide another route to customizable biological microfluidic systems (PMID: 40667579). Parylene C coatings have also been shown to limit material degradation and cytotoxic leaching from printed resins in primary-cell applications (PMID: 37534979). Particularly relevant to neuronal compartmentalization, Ristola and colleagues combined SU-8 microfeatures with individually positioned 3D-printed metal inserts to fabricate neuron–oligodendrocyte culture devices (Ristola et al., 2019; DOI: 10.1088/1361-6439/ab16a7). This precedent demonstrates the feasibility of combining patterned wafers with tall inserts, while highlighting the importance of practical alignment and assembly procedures when extending such designs to larger arrays.*

V9 P85 **Table row R08**

> *Previous approaches have demonstrated the feasibility of combining micrometer- and millimeter-scale features using hybrid soft lithography, alternative molding strategies, and additive manufacturing (PMIDs: 20049640; 32832365). Particularly relevant to our approach, Ristola and colleagues incorporated individually aligned 3D-printed metal inserts into SU-8 molds to fabricate compartmentalized neuron–oligodendrocyte culture devices (DOI: 10.1088/1361-6439/ab16a7). Our method extends this principle through a common parametric design framework that generates matching photomasks, alignment features, and resin-insert arrays, permitting all 96 inserts to be positioned in a single alignment operation. The use of corresponding square alignment pins and SUEX openings constrains lateral displacement during assembly, while a compressible backing accommodates variations in insert height. In the present design, the nominal lateral clearance was 75 µm per side, substantially smaller than the 1.275 mm margin between the microchannel array and surrounding chamber boundaries. Although these geometrical relationships provide tolerance for insert positioning, the actual alignment accuracy remains dependent on photolithographic registration and resin-printing precision and was not directly quantified. Importantly, the same array-transfer procedure can be applied to different numbers and arrangements of inserts, provided the complete array remains compatible with the wafer and printer dimensions.*

R09. Introduction: what this study tests Table row R09

Both describe the same fabrication and neuronal demonstration; V9 retains the computational-only status in Methods and Results but expands the overview and repeats dimensions.

My suggestion: Concise study overview; retain the untested-layout qualification

Related feedback: Feedback L84 Feedback L327

V7 P10 **Table row R09**

> *We evaluated the method using a compartmentalized neuronal culture device, from file generation and mold fabrication through repeated casting, plate handling, imaging, tracer-movement testing, and long-term culture. We also examined how the same design method generates matching masks, inserts, and frames for three literature-inspired layouts.*

V7 P16 **Table row R09**

> *OpenMFD also generated matched photomasks, inserts, wafer outlines, and frames for three literature-inspired layouts: oligodendrocyte myelination (Kerman et al., 2015), axon guidance with gradients at right angles to each other (Taylor et al., 2015), and a three-compartment assay (Coquinco et al., 2014). These devices were not fabricated or biologically tested (Supplementary Table S3).*

V7 P66 **Table row R09**

> *The same design method generated matching masks and insert arrays for the myelination, axon-guidance, and three-compartment layouts (Figure 5). Changes in device spacing, well arrangement, and channel design were carried through to the corresponding mask and insert files. These examples show how the relationships between the mold components are retained when the layout changes. Physical fabrication and assay testing of these layouts remain future work. Supplementary Table S3 lists their dimensions and the checks before fabricating a new layout.*

V9 P17 **Table row R09**

> *Here, we introduce OpenMFD, an open-source parametric design and hybrid fabrication workflow for producing reusable wafer-scale molds for medium-throughput microfluidic culture devices. Shared design coordinates generate matched photomasks, resin-insert arrays and plate components, enabling the simultaneous alignment and bonding of well-forming inserts to SU-8/SUEX-patterned silicon wafers. Photolithography defines the fine microchannels, while the retained printed inserts form the culture wells directly during each PDMS cast, removing the need for individual well punching in successive production runs. We demonstrate the approach using an array of 48 two-compartment neuronal devices containing 96 wells on a standard 9 mm plate grid. We evaluate repeated casting, compatibility with multichannel pipetting and automated imaging, directional tracer movement, and primary cortical neuron culture with retrograde axonal labeling. We further demonstrate that the parametric workflow generates coordinated fabrication files for alternative compartment geometries, providing a customizable strategy for manufacturing plate-format microfluidic devices.*

V9 P27 **Table row R09**

> *OpenMFD also generated matched photomasks, inserts, wafer outlines, and frames for three literature-inspired layouts: oligodendrocyte myelination (PMID: 26015546), axon guidance with gradients at right angles to each other (PMID: 26000554), and a three-compartment assay (PMID: 24662423). These devices were not fabricated or biologically tested (Supplementary Table S3).*

V9 P60 **Table row R09**

> *To examine the adaptability of OpenMFD, we generated matching photomasks and resin-insert arrays for three additional microfluidic configurations: an oligodendrocyte myelination platform, an orthogonal-gradient axon-guidance platform, and a three-compartment neuronal culture platform (Fig. 5). Modifications to compartment number, well arrangement, and microchannel geometry were incorporated into the corresponding fabrication files while maintaining the alignment relationships between photolithographic and printed components. These examples establish the capacity of the design framework to generate coordinated fabrication files for distinct experimental configurations, although the additional designs were not physically fabricated or biologically evaluated.*

R10. Methods: one-time fabrication and repeated casting Table row R10

V9 removes the opening Methods sentence distinguishing routine production in Protocol S1 from one-time mold fabrication in Protocol S2; the two protocols are unchanged, and Discussion retains the underlying distinction.

My suggestion: Restore the short Protocol S1/S2 workflow explanation

Related feedback: Feedback L539

V7 P12 **Table row R10**

> *The two-compartment design was fabricated and evaluated through casting, assembly, directional tracer movement, neuronal culture, and imaging. Supplementary Protocol S1 describes routine device production and assays; Supplementary Protocol S2 describes one-time hybrid mold fabrication.*

V9 P24 **Table row R10**

> *Materials and Methods*

V9 P87

R11. Methods: design and generated files Table row R11

V9 preserves the software/export description, shared coordinates, device dimensions and untested layouts in three paragraphs; its leading heading is incorporated into the first paragraph rather than standing separately.

My suggestion: Keep separate design, tested-geometry and generated-layout paragraphs

Related feedback: Feedback L19 Feedback L62

V7 P13 **Table row R11**

> *OpenMFD design and fabrication-file generation*

V7 P14 **Table row R11**

> *OpenMFD (https://github.com/trissim/OpenMFD) generated photomasks, insert arrays, wafer outlines, cutting guides, and plate frames from shared well, chamber, channel, and array dimensions (Figure 1). Device shapes were defined in OpenSCAD design software using the SolidPython library and exported as two-dimensional DXF mask drawings for each layer and three-dimensional STL models of the inserts and frames. Matching solid and hollow marks were used to align the photolithographic layers. The pin positions in the insert STL and the hole positions in the SUEX photomask came from the same coordinates, allowing the printed array to be aligned with the patterned wafer as a unit.*

V7 P15 **Table row R11**

> *The tested design contained a 6 × 8 array of two-well units spaced 18 × 9 mm apart, giving 48 devices and 96 wells on a 9 mm grid. Each device had 62 microchannels with a design width of 10 µm and an interchannel gap of 30 µm. Matching masks and inserts received a horizontal scale factor of 1.0226 (2.26% enlargement) for the 100°C PDMS cure. Supplementary Table S3 gives the nominal device, channel, pin, taper, and frame dimensions used to generate the fabrication files.*

V7 P16

V9 P25 **Table row R11**

> *Parametric design and fabrication-file generation - OpenMFD (https://github.com/trissim/OpenMFD) generated photomasks, insert arrays, wafer outlines, cutting guides, and plate frames from shared well, chamber, channel, and array dimensions (Figure 1). Device shapes were defined in OpenSCAD design software using the SolidPython library and exported as two-dimensional DXF mask drawings for each layer and three-dimensional STL models of the inserts and frames. Matching solid and hollow marks were used to align the photolithographic layers. The pin positions in the insert STL and the hole positions in the SUEX photomask came from the same coordinates, allowing the printed array to be aligned with the patterned wafer as a unit.*

V9 P26 **Table row R11**

> *The tested design contained a 6 × 8 array of two-well units spaced 18 × 9 mm apart, giving 48 devices and 96 wells on a 9 mm grid. Each device had 62 microchannels with a design width of 10 µm and an interchannel gap of 30 µm. Matching masks and inserts received a horizontal scale factor of 1.0226 (2.26% enlargement) for the 100°C PDMS cure. Supplementary Table S3 gives the nominal device, channel, pin, taper, and frame dimensions used to generate the fabrication files.*

V9 P27

R12. Methods: SU-8 and SUEX layers Table row R12

V9 retains the base, microchannel and well layers, developer replacement, heating/cooling conditions and direct oven placement without controlled ramps; the procedural losses identified in V8 are absent here.

My suggestion: Keep the layer and bake details

Related feedback: Feedback L39 Feedback L96

V7 P17 **Table row R12**

> *Hybrid mold fabrication*

V7 P18 **Table row R12**

> *The mold comprised two 5 µm layers of SU-8 2005 liquid photoresist (Kayaku Advanced Materials), a 200 µm layer of SUEX K200 dry-film photoresist (DJ MicroLaminates), and a printed resin-insert array on a 6-inch silicon wafer (UniversityWafer ID857). These photoresists are light-sensitive resins used to pattern the mold features. SU-8 was spread by spinning the wafer (spin-coating) using a Laurell WS-650-8B, SUEX was laminated using an SK 335R6, and photolithographic layers were aligned and exposed using an EVG620. Unless otherwise noted, materials were processed according to the manufacturers' instructions; the SU-8 2005 procedure targeted a 5 µm layer. Changes to the manufacturers' procedures are described below and in Supplementary Protocol S2. No dehydration bake or HMDS adhesion-promoter treatment was used. Facility-supplied propylene glycol monomethyl ether acetate (PGMEA) was the developer solvent used to remove unexposed material from both photoresists.*

V7 P19 **Table row R12**

> *Layer 1 (base): SU-8 2005 was spin-coated to a 5 µm target thickness, flood-exposed without a mask, and left undeveloped beneath the patterned microchannel layer (Supplementary Note S2).*

V7 P20 **Table row R12**

> *Layer 2 (microchannels): A second 5 µm SU-8 2005 layer was exposed through the microchannel photomask and developed. Before SUEX lamination, the wafer bearing the approximately 10 µm SU-8 stack was placed directly in an oven at 150°C for 10 min, then removed without controlled heating or cooling.*

V7 P21 **Table row R12**

> *Layer 3 (wells): SUEX K200 dry film (200 µm) defined the well features and insert-alignment holes. The SUEX was exposed at 2800 mJ/cm² through an LP360 long-pass optical filter to block shorter-wavelength ultraviolet light. After exposure, the oven was heated from room temperature to 50°C at 1°C/min, held at 50°C for at least 12 h overnight, then switched off and allowed to cool. The wafer was submerged in developer and mixed for 20 min; the developer was replaced at 15 min. For the hard bake, the oven was heated to 120°C at 3°C/min, held for 30 min, then switched off and allowed to cool.*

V9 P28 **Table row R12**

> *Fabrication of hybrid SU-8/SUEX and 3D-printed wafer molds - The mold comprised two 5 µm layers of SU-8 2005 liquid photoresist (Kayaku Advanced Materials), a 200 µm layer of SUEX K200 dry-film photoresist (DJ MicroLaminates), and a printed resin-insert array on a 6-inch silicon wafer (UniversityWafer ID857). These photoresists are light-sensitive resins*

> *used to pattern the mold features. SU-8 was spread by spinning the wafer (spin-coating) using a Laurell WS-650-8B, SUEX was laminated using an SK 335R6, and photolithographic layers were aligned and exposed using an EVG620. Unless otherwise noted, materials were processed according to the manufacturers' instructions; the SU-8 2005 procedure targeted a 5 µm layer. Changes to the manufacturers' procedures are described below and in Supplementary Protocol S2. No dehydration bake or HMDS adhesion-promoter treatment was used. Facility-supplied propylene glycol monomethyl ether acetate (PGMEA) was the developer solvent used to remove unexposed material from both photoresists.*

V9 P29 **Table row R12**

> *Layer 1 (base): SU-8 2005 was spin-coated to a 5 µm target thickness, flood-exposed without a mask, and left undeveloped beneath the patterned microchannel layer (Supplementary Note S2).*

V9 P30 **Table row R12**

> *Layer 2 (microchannels): A second 5 µm SU-8 2005 layer was exposed through the microchannel photomask and developed. Before SUEX lamination, the wafer bearing the approximately 10 µm SU-8 stack was placed directly in an oven at 150°C for 10 min, then removed without controlled heating or cooling.*

V9 P31 **Table row R12**

> *Layer 3 (wells): SUEX K200 dry film (200 µm) defined the well features and insert-alignment holes. The SUEX was exposed at 2800 mJ/cm² through an LP360 long-pass optical filter to block shorter-wavelength ultraviolet light. After exposure, the oven was heated from room temperature to 50°C at 1°C/min, held at 50°C for at least 12 h overnight, then switched off and allowed to cool. The wafer was submerged in developer and mixed for 20 min; the developer was replaced at 15 min. For the hard bake, the oven was heated to 120°C at 3°C/min, held for 30 min, then switched off and allowed to cool.*

R13. Methods: resin printing Table row R13

V9 retains the 2.26% design enlargement and explicitly says no shrinkage or tolerance compensation was applied in the print-preparation software.

My suggestion: Keep design scaling separate from print-preparation compensation

V7 P15

V7 P22 **Table row R13**

> *Well inserts were printed at their final relative positions using an Elegoo Mars 3 Pro and Siraya Tech Sculpt Clear resin. The profile used 20 µm layers, 5.5 s normal exposure, 50 s bottom exposure, 7 bottom layers, edge smoothing (anti-aliasing) at grey level 2, and no shrinkage or tolerance compensation in the print-preparation software. Supplementary Table S4 provides the remaining motion, timing, washing, and UV post-cure settings. Each insert had a 1.85 × 1.85 mm alignment pin corresponding to a 2.0 × 2.0 mm SUEX hole, giving 150 µm total clearance (75 µm per side).*

V9 P26

V9 P32 **Table row R13**

> *Well inserts were printed at their final relative positions using an Elegoo Mars 3 Pro and Siraya Tech Sculpt Clear resin. The profile used 20 µm layers, 5.5 s normal exposure, 50 s bottom exposure, 7 bottom layers, edge smoothing (anti-aliasing) at grey level 2, and no*

> *shrinkage or tolerance compensation in the print-preparation software. Supplementary Table S4 provides the remaining motion, timing, washing, and UV post-cure settings. Each insert had a 1.85 × 1.85 mm alignment pin corresponding to a 2.0 × 2.0 mm SUEX hole, giving 150 µm total clearance (75 µm per side).*

R14. Methods: array transfer and bonding Table row R14

V9 retains the rubber position, thickness and hardness, clamps, epoxy, washes, cure and release sequence; the corresponding height variation and seating outcome remain in Results.

My suggestion: Keep the procedure in Methods

Related feedback: Feedback L11 Feedback L39

V7 P23 Table row R14

> *The inserts were kept on a detachable flexible metal build plate so the whole array could be transferred at once. EPO-TEK 301-2 epoxy adhesive was applied around each pin, and the mold was positioned until the pins seated in their SUEX holes. An EPDM rubber sheet (0.03125 inch thick; Shore A hardness 60) between the fixed magnet and removable build plate provided a compressible backing; a second flat plate and clamps distributed pressure across the assembly (Figure 2). The clamped assembly was washed in two successive 1 min acetone baths, thoroughly blow-dried, and cured at room temperature for 48 h. After curing, the assembly was cleaned in an ultrasonic acetone bath for 10 min to remove residual material. The flexible metal build plate was flexed to release the bonded array (Supplementary Protocol S2).*

V9 P33 Table row R14

> *The inserts were kept on a detachable flexible metal build plate so the whole array could be transferred at once. EPO-TEK 301-2 epoxy adhesive was applied around each pin, and the mold was positioned until the pins seated in their SUEX holes. An EPDM rubber sheet (0.03125 inch thick; Shore A hardness 60) between the fixed magnet and removable build plate provided a compressible backing; a second flat plate and clamps distributed pressure across the assembly (Figure 2). The clamped assembly was washed in two successive 1 min acetone baths, thoroughly blow-dried, and cured at room temperature for 48 h. After curing, the assembly was cleaned in an ultrasonic acetone bath for 10 min to remove residual material. The flexible metal build plate was flexed to release the bonded array (Supplementary Protocol S2).*

R15. Methods: parylene coating Table row R15

V9 preserves the coating thickness, facility, equipment and pre-coating design dimensions; its added PMID should be verified and formatted with the other references.

My suggestion: Keep the separate coating paragraph

Related feedback: Feedback L96

V7 P24 Table row R15

> *After bonding, the hybrid mold received a 1 µm coating of the polymer parylene C to separate the printed resin and adhesive from the cast PDMS and aid release during demolding (Musgrove et al., 2023; Chen et al., 2013). Parylene was deposited using a Specialty Coating Systems SCS 200 coater at the McGill Nanotools Microfab facility. The specified channel dimensions are the design dimensions before parylene coating.*

V9 P34 **Table row R15**

> *After bonding, the hybrid mold received a 1 µm coating of the polymer parylene C to separate the printed resin and adhesive from the cast PDMS and aid release during demolding (PMID: 37534979; DOI: 10.1016/j.sna.2012.09.024). Parylene was deposited using a Specialty Coating Systems SCS 200 coater at the McGill Nanotools Microfab facility. The specified channel dimensions are the design dimensions before parylene coating.*

R16. Methods: frame printing Table row R16

V9 retains HIPS printing conditions, Table S6 and the supplied Creality Print 6.0.4 G-code file.

My suggestion: Keep the supplied printing-file details

Related feedback: Feedback L562

V7 P26 **Table row R16**

> *Frames were generated with the other fabrication files and printed in high-impact polystyrene (HIPS), a rubber-modified plastic, on an enclosed Creality K1C using a 0.4 mm nozzle, 0.20 mm layers, a 220°C nozzle, and a 100°C build plate. The remaining printing settings are listed in Supplementary Table S6 and included in the supplied Creality Print 6.0.4 G-code printer-instruction file.*

V9 P35 **Table row R16**

> *PDMS casting and plate assembly - Frames were generated with the other fabrication files and printed in high-impact polystyrene (HIPS), a rubber-modified plastic, on an enclosed Creality K1C using a 0.4 mm nozzle, 0.20 mm layers, a 220°C nozzle, and a 100°C build plate. The remaining printing settings are listed in Supplementary Table S6 and included in the supplied Creality Print 6.0.4 G-code printer-instruction file.*

R17. Methods: casting, demolding and glass bonding Table row R17

V9 retains the rack, parallel processing, wafer support, peeling direction, guillotine, glass dimensions and bonding parameters; it adds prior-work references to the tapered-sidewall paragraph.

My suggestion: Keep the detailed casting and bonding sequence

Related feedback: Feedback L21 Feedback L96

V7 P25 **Table row R17**

> *PDMS casting and plate assembly*

V7 P27 **Table row R17**

> *PDMS (Sylgard 184) was mixed at a 10:1 base:curing-agent ratio and poured over the full wafer within an aluminum-foil reservoir. The fill level remained below the insert tops to leave the wells open. Filled molds were supported in a six-shelf metal wafer rack, degassed until visible bubbles were removed (typically 10–30 min; approximately −15 inHg in our setup), and cured for 1 h at 100°C. Multiple molds could be degassed and cured in parallel.*

V7 P28 **Table row R17**

> *The large mold features had tapered sidewalls to aid demolding, following the same design principle used in earlier multiscale PDMS molds (Park et al., 2010; Ristola et al., 2019). The wafer was supported on a flat surface, and peeling proceeded from the corners toward the center while the PDMS was pulled backward to reduce local stress on the wafer.*

V7 P29 **Table row R17**

> *A rectangular guide molded around the array was aligned to an industrial paper guillotine to trim each cast to 108 × 72 mm. The trimmed PDMS and a 110 × 74 × 0.17 mm coverslip-glass sheet were activated with dry-air plasma for 1 min at 200–600 mTorr in a Harrick PDC-001, brought into contact, and heated at 100°C for at least 1 min under uniform pressure (Supplementary Protocol S1).*

V9 P36 **Table row R17**

> *PDMS (Sylgard 184) was mixed at a 10:1 base:curing-agent ratio and poured over the full wafer within an aluminum-foil reservoir. The fill level remained below the insert tops to leave the wells open. Filled molds were supported in a six-shelf metal wafer rack, degassed until visible bubbles were removed (typically 10–30 min; approximately −15 inHg in our setup), and cured for 1 h at 100°C. Multiple molds could be degassed and cured in parallel.*

V9 P37 **Table row R17**

> *The large mold features had tapered sidewalls to aid demolding, following the same design principle used in earlier multiscale PDMS molds (PMID: 20049640; DOI: 10.1088/1361-6439/ab16a7). The wafer was supported on a flat surface, and peeling proceeded from the corners toward the center while the PDMS was pulled backward to reduce local stress on the wafer.*

V9 P38 **Table row R17**

> *A rectangular guide molded around the array was aligned to an industrial paper guillotine to trim each cast to 108 × 72 mm. The trimmed PDMS and a 110 × 74 × 0.17 mm coverslip-glass sheet were activated with dry-air plasma for 1 min at 200–600 mTorr in a Harrick PDC-001, brought into contact, and heated at 100°C for at least 1 min under uniform pressure (Supplementary Protocol S1).*

R18. Methods: frame assembly and surface treatment Table row R18

V9 retains frame/glass orientation, adhesive volume and cure, trimming, and the separate second plasma treatment; its explanatory final sentence repeats a distinction already made by the sequence.

My suggestion: Keep the assembly sequence and distinct plasma steps

Related feedback: Feedback L96

V7 P30 **Table row R18**

> *The bonded PDMS-on-glass assembly was autoclaved in a sterilization pouch at 121°C for 15 min before frame assembly. The frame was decontaminated with 70% ethanol and inverted so that its adhesive groove faced upward. Henkel LOCTITE SI 5140 (3 oz tube, IDH 135264), a one-part silicone sealant that cures at room temperature, was dispensed into the groove (typically 0.5–1 mL per frame) (Henkel, 2023). The bonded device was inverted with the glass facing upward and seated into the frame. After a 3-day room-temperature cure, the assembly was turned upright and excess adhesive was trimmed with a razor blade.*

V7 P31 **Table row R18**

> *After adhesive curing, the assembled device received a second dry-air plasma treatment (Harrick PDC-001; high power, 30 W, 10 min, 400 mTorr) before aqueous coating and*

> *culture. This surface-treatment step was distinct from the earlier 1-min PDMS-glass bonding treatment.*

V9 P39 **Table row R18**

> *The bonded PDMS-on-glass assembly was autoclaved in a sterilization pouch at 121°C for 15 min before frame assembly. The frame was decontaminated with 70% ethanol and inverted so that its adhesive groove faced upward. Henkel LOCTITE SI 5140 (3 oz tube, IDH 135264), a one-part silicone sealant that cures at room temperature, was dispensed into the groove (typically 0.5–1 mL per frame) (manufacturer technical data sheet). The bonded device was inverted with the glass facing upward and seated into the frame. After a 3-day room-temperature cure, the assembly was turned upright and excess adhesive was trimmed with a razor blade.*

V9 P40 **Table row R18**

> *After adhesive curing, the assembled device received a second dry-air plasma treatment (Harrick PDC-001; high power, 30 W, 10 min, 400 mTorr) before aqueous coating and culture. This surface-treatment step was distinct from the earlier 1-min PDMS-glass bonding treatment.*

R19. Methods: mold and device assessment Table row R19

V9 retains individual pin measurements, inspection criteria, usable-cast definition, adhesive culture screening and the single unframed control; it also retains the statement that toxicity was not quantified.

My suggestion: Keep the inspection criteria and unframed control

Related feedback: Feedback L46 Feedback L528

V7 P32 **Table row R19**

> *Mold and device characterization*

V7 P33 **Table row R19**

> *Each pin height was measured individually with a micrometer at all 96 positions of one 8 × 12 printed array before bonding. Supplementary Table S1 reports the individual values, mean, standard deviation, and difference between the tallest and shortest pins. Molds and casts were inspected to check that the SU-8/SUEX features remained attached, the inserts were seated and bonded, and the cast wells and microchannels were open and intact. Microscopy was used to inspect the channels after coating and casting. Supplementary Table S5 reports mold numbers and usable-cast yield separately and defines a usable cast.*

V7 P34 **Table row R19**

> *Printed frames and assembled joints were inspected visually while selecting frame materials. Frame adhesives were screened by whether primary cortical-neuron cultures survived to the end of the experiment. Cell toxicity was not quantified.*

V7 P35 **Table row R19**

> *To test whether cells could grow at the outer positions without the frame, one PDMS device was bonded to glass without a frame or frame adhesive and placed in a 14.5 cm-diameter polystyrene culture dish. This control removed both the frame assembly and the sealing effect of the frame adhesive.*

V9 P41 **Table row R19**

> *Mold and device characterization - Each pin height was measured individually with a micrometer at all 96 positions of one 8 × 12 printed array before bonding. Supplementary Table S1 reports the individual values, mean, standard deviation, and difference between the tallest and shortest pins. Molds and casts were inspected to check that the SU-8/SUEX features remained attached, the inserts were seated and bonded, and the cast wells and microchannels were open and intact. Microscopy was used to inspect the channels after coating and casting. Supplementary Table S5 reports mold numbers and usable-cast yield separately and defines a usable cast.*

V9 P42 **Table row R19**

> *Printed frames and assembled joints were inspected visually while selecting frame materials. Frame adhesives were screened by whether primary cortical-neuron cultures survived to the end of the experiment. Cell toxicity was not quantified.*

V9 P43 **Table row R19**

> *To test whether cells could grow at the outer positions without the frame, one PDMS device was bonded to glass without a frame or frame adhesive and placed in a 14.5 cm-diameter polystyrene culture dish. This control removed both the frame assembly and the sealing effect of the frame adhesive.*

R20. Methods: pipetting and microscopy Table row R20

Both identify eight fitted tips and imaging on two unmodified high-content systems.

My suggestion: Keep the experiment-specific procedure

Related feedback: Feedback L74

V7 P36 **Table row R20**

> *Eight tips were fitted to a 12-channel pipette and aligned with one plate column to fill eight wells in parallel. Compatibility with plate-format imaging was tested by focusing and collecting images on Opera Phenix Plus and ImageXpress Micro Confocal microscopes at the Early Drug Discovery Unit (EDDU), The Neuro, McGill University, without custom plate handling or instrument modification.*

V9 P44 **Table row R20**

> *Eight tips were fitted to a 12-channel pipette and aligned with one plate column to fill eight wells in parallel. Compatibility with plate-format imaging was tested by focusing and collecting images on Opera Phenix Plus and ImageXpress Micro Confocal microscopes at the Early Drug Discovery Unit (EDDU), The Neuro, McGill University, without custom plate handling or instrument modification.*

R21. Methods: fluorescent tracers Table row R21

V9 retains the secondary antibodies, volumes, dilution, incubation, objective, binning, exposure and visual assessment; it also retains the unmeasured flow-rate and duration limit.

My suggestion: Keep the tracer procedure and its stated limit

Related feedback: Feedback L78 Feedback L432

V7 P37 **Table row R21**

> *Fluorescent tracer assay*

V7 P38 **Table row R21**

> *Tracer distributions were compared immediately after loading and after 72 h in 24 two-compartment devices. Alexa Fluor 568-conjugated secondary antibody diluted 1:1000 in phosphate-buffered saline (PBS) was loaded at 25 µL into the left compartment, and Alexa Fluor 488-conjugated secondary antibody diluted 1:1000 in PBS was loaded at 50 µL into the right compartment. The volume difference was established once, and the devices were kept in a cell-culture incubator for 72 h without replenishment or volume adjustment. Images were acquired in widefield mode using a 10× objective with a numerical aperture (NA) of 0.30, 2 × 2 camera binning, a 100 ms exposure, and red and green channels suited to the two fluorophores. Tracer movement was assessed visually by comparing green fluorescence appearing in the initially lower-volume compartment with red fluorescence appearing in the opposite compartment. Flow rate and the duration of the pressure difference were not measured.*

V9 P45 **Table row R21**

> *Assessment of directional fluorescent tracer transport - Tracer distributions were compared immediately after loading and after 72 h in 24 two-compartment devices. Alexa Fluor 568-conjugated secondary antibody diluted 1:1000 in phosphate-buffered saline (PBS) was loaded at 25 µL into the left compartment, and Alexa Fluor 488-conjugated secondary antibody diluted 1:1000 in PBS was loaded at 50 µL into the right compartment. The volume difference was established once, and the devices were kept in a cell-culture incubator for 72 h without replenishment or volume adjustment. Images were acquired in widefield mode using a 10× objective with a numerical aperture (NA) of 0.30, 2 × 2 camera binning, a 100 ms exposure, and red and green channels suited to the two fluorophores. Tracer movement was assessed visually by comparing green fluorescence appearing in the initially lower-volume compartment with red fluorescence appearing in the opposite compartment. Flow rate and the duration of the pressure difference were not measured.*

R22. Methods: neuronal culture and CTβ **Table row R22**

V9 retains the one-embryo preparation, coating, plating, medium, CTβ timing and exclusion rule; it moves the technical-replicate statement into the culture paragraph and explains soma/axon terminology.

My suggestion: Keep V7's terminology; reconcile the exclusions in R53

Related feedback: Feedback L51

V7 P39 **Table row R22**

> *Primary cortical neuron culture and CTβ retrograde axon tracing*

V7 P40 **Table row R22**

> *Primary cortical neurons were isolated from one embryonic day 18 (E18) Sprague Dawley rat embryo following Harris et al. (2007), then distributed across three separately assembled plates. Routine culture maintenance followed the cited protocol, with the device-specific conditions described below. The three plates were technical replicates of a single neuronal preparation. Dendritic polyglycerol amine (dPGA; DendroTEK Biosciences) was selected as the cell-attachment coating based on Clément et al. (2022). Wells were coated with 30 µL of 10 µg/mL dPGA for 10 min and washed once with water. Neurons were seeded into the cell-body (soma) compartment in 50 µL Dulbecco's modified Eagle medium (DMEM) containing 10% fetal bovine serum (FBS) at 5,000–10,000 cells per well, allowed to attach for at least 30 min, and then switched to supplemented Neurobasal Plus medium*

> *(1% N2, 2% B27, 1% glutamine). Volumes of 50 µL in the soma compartment and 25 µL in the axon compartment created a small pressure difference that directed fluid from the soma compartment toward the axon compartment during culture.*

V7 P41 **Table row R22**

> *Cultures were maintained for 11 days in vitro (DIV11). At DIV10, Alexa Fluor-conjugated cholera toxin subunit B (CTβ-647) was applied to the lower-volume axon compartment at 1 µg/mL. CTβ-positive neuronal cell bodies were counted 24 h later, with one count per analyzed device. Analysis was restricted to the 24 interior devices on each plate because culture performance was poorer at the plate edges. Devices were excluded only when contamination or a technical failure prevented cell counting in the final image; high or low cell counts alone were not grounds for exclusion. Supplementary Protocol S1 provides the complete procedure.*

V9 P46 **Table row R22**

> *Primary cortical neuron culture and retrograde axonal tracing - Primary cortical neurons were isolated from one embryonic day 18 (E18) Sprague Dawley rat embryo following Harris et al. (2007), then distributed across three separately assembled plates. Routine culture maintenance followed the cited protocol, with the device-specific conditions described below. The three plates were technical replicates of a single neuronal preparation. Dendritic polyglycerol amine (dPGA; DendroTEK Biosciences) was selected as the cell-attachment coating based on Clément et al. (2022). Wells were coated with 30 µL of 10 µg/mL dPGA for 10 min and washed once with water. Neurons were seeded into the cell-body (soma) compartment in 50 µL Dulbecco's modified Eagle medium (DMEM) containing 10% fetal bovine serum (FBS) at 5,000–10,000 cells per well, allowed to attach for at least 30 min, and then switched to supplemented Neurobasal Plus medium (1% N2, 2% B27, 1% glutamine). Volumes of 50 µL in the soma compartment and 25 µL in the axon compartment created a small pressure difference that directed fluid from the soma compartment toward the axon compartment during culture.*

V9 P48 **Table row R22**

> *Cultures were maintained for 11 days in vitro (DIV11). At DIV10, Alexa Fluor-conjugated cholera toxin subunit B (CTβ-647) was applied to the lower-volume axon compartment at 1 µg/mL. CTβ-positive neuronal cell bodies were counted 24 h later, with one count per analyzed device. Analysis was restricted to the 24 interior devices on each plate because culture performance was poorer at the plate edges. Devices were excluded only when contamination or a technical failure prevented cell counting in the final image; high or low cell counts alone were not grounds for exclusion. Supplementary Protocol S1 provides the complete procedure.*

R23. Methods: imaging and counting **Table row R23**

V9 retains the pixel scale, exposures, stitching, common counting area, reason for Plate 3's larger grid and the pre-axotomy baseline sentence; the baseline clarification requested in the saved discussion is already present in both V7 and V9.

My suggestion: Keep the imaging and baseline explanation

Related feedback: Feedback L25 Feedback L543 Feedback L608

V7 P42 **Table row R23**

> *Imaging and cell counting*

V7 P43 **Table row R23**

> *CTβ images were collected at DIV11 in widefield mode on a PerkinElmer Opera Phenix Plus at EDDU using a 10×/0.30 NA objective, 2 × 2 camera binning, and laser autofocus. The effective pixel size was 1.187 µm. Brightfield images used a 10 ms exposure; Alexa 647 images used 640 nm excitation, 706 nm emission, and a 100 ms exposure.*

V7 P44 **Table row R23**

> *For each device, images were collected at 35 positions (7 × 5 grid) on Plates 1 and 2 and 42 positions (7 × 6 grid) on Plate 3, with approximately 10% overlap. These images were stitched into one image per device. The PDMS array on Plate 3 was shifted slightly rightward during gluing, so a larger area was imaged to capture more microchannel exits. Each stitched image was aligned to a reference brightfield image and cropped to the same area for cell counting, regardless of the number of imaging positions. One cell count was recorded per analyzed device. These were baseline images collected before axotomy; Supplementary Protocol S1 describes the dataset. The supplementary files include the microscope's saved settings and image positions for Plate 1 in XML format.*

V9 P49 **Table row R23**

> *High-content imaging and cell counting - CTβ images were collected at DIV11 in widefield mode on a PerkinElmer Opera Phenix Plus at EDDU using a 10×/0.30 NA objective, 2 × 2 camera binning, and laser autofocus. The effective pixel size was 1.187 µm. Brightfield images used a 10 ms exposure; Alexa 647 images used 640 nm excitation, 706 nm emission, and a 100 ms exposure.*

V9 P50 **Table row R23**

> *For each device, images were collected at 35 positions (7 × 5 grid) on Plates 1 and 2 and 42 positions (7 × 6 grid) on Plate 3, with approximately 10% overlap. These images were stitched into one image per device. The PDMS array on Plate 3 was shifted slightly rightward during gluing, so a larger area was imaged to capture more microchannel exits. Each stitched image was aligned to a reference brightfield image and cropped to the same area for cell counting, regardless of the number of imaging positions. One cell count was recorded per analyzed device. These were baseline images collected before axotomy; Supplementary Protocol S1 describes the dataset. The supplementary files include the microscope's saved settings and image positions for Plate 1 in XML format.*

R24. Methods: statistics **Table row R24**

V9 retains the within-plate SD and explicit technical-replicate explanation without V8's extra warning about inferring other preparations.

My suggestion: Keep the descriptive summaries; update them with R53

Related feedback: Feedback L26 Feedback L432

V7 P45 **Table row R24**

> *Statistical analysis and reporting*

V7 P46 **Table row R24**

> *Supplementary Table S5 lists the numbers of molds and usable casts. All three culture plates contained neurons isolated from the same E18 rat embryo. The plates and the devices within them are therefore technical replicates. Figure 4C shows the mean number*

> *of CTβ-positive cell bodies per device for each plate, with the standard deviation across the analyzed interior devices. These summaries describe variation within this experiment.*

V9 P51 **Table row R24**

> *Statistical analysis - Supplementary Table S5 lists the numbers of molds and usable casts. All three culture plates contained neurons isolated from the same E18 rat embryo. The plates and the devices within them are therefore technical replicates. Figure 4C shows the mean number of CTβ-positive cell bodies per device for each plate, with the standard deviation across the analyzed interior devices. These summaries describe variation within this experiment.*

R25. Results: four groups become six **Table row R25**

V9 separates design, mold development, casting, compatibility, tracer transport and neuronal testing; V7 covers these topics in four groups, and both already separate procedures from findings.

My suggestion: Discuss four versus six Results groups

Related feedback: Feedback L58

V7 P48 **Table row R25**

> *Hybrid molds form all 96 wells in a single cast*

V7 P55 **Table row R25**

> *Cast devices are compatible with plate-format handling and imaging*

V7 P62 **Table row R25**

> *Neuronal compatibility demonstrated by CTβ retrograde axon tracing*

V7 P65 **Table row R25**

> *Shared dimensions support changes in device layout*

V9 P58 **Table row R25**

> *Parametric design enables coordinated fabrication of medium-throughput microfluidic arrays - We developed OpenMFD, a parametric design framework that integrates the generation of photolithographic masks, three-dimensional resin inserts, and supporting plate components within a common coordinate system (Fig. 1). The framework enables microchannel geometries, culture compartment dimensions, well positions, and device arrangements to be modified while maintaining the spatial relationships required for mold fabrication. In particular, the positions of the alignment pins on the printed resin inserts are matched to corresponding openings patterned in the SUEX layer, allowing complete insert arrays to be aligned with the photolithographically patterned silicon wafer.*

V9 P61 **Table row R25**

> *Development and optimization of reusable hybrid wafer molds - The hybrid mold fabrication strategy combined photolithographically defined SU-8 microchannels and SUEX compartment structures with 3D-printed resin inserts that form the culture wells during PDMS casting (Fig. 2). A central requirement was the integration of micrometer-scale features with millimeter-scale structures while maintaining the integrity of the patterned wafer and enabling simultaneous positioning of the well-forming inserts.*

V9 P67 **Table row R25**

> *Repeated casting and assembly of plate-format microfluidic devices - The completed hybrid molds enabled the fabrication of PDMS devices containing all 48 compartmentalized culture units and 96 wells in a single casting operation (Fig. 3). Because the well-forming inserts remained attached to the mold, culture wells were generated directly during PDMS curing, eliminating the need for individual reservoir punching in subsequent production cycles. Multiple molds could be filled, degassed, and cured simultaneously, allowing parallel production of complete device arrays.*

V9 P71 **Table row R25**

> *Compatibility with multichannel liquid handling and automated imaging - The arrangement of the 96 culture wells on a standard 9 mm microplate grid enabled liquid handling using conventional multichannel pipettes (Fig. 1). Eight wells within a plate column could be filled simultaneously using a multichannel pipette, providing an alternative to the individual loading procedures commonly required for conventional compartmentalized microfluidic devices.*

V9 P74 **Table row R25**

> *Directional fluid transport in compartmentalized devices - To assess fluid movement between the two culture compartments, we examined the redistribution of fluorescent tracers following the establishment of an initial volume difference between wells (Fig. 4A). Alexa Fluor 568-conjugated antibody was introduced into the lower-volume compartment (25 µL), while Alexa Fluor 488-conjugated antibody was introduced into the opposite compartment (50 µL). Tracer distributions were examined in 24 devices immediately after loading and following 72 h of incubation without further volume adjustment.*

V9 P77 **Table row R25**

> *Validation of primary neuronal culture and retrograde axonal labeling - We next evaluated the suitability of the fabricated microfluidic devices for compartmentalized primary neuronal culture. Embryonic rat cortical neurons were seeded into the somatic compartments of three independently assembled plates and maintained for 11 days in vitro (DIV11). The three plates contained neurons derived from a single embryonic preparation and therefore represented technical replicates.*

R26. Results: coordinated design and plate geometry Table row R26

V9 explains the same coordinated components and dimensions over two paragraphs; these dimensions are already specified in Methods, so the Results can concentrate on what the coordinated design enables.

My suggestion: V7's concise account under the proposed design heading

Related feedback: Feedback L62 Feedback L136

V7 P49 **Table row R26**

> *OpenMFD generated the photomasks and printed insert array from shared device dimensions, keeping the microfeatures and wells aligned (Figure 1). Matching pins and alignment holes allowed all 96 inserts to be positioned together on the patterned wafer. The completed mold combined SU-8 microchannels, a SUEX well layer, and millimeter-tall resin inserts. During PDMS casting, the inserts formed all 96 wells at once, eliminating individual well punching.*

V9 P58

V9 P59 **Table row R26**

> *To demonstrate this approach, we designed a medium-throughput platform comprising 48 independent, two-compartment microfluidic devices arranged in a 6 × 8 array (Fig. 1). Each device contained two culture wells connected by 62 parallel microchannels with nominal widths of 10 µm and interchannel gaps of 30 µm. The resulting array contained 96 wells positioned on a standard 9 mm microplate grid, providing a format compatible with multichannel liquid handling and automated plate-based microscopy. The coordinated design framework also incorporated a common dimensional scaling correction to account for PDMS shrinkage during curing.*

R27. Results: alternative layouts moved earlier Table row R27

V9 places alternative fabrication files beside the design result and retains the untested status; V7 explains the same coordinated changes in its final Results group.

My suggestion: Earlier placement with concise generated-layout wording

Related feedback: Feedback L84 Feedback L327

V7 P65

V7 P66

V9 P60

R28. Results: general mold-development statements Table row R28

V9 adds an opening requirement statement and a closing summary around the existing filter, adhesion and clamping observations; the intervening paragraphs already show how the mold was developed.

My suggestion: Begin with the failures and solutions; omit the recap

V7 P49

V7 P50 **Table row R28**

> *Flat SUEX surfaces were needed for the inserts to form a continuous epoxy seal. Exposure without an LP360 long-pass filter produced raised edges that interfered with sealing. Filtered exposure substantially reduced these ridges and provided a flatter bonding surface (Supplementary Note S1).*

V7 P51 **Table row R28**

> *The thin SU-8 features also needed to remain attached through SUEX processing and insert bonding. Microchannel features patterned directly on bare silicon detached from the wafer. An SU-8 base layer exposed uniformly without a mask prevented this observed failure. A high-temperature bake before SUEX lamination further improved layer adhesion. In an eight-wafer development batch without this bake, some wafers showed warping and layer separation around the well features during cooling after SUEX processing. Both wafers in a subsequent batch with the 150°C, 10 min bake remained intact (Supplementary Note S2).*

V9 P61

V9 P66 **Table row R28**

> *Together, these optimizations enabled the integration of 96 well-forming resin inserts with wafer-scale photolithographically patterned microstructures, creating reusable hybrid molds suitable for repeated PDMS casting.*

R29. Results: filter and base-layer observations Table row R29

Both report the raised edges, detachment and successful process changes in Results; V9 still cites only Notes S1/S2 rather than the figures themselves.

My suggestion: Existing observations with direct Figure S1/S2 links

Related feedback: Feedback L33 Feedback L34 Feedback L467 Feedback L471

V7 P50

V7 P51

V9 P62 Table row R29

> *Several fabrication challenges were identified and addressed during process development. Initial SUEX processing without an LP360 long-pass optical filter produced raised edges around patterned structures that interfered with contact between the resin inserts and their bonding surfaces. Incorporation of the long-pass filter reduced these surface irregularities and improved the apparent flatness of the SUEX structures, providing more suitable surfaces for insert attachment (Supplementary Note S1).*

V9 P63 Table row R29

> *Adhesion of the photolithographically patterned microchannels presented a second challenge. SU-8 microchannel structures patterned directly onto bare silicon detached during subsequent processing. Introducing an underlying, uniformly exposed SU-8 layer prevented this observed detachment. Additional optimization was required to maintain layer integrity during SUEX lamination and thermal processing. In an initial development batch of eight wafers fabricated without a high-temperature SU-8 bake, some wafers exhibited warping and separation of patterned layers during cooling after SUEX processing. In contrast, both wafers fabricated in a subsequent batch incorporating a 150 °C bake for 10 min remained intact (Supplementary Note S2).*

R30. Results: additional SU-8 bake Table row R30

V9 keeps the eight-wafer and subsequent two-wafer observations; “optimization” and “optimized” also appear in its mold-development account, although the sequential batches do not isolate variables or establish an optimum.

My suggestion: Keep the batch observations without calling the process optimized

V7 P51

V9 P63

R31. Results: pin heights and clamping Table row R31

V9 brings the mean 5.551 mm and SD 26.7 µm from Table S1 into main Results alongside V7’s existing 116 µm range, uneven rigid clamping and successful EPDM seating.

My suggestion: Keep the added mean and SD with the existing clamping result

Related feedback: Feedback L11 Feedback L127

V7 P52 Table row R31

> *Bonding the entire insert array required contact across all inserts despite differences in pin height. The measured height range was 116 µm across one 8 × 12 array (Supplementary Table S1). With rigid clamping, pressure was uneven across the array. A compressible EPDM rubber sheet backing accommodated the height differences and allowed all pins to seat together. Slow heating at 1–3°C/min during SUEX processing, release from a flexible*

> *metal build plate, and room-temperature epoxy curing were used to reduce wafer stress and insert detachment.*

V9 P64 **Table row R31**

> *Simultaneous bonding of the resin inserts required accommodation of dimensional variation across the printed array. Measurements of 96 printed alignment pins revealed a mean height of 5.551 mm, a standard deviation of 26.7 µm, and a maximum height difference of 116 µm (Supplementary Table S1). Rigid clamping did not provide uniform contact across the insert array. Incorporating a compressible EPDM backing distributed the applied pressure and accommodated differences in insert height, enabling the pins to engage their corresponding SUEX alignment openings during a single bonding operation (Fig. 2).*

R32. Results: open channels and mold release Table row R32

Both report open cast channels and unmeasured channel dimensions; V9 retains the limit beside that result, while its later Discussion repeats it.

My suggestion: Report the observation; integrate the width check in R59

Related feedback: Feedback L136 Feedback L149

V7 P53 **Table row R32**

> *With the final process, bonding surfaces were visibly flat, SU-8 features remained attached, and inserts seated in their alignment holes. The parylene-coated molds released cured PDMS without a silane release treatment (silanization). Microscopy showed open channels in the resulting casts. Channel dimensions are reported as nominal values before parylene coating. Dimensional accuracy in the casts remains to be measured.*

V9 P65 **Table row R32**

> *The optimized process produced hybrid molds in which the photolithographic structures remained attached to the silicon substrate and the printed inserts were bonded in their designated positions. Following parylene C coating, cured PDMS could be released from the molds without additional silane-based release treatment. Microscopic inspection of the resulting PDMS casts confirmed the presence of open microchannels. However, the dimensional accuracy of the cast microchannels relative to their nominal photolithographic dimensions has not yet been quantitatively established.*

R33. Results: repeated casting and mold use Table row R33

V9 retains five assembled molds, three used routinely, more than 50 casts, both losses and more than eight months of use; it adds a direct Figure 3 callout and does not retain V8's standardized-durability warning.

My suggestion: Keep the reuse observations and V9's Figure 3 link

Related feedback: Feedback L48 Feedback L138 Feedback L522

V7 P54 **Table row R33**

> *Five wafer-scale hybrid molds were assembled using the final protocol. Three entered routine use and together produced more than 50 usable casts. One wafer broke during demolding; another was lost when PDMS leaked beneath it and cured in the casting rack. Supplementary Table S5 defines a usable cast and reports the mold and cast outcomes. The most recent molds remained in active use for more than 8 months without visible degradation.*

V7 P56 **Table row R33**

> *Each PDMS cast contained the complete well array. Multiple molds could be poured, degassed, and cured in parallel. The 96 wells follow the standard 9 mm grid. Eight wells were filled together with a multichannel pipette.*

V9 P67

V9 P68 **Table row R33**

> *Five wafer-scale hybrid molds were assembled using the optimized fabrication procedure. Three molds entered routine use and collectively produced more than 50 usable PDMS casts (Supplementary Table S5). Of the remaining molds, one wafer fractured during demolding, while another was lost following PDMS leakage beneath the wafer and subsequent curing within the casting rack. The most recently fabricated molds remained in active use for more than eight months without visible deterioration of their structures.*

R34. Results: frame materials and adhesives Table row R34

V9 retains lower visible HIPS warping and the surviving SI 5140 cultures but changes the failed-adhesive wording and adds an unperformed mechanical-strength test; Methods already states that toxicity was not quantified.

My suggestion: V7's direct observations without a new strength warning

V7 P57 **Table row R34**

> *HIPS and polypropylene plastics were evaluated as frame materials. HIPS showed less visible warping in our printing trials and was selected for assembly. Polypropylene warpage during extrusion printing has been described previously (Spoerk et al., 2020).*

V7 P58 **Table row R34**

> *LOCTITE SI 5140 was the only frame adhesive tested that supported primary cortical-neuron cultures to the end of the experiment; cultures did not survive with any of the other screened adhesives. SI 5140 also formed a flexible, visibly leak-tight glass-to-HIPS joint that remained intact during repeated incubator transfers.*

V9 P69 **Table row R34**

> *PDMS casts were trimmed, plasma-bonded to glass, and incorporated into 3D-printed supporting frames to generate complete plate-format devices (Fig. 3). During development, both high-impact polystyrene (HIPS) and polypropylene were evaluated as frame materials. HIPS exhibited less visible warping under the printing conditions tested and was selected for subsequent device assembly.*

V9 P70 **Table row R34**

> *The adhesive used to attach the glass-backed PDMS devices to the supporting frames was also found to influence biological compatibility. Among the adhesives evaluated, LOCTITE SI 5140 was the only formulation that supported primary cortical neuronal cultures through the experimental endpoint. The resulting adhesive joint remained visibly intact and leak-tight during routine handling and repeated incubator transfers. These observations were qualitative, and systematic measurements of adhesive cytotoxicity or joint mechanical strength were not performed.*

R35. Results: multichannel loading Table row R35

V9 reports the manual eight-well operation without V8's robotic-dispensing warning; its closing compatibility paragraph repeats the preceding loading and imaging observations.

My suggestion: Keep the eight-well loading observation

Related feedback: Feedback L74

V7 P55

V7 P56

V9 P71

V9 P73 Table row R35

> *These observations demonstrate compatibility of the fabricated devices with standard multichannel liquid handling and automated image acquisition, providing the basis for parallelized experimental measurements across multiple compartmentalized cultures.*

R36. Results: microscope use Table row R36

Both report use of two unmodified systems; V9 omits V7's explanation that Opera Phenix was faster for these experiments.

My suggestion: Restore the reason for choosing Opera Phenix

V7 P60 Table row R36

> *Both the Opera Phenix Plus and ImageXpress Micro Confocal focused and collected images without custom plate handling or instrument modification. The dye and CTβ images shown were collected on the Opera Phenix Plus, which was faster for these experiments.*

V9 P72 Table row R36

> *The assembled plates were compatible with two automated high-content imaging platforms, the Opera Phenix Plus and ImageXpress Micro Confocal. Both instruments successfully focused on and acquired images of the microfluidic arrays without modification to their plate-handling or imaging hardware. The Opera Phenix Plus was subsequently used for fluorescent tracer imaging and quantitative neuronal assays.*

R37. Results: tracer movement Table row R37

V9 spreads loading, redistribution and the existing flow/isolation limit over three paragraphs; V7 already reports the two-time-point result and its limits in one Results paragraph.

My suggestion: V7's concise result and limit

Related feedback: Feedback L78 Feedback L432

V7 P61 Table row R37

> *Images of 24 two-compartment devices acquired immediately after loading and at 72 h showed green tracer redistribution toward the initially lower-volume left compartment, with little visible red signal in the opposite direction (Figure 4A). The observed redistribution is consistent with the initial volume difference driving tracer movement. Flow duration and compartmental isolation between the two imaging time points remain uncharacterized.*

V9 P74

V9 P75 **Table row R37**

> *At the initial imaging time point, the two fluorophores were predominantly localized to their respective loading compartments. After 72 h, green fluorescence was observed in the initially lower-volume compartment, whereas comparatively little red fluorescence was visible in the opposite compartment (Fig. 4A). This redistribution was consistent with net tracer movement in the direction established by the initial hydrostatic volume difference.*

V9 P76 **Table row R37**

> *These observations demonstrate that the connecting microchannels permit tracer transport between compartments and that an initial difference in fluid volume can produce directional redistribution. However, the experimental design did not determine the magnitude or duration of fluid flow, and compartmental isolation throughout the 72 h interval was not established.*

R38. Results: neuronal labeling Table row R38

V9 adds the direct Figure 4B link and retains the inference of uptake by axons and retrograde transport; its abstract uses stronger “compartmentalized axonal growth” wording than this result supports directly.

My suggestion: Concise neuronal result with the Figure 4B link

Related feedback: Feedback L430

V7 P62

V7 P63 **Table row R38**

> *Primary cortical neurons isolated from one E18 rat embryo were distributed across three assembled plates and cultured to DIV11. CTβ-647 added to the axon compartment on DIV10 labeled neuronal cell bodies in the soma compartment, consistent with tracer uptake by axons extending into the axon well and retrograde transport.*

V9 P77

V9 P78 **Table row R38**

> *At DIV10, fluorescently conjugated cholera toxin subunit B (CTβ-647) was applied selectively to the axonal compartments. Following 24 h of incubation, fluorescently labeled neuronal cell bodies were detected within the somatic compartments (Fig. 4B). The appearance of CTβ-positive cell bodies following tracer application to the axonal compartment was consistent with uptake by axons extending through the microchannels and subsequent retrograde transport to neuronal somata.*

R39. Results: assessed and counted devices Table row R39

Both report 72 interior devices, three exclusions and 69 analyzed devices, with plate means and within-plate SD; the later DMSO dataset needs a reconciled exclusion account before replacing those totals.

My suggestion: Integrate the updated DMSO counts and plot in R53

Related feedback: Feedback L53 Feedback L551

V7 P64 **Table row R39**

> *Biological analysis used the 24 interior devices per plate because culture performance was poorer at the plate edges, giving 72 devices across three plates. Three devices were excluded because contamination or a technical failure prevented cell counting in their final*

> *images, leaving 69 analyzed devices. Figure 4C shows the mean CTβ-positive cell-body count per device and the standard deviation within each plate.*

V9 P79 **Table row R39**

> *Neuronal culture performance varied with position within the assembled plates, with poorer growth observed in peripheral devices. Quantitative analysis was therefore restricted to the 24 interior devices on each plate. Of the 72 interior devices across the three plates, three were excluded because contamination or technical failures prevented reliable cell counting, leaving 69 devices for analysis.*

V9 P80 **Table row R39**

> *CTβ-positive neuronal cell bodies were quantified from standardized image regions acquired by automated high-content microscopy. The mean number of labeled cell bodies per device and the corresponding within-plate standard deviations are presented for each of the three plates in Fig. 4C. These measurements demonstrate the feasibility of collecting quantitative retrograde labeling data across multiple compartmentalized neuronal cultures using an automated imaging workflow.*

R40. Results: unframed control and edge growth Table row R40

V9 moves the unframed control into neuronal Results and explicitly lists the unresolved frame, adhesive, sealing and culture-environment contributions; V7 already reports the control and unresolved cause.

My suggestion: Keep the control observation with its limits

Related feedback: Feedback L55

V7 P59 **Table row R40**

> *In the single unframed PDMS-on-glass control in a 14.5 cm polystyrene dish, cells at the outer positions appeared to grow as well as those elsewhere in the device. This observation shows that the outer positions can support culture in the unframed arrangement. The separate contributions of frame geometry, adhesive, sealing, and the surrounding dish environment remain unresolved.*

V7 P64

V7 P73 **Table row R40**

> *The printed plate frame and its glass-to-frame seal are practical targets for further development. The assembled plates supported multichannel handling, automated imaging, and neuronal culture in the interior devices, while cell growth was poorer at the plate edges. In one unframed PDMS-on-glass device placed in a culture dish, cells at the outer positions appeared to grow as well as those elsewhere. The cause of the poorer edge growth in the assembled plates remains unresolved. Frame geometry and sealing can be revised independently of the PDMS device and reusable mold.*

V9 P81 **Table row R40**

> *To investigate the reduced neuronal growth at peripheral positions, an additional PDMS device was bonded directly to glass and maintained in a conventional culture dish without the supporting frame or frame adhesive. In this configuration, neuronal growth at peripheral positions appeared comparable to that in more central regions. Although this observation suggests that peripheral devices can support neuronal culture, the relative contributions of frame geometry, adhesive exposure, sealing, and culture environment to the reduced growth observed in assembled plates remain unresolved.*

R41. Results: repeated neuronal-summary paragraph Table row R41

V9 adds a closing neuronal-validation summary and future independent-preparation sentence, then repeats the same limit in Discussion; the one-preparation design is already explicit in Methods.

My suggestion: Keep the result; state the independent-preparation limit once

V7 P75 **Table row R41**

> *We assessed repeated casting, routine device use, tracer movement, and neuronal culture from one embryo. Further testing across independent neuronal preparations would assess biological reproducibility. Time-resolved transport measurements and standardized durability and toxicity tests would extend the assessment of device performance.*

V9 P82 **Table row R41**

> *Collectively, these experiments establish the compatibility of the microfluidic platform with primary cortical neuronal culture, compartment-specific axonal labeling, and automated quantitative imaging. Further experiments using independent neuronal preparations will be required to determine biological reproducibility across cultures.*

V9 P88 **Table row R41**

> *The compatibility of the resulting devices with conventional laboratory workflows is an important practical outcome. The 48-device array provides 96 culture wells on a standard 9 mm microplate grid, enabling multichannel pipetting and automated high-content imaging without instrument modification. Previous studies have demonstrated the value of parallelized and modular microfluidic systems for neuronal experimentation (PMIDs: 38150531; 35466333; 38841815). Our approach complements these platforms by providing an alternative fabrication strategy rather than introducing neuronal compartmentalization or medium-throughput culture as new capabilities. The demonstration of directional fluorescent tracer redistribution and retrograde CTβ labeling confirms that the fabricated devices support fluid movement through the connecting microchannels and compartment-specific neuronal manipulations. However, the tracer experiment did not establish the duration of hydrostatic flow or quantitatively measure fluidic isolation. Similarly, neuronal experiments using 69 interior devices across three plates demonstrate assay feasibility but were performed using a single embryonic neuronal preparation. Validation across independent preparations and additional functional assays will be necessary to establish biological reproducibility and suitability for screening applications.*

R42. Discussion: contribution and prior inserts Table row R42

V9 retains simultaneous positioning and transfer across other array layouts but expands the contribution and adds an alignment-accuracy warning; its introductory statement remains centered on the fabrication strategy.

My suggestion: V7's process-centered account with the earlier prior-work citation

V7 P67 **Table row R42**

> *Discussion*

V7 P68

V9 P84 **Table row R42**

> *The hybrid fabrication strategy developed in this study provides a scalable approach for producing compartmentalized microfluidic culture devices by integrating*

> *photolithographically defined microfeatures with 3D-printed, millimeter-scale structures on reusable silicon-wafer molds. Building on previous advances in multiscale molding and the elimination of manual reservoir punching, our approach introduces a coordinated parametric design framework that enables the simultaneous alignment and attachment of large arrays of well-forming inserts to patterned wafers. This strategy addresses a practical limitation of conventional PDMS microfluidic fabrication, in which high-resolution microchannels can be reproduced by soft lithography, but the formation of individual culture wells frequently requires additional processing after casting. By incorporating all well-forming structures into a reusable mold, the method separates the specialized fabrication steps required to construct the mold from subsequent production of complete microfluidic devices. The resulting platform combines customizable microfluidic geometries with a standardized plate format suitable for medium-throughput biological experimentation.*

V9 P85

R43. Discussion: printer resolution and parylene Table row R43

V9 retains the distinction between resin in the mold and PDMS/glass in the finished device, adds printed-mold/coating references, and appends a future configuration-compatibility qualification.

My suggestion: Keep printed parts confined to the mold; vet the added references

Related feedback: Feedback L260 Feedback L500

V7 P69 Table row R43

> *Recent resin-printing methods use both printed molds and printed parts retained in the finished culture device (Meynard et al., 2025). Post-treatment can improve PDMS curing and release from resin molds, but reproducibly printing features near 10 µm across a plate-scale mold remains difficult with accessible printers (Venzac, 2025). Combining SU-8 photolithography for the channels with resin printing for the wells accommodates both scales. Here, the resin inserts remain in the parylene-coated mold, and each finished culture device consists of PDMS bonded to glass.*

V9 P86 Table row R43

> *The combination of photolithography and resin printing also exploits the complementary capabilities of these fabrication technologies. Photolithography remains well suited to defining micrometer-scale structures, whereas resin printing provides an accessible means of producing larger three-dimensional features. Recent developments in printed molds and hybrid fabrication have expanded the range of structures that can be manufactured for biological microfluidic applications, although achievable resolution, PDMS curing compatibility, and surface treatment remain important considerations (PMIDs: 40190179; 40667579). In our approach, the printed resin inserts remain permanently attached to the silicon-wafer mold rather than becoming components of the finished culture device. A conformal parylene C coating separates the printed resin and adhesive from the cast PDMS and facilitates repeated demolding. The suitability of parylene C as a protective coating for resin-printed components has also been demonstrated in biological applications (PMID: 37534979). Consequently, the finished culture surfaces comprise PDMS bonded to glass, while the printed components are confined to the reusable manufacturing tool. This distinction is potentially advantageous for applications requiring conventional PDMS surface treatments and established neuronal culture protocols, although the biological compatibility of each assembled device configuration must still be established experimentally.*

R44. Discussion: repeated use and bench production Table row R44

V9 retains the eight-month use observation and later bench production, then adds unmeasured time, labor and cost comparisons; V7 describes avoided processing steps rather than claiming measured savings.

My suggestion: Keep the one-time mold/repeated-casting explanation

Related feedback: Feedback L140 Feedback L432

V7 P70 Table row R44

> *Three molds produced more than 50 usable casts, and the most recent molds remained in use for more than 8 months without visible degradation. Reuse depended on flat bonding surfaces, attached SU-8 features, and evenly seated inserts. Filtered SUEX exposure, the SU-8 base layer and hard bake, compressible clamping layers, and the flexible transfer plate addressed the failures observed during fabrication. Once the mold is complete, laboratories can cast and assemble further devices without repeating cleanroom processing. Parylene coating can be performed locally or by an external service.*

V7 P71 Table row R44

> *Changing device dimensions updates the matching masks, inserts, wafer layout, and frame together. The additional designs in Figure 5 extend this approach to an oligodendrocyte myelination layout, an axon-guidance layout with gradients at right angles to each other, and a three-compartment assay. These examples change the number and arrangement of compartments and the channel layout while retaining matching positions in the photomasks and printed insert arrays. This allows the culture arrangement to be adapted without separately redrawing the alignment features for each mold component. The generated files provide starting points for these assays; fabrication must still establish alignment, successful casting and release, and suitable fluid movement before biological testing. Several dimensions must work together: pin and hole sizes determine alignment clearance, the insert skirt (the rim around its base) provides space for adhesive, and the sidewall slope affects demolding and well capacity. The same 2.26% shrinkage correction is applied to the masks and inserts.*

V9 P87

R45. Discussion: coating and working volume Table row R45

Unlike V8, V9 retains the relative-height explanation: coating raises both ridge and surrounding surface; it also retains the expected width increase and 50 µL working-volume constraint, while adding dimensional-fidelity and scaling-refinement proposals.

My suggestion: Keep the coating explanation and integrate R59

V7 P72 Table row R45

> *Parylene coating also changes the mold geometry. A uniform 1 µm coating would add approximately 2 µm to the width of a channel-forming ridge, a substantial change for the nominal 10 µm channels. The ridge height relative to the surrounding surface would remain approximately unchanged because the coating raises both the ridge top and the flood-exposed base. This predicts wider molded channels, with final dimensions also affected by PDMS shrinkage. Measuring the coated molds and casts would allow a coating-thickness correction to be added to the design parameters. Supplementary Table S3 summarizes these relationships and the checks needed when adapting the design.*

V9 P89 **Table row R45**

> *Several design considerations remain important when adapting the platform to other experimental configurations. The current mold incorporates a 2.26% dimensional scaling correction to compensate for PDMS shrinkage, but the dimensions of the finished microchannels have not been quantitatively characterized. In addition, the 1 µm parylene coating is expected to increase the width of a channel-forming ridge by approximately 2 µm, assuming uniform deposition, while leaving its height relative to the surrounding coated surface approximately unchanged. These geometrical effects may be significant for microchannels with nominal widths of 10 µm. Direct measurements of coated mold features and the resulting PDMS channels would establish dimensional fidelity and permit refinement of the scaling parameters. The geometry of the well-forming inserts also introduces practical constraints. Tapered sidewalls facilitate demolding but reduce the available reservoir dimensions at fixed well spacing, limiting the present design to a maximum working volume of approximately 50 µL per well. Increasing reservoir capacity would require balancing well dimensions, wall thickness, demolding performance, and the dimensions and cost of the supporting glass substrate.*

R46. Discussion: edge growth Table row R46

Both report the poorer peripheral growth and unframed control; V9 expands the same unresolved mechanism into a separate Discussion paragraph after describing it in Results.

My suggestion: V7's concise wording with the unresolved frame factors

Related feedback: Feedback L55

V7 P73

V9 P90 **Table row R46**

> *The supporting plate frame represents another opportunity for optimization. Although assembled devices supported routine handling, automated imaging, and neuronal culture, neuronal growth was reduced at peripheral positions. In an unframed PDMS-on-glass device maintained in a conventional culture dish, peripheral cultures appeared comparable to those at interior positions, suggesting that this limitation is not necessarily intrinsic to the compartmentalized PDMS geometry. However, this observation does not distinguish the potential contributions of frame geometry, adhesive exposure, sealing, or the surrounding culture environment. Modifying the frame and its assembly procedure independently of the reusable mold should allow these factors to be investigated without redesigning the microfluidic structures. Resolving this limitation will be important for achieving reliable biological performance across the complete 48-device array.*

R47. Discussion: adapting other layouts Table row R47

Both explain changes to pins, holes, adhesive space and layout; V9 retains the shared-file logic but adds a list of future alignment, casting, transport and biological checks for the untested layouts.

My suggestion: Keep the coordinated-design explanation and tested/generated distinction

Related feedback: Feedback L84 Feedback L327

V7 P74 **Table row R47**

> *Well geometry and the selected glass size also constrain the plate design. The tapered sidewalls needed for demolding leave less space for wells than in a conventional 96-well plate. At the fixed 9 mm well spacing, taper and the wall thickness between adjacent wells*

> *constrain both well width and depth (Supplementary Table S3). The safe maximum working volume in the present plate was 50 µL per well. Slightly larger wells were possible, but would have required a larger glass sheet to accommodate the array. Custom-cut glass would have substantially increased fabrication costs. Applications requiring more than 50 µL per well would therefore need a revised design that balances well capacity, demolding, plate dimensions, and glass cost.*

V9 P91 **Table row R47**

> *Finally, the open-source parametric framework provides a means of extending the fabrication strategy beyond the two-compartment neuronal platform demonstrated here. OpenMFD generated coordinated fabrication files for oligodendrocyte myelination, axon-guidance, and three-compartment neuronal culture configurations, illustrating how compartment geometry and microchannel organization can be modified while preserving alignment between photolithographic and printed mold components. These designs have not yet been fabricated, and changes in insert geometry, well spacing, or compartment arrangement will require validation of alignment, casting, demolding, fluid transport, and biological compatibility. Nevertheless, the ability to generate coordinated fabrication components from shared design parameters reduces the need to redesign individual mold elements independently. By combining high-resolution photolithography, array-scale integration of 3D-printed inserts, and repeated PDMS casting, this approach provides a flexible foundation for developing customized medium-throughput microfluidic devices while retaining compatibility with established laboratory workflows.*

R48. Conclusion **Table row R48**

V9 repeats the contribution and adds future quantitative reproducibility and peripheral-culture work; its “48 independent” devices should mean separate device units rather than independent biological replicates.

My suggestion: V7 conclusion with concise observed reuse and assay results

Related feedback: Feedback L44 Feedback L140 Feedback L432

V7 P76 **Table row R48**

> *Conclusion*

V7 P77 **Table row R48**

> *OpenMFD links the design of matching photomasks and printed inserts to alignment and bonding of the entire insert array. The resulting reusable mold forms the microfeatures and wells in each PDMS cast, removing individual well punching from repeated device production. The demonstrated devices combine compartmentalized neuronal culture with multichannel pipetting and automated imaging, although culture performance at the plate edges requires improvement. By separating specialized mold fabrication from repeated device production, this approach allows laboratories to make subsequent batches by benchtop casting and assembly. The open-source design tools and fabrication procedure provide a starting point for adapting this approach to other plate-format culture devices.*

V9 P92 **Table row R48**

> *Conclusions*

V9 P93 **Table row R48**

> *We developed an open-source hybrid fabrication strategy that combines high-resolution photolithography with 3D-printed resin inserts to produce reusable silicon-wafer molds for*

> *medium-throughput microfluidic devices. By integrating parametric design with simultaneous alignment of well-forming inserts, this approach enables repeated fabrication of compartmentalized PDMS devices without individual reservoir punching. The resulting platform accommodates 48 independent microfluidic devices with 96 wells arranged in a standard microplate format, supporting multichannel liquid handling, automated high-content imaging, and compartmentalized neuronal culture. Repeated casting demonstrated mold durability, while retrograde axonal labeling established compatibility with neuronal assays. Although further optimization is required to improve peripheral culture performance and establish quantitative fabrication reproducibility, this approach provides an adaptable and accessible framework for producing customized microfluidic platforms for neuroscience and other applications requiring spatially controlled cell culture.*

R49. Code, files and data statement Table row R49

V9 compresses the file list and states that experimental data are provided in the supplement or repository; the embedded supplement is unchanged and lacks individual counts, so name the actual supplied dataset and reconcile it through R53, while completing the promised versioned archive.

My suggestion: Accurate file list and dataset availability, with a completed study archive

Related feedback: Feedback L551 Feedback L560 Feedback L562 Feedback L564

V7 P78 Table row R49

> *Code and data availability*

V7 P79 Table row R49

> *OpenMFD, including the layout-generation code, photomask DXFs, resin-insert models, plate-frame models, and manuscript source, is available at https://github.com/trissim/OpenMFD. A versioned archive of the reported files will be deposited with the submission.*

V7 P80 Table row R49

> *The Supplementary Information contains Tables S1-S6, Protocols S1-S2, and Notes S1-S2. These contain pin-height measurements, comparisons of fabrication methods, checks for new designs, resin and HIPS printing settings, numbers of molds and casts produced, device and mold protocols, and observations of LP360-filtered exposure and SU-8 adhesion. The supplementary files also include a G-code file containing the HIPS frame-printing settings and an XML file containing the Opera Phenix microscope settings and image positions for Plate 1.*

V7 P81 Table row R49

> *Source images for the dye and CTβ panels and the script used to assemble Figure 4 are included in the source repository. The per-device CTβ count table will be deposited with the submission. The cell counts are from neurons isolated from one embryo and cultured across three plates.*

V9 P94 Table row R49

> *Data availability*

V9 P95 Table row R49

> *The OpenMFD software and associated fabrication files, including photomasks, 3D-printable resin inserts, and plate-frame designs, are available at*

> *https://github.com/trissim/OpenMFD. A versioned archive of the software and fabrication files will be deposited in a public repository before publication. Experimental data supporting the findings, including mold fabrication and casting outcomes, resin-insert measurements, neuronal cell counts, and source images used for quantitative analyses, are provided in the Supplementary Information or associated repository. Detailed fabrication protocols, printing parameters, and imaging configurations are also provided to facilitate reproduction of the devices and experimental procedures.*

R50. Conflict-of-interest statement Table row R50

V9 adds the conflicts declaration after Data availability; the required order is conflicts, data availability, acknowledgements.

My suggestion: Keep the declaration and put it before Data availability

Related feedback: Feedback L581 Feedback L583

V7 P78

V7 P82 **Table row R50**

> *Acknowledgements*

V9 P94

V9 P96 **Table row R50**

> *Conflict of Interest Statement*

V9 P97 **Table row R50**

> *There are no conflicts to declare.*

R51. Acknowledgements and funding Table row R51

V9 adds CIHR and MSCanada funding but removes V7's separate thanks; verify the funding and retain the existing acknowledgements as appropriate.

My suggestion: V7 acknowledgements plus accurate funding details

V7 P82

V7 P83 **Table row R51**

> *We acknowledge the McGill Makerspace for access to filament-extrusion (FDM) 3D printers used to develop the device frame. We also acknowledge the McGill Nanotools Microfab facility for access to microfabrication equipment and parylene coating, and for technical collaboration during process development.*

V9 P98 **Table row R51**

> *Acknowledgements - We acknowledge the McGill Makerspace for access to filament-extrusion (FDM) 3D printers used to develop the device frame. We also acknowledge the McGill Nanotools Microfab facility for access to microfabrication equipment and parylene coating, and for technical collaboration during process development. AEF is funded by grants from the Canadian Institutes of Health Research and MSCanada.*

R52. References and citation style Table row R52

V7 has a 24-entry bibliography and author–year citations; V9 omits that list and adds 20 PMID/PMIDs labels while retaining author–year text in the legends, supplement and some

Methods paragraphs; include the complete references, vet the additions, and apply the final numbered journal style.

My suggestion: V7 bibliography with vetted new entries and consistent citations

Related feedback: Feedback L436 Feedback L569 Feedback L571

V7 P84 **Table row R52**

> *References*

V7 P85 **Table row R52**

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V7 P86 **Table row R52**

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V7 P87 **Table row R52**

> *Taylor, A. M., et al. A microfluidic culture platform for CNS axonal injury, regeneration and transport. Nat. Methods 2, 599-605 (2005). https://doi.org/10.1038/nmeth777*

V7 P88 **Table row R52**

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V7 P89 **Table row R52**

> *Kerman, B. E., et al. In vitro myelin formation using embryonic stem cells. Development 142, 2213-2225 (2015). https://doi.org/10.1242/dev.116517*

V7 P90 **Table row R52**

> *Wang, Z. Z., Wood, M. D., Mackinnon, S. E., Sakiyama-Elbert, S. E. A microfluidic platform to study the effects of GDNF on neuronal axon entrapment. J. Neurosci. Methods 308, 183-191 (2018). https://doi.org/10.1016/j.jneumeth.2018.08.002*

V7 P91 **Table row R52**

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V7 P92 **Table row R52**

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V7 P93 **Table row R52**

> *Epoxy Technology. EPO-TEK 301-2 technical data sheet. Rev. XVIII (2021). https://products.meridianadhesives.com/storage/downloads/uennuicaaxt1bodtipkbpydtg0u8qs33/301-2.pdf*

V7 P94 **Table row R52**

> *Henkel. LOCTITE SI 5140 technical data sheet. January 2023. https://datasheets.tdx.henkel.com/LOCTITE-SI-5140-en_GL.pdf*

V7 P95 **Table row R52**

> *Li, R., et al. A rapidly fabricated microfluidic chip for cell culture. J. Chromatogr. Sci. 54, 523-530 (2016). https://doi.org/10.1093/chromsci/bmv176*

V7 P96 **Table row R52**

> *Siraya Tech. Sculpt High-Temperature Resistant Resin product information. n.d. Accessed 12 August 2026. https://siraya.tech/products/sculpt-resin-high-temperature-resistant-resin*

V7 P97 **Table row R52**

> *Park, J., Li, J., Han, A. Micro-macro hybrid soft-lithography master (MMHSM) fabrication for lab-on-a-chip applications. Biomed. Microdevices 12, 345-351 (2010). https://doi.org/10.1007/s10544-009-9390-9*

V7 P98 **Table row R52**

> *Tamura, T., Suzuki, T. Seamless fabrication technique for micro to millimeter structures by combining 3D printing and photolithography. Jpn. J. Appl. Phys. 58, SDDL10 (2019). https://doi.org/10.7567/1347-4065/ab09d0*

V7 P99 **Table row R52**

> *Kajtez, J., et al. 3D-Printed Soft Lithography for Complex Compartmentalized Microfluidic Neural Devices. Adv. Sci. 7, 2001150 (2020). https://doi.org/10.1002/advs.202001150*

V7 P100 **Table row R52**

> *Pan, J. Y., et al. Hybrid microfabrication of 3D pyrolytic carbon electrodes by photolithography and additive manufacturing. Micro Nano Eng. 15, 100124 (2022). https://doi.org/10.1016/j.mne.2022.100124*

V7 P101 **Table row R52**

> *Chen, Y., Pei, W., Tang, R., Chen, S., Chen, H. Conformal coating of parylene for surface anti-adhesion in polydimethylsiloxane (PDMS) double casting technique. Sens. Actuators A Phys. 189, 143-150 (2013). https://doi.org/10.1016/j.sna.2012.09.024*

V7 P102 **Table row R52**

> *Musgrove, H. B., Cook, S. R., Pompano, R. R. Parylene-C Coating Protects Resin-3D-Printed Devices from Material Erosion and Prevents Cytotoxicity toward Primary Cells. ACS Appl. Bio Mater. 6, 3079-3083 (2023). https://doi.org/10.1021/acsabm.3c00444*

V7 P103 **Table row R52**

> *Ristola, M., et al. A compartmentalized neuron-oligodendrocyte co-culture device for myelin research: design, fabrication and functionality testing. J. Micromech. Microeng. 29, 065009 (2019). https://doi.org/10.1088/1361-6439/ab16a7*

V7 P104 **Table row R52**

> *Moll, L., et al. A Microfluidic High-Capacity Screening Platform for Neurological Disorders. ACS Chem. Neurosci. 15, 236-244 (2024). https://doi.org/10.1021/acschemneuro.3c00409*

V7 P105 **Table row R52**

> *Spijkers, X. M., et al. A directional 3D neurite outgrowth model for studying motor axon biology and disease. Sci. Rep. 11, 2080 (2021). https://doi.org/10.1038/s41598-021-81335-z*

V7 P106 **Table row R52**

> *Spoerk, M., Holzer, C., González-Gutiérrez, J. Material extrusion-based additive manufacturing of polypropylene: A review on how to improve dimensional inaccuracy and warpage. J. Appl. Polym. Sci. 137, 48545 (2020). https://doi.org/10.1002/app.48545*

V7 P107 **Table row R52**

> *Meynard, T., et al. Fabrication of a bioreactor combining soft lithography and vat photopolymerisation to study tissues and multicellular organisms under dynamic culture conditions. Lab Chip 25, 4119-4137 (2025). https://doi.org/10.1039/d5lc00172b*

V7 P108 **Table row R52**

> *Venzac, B. Light-based 3D printing and post-treatments of moulds for PDMS soft lithography. Lab Chip 25, 2129-2147 (2025). https://doi.org/10.1039/d4lc00836g*

V9 P13

V9 P16

V9 P27

V9 P34

V9 P108 **Table row R52**

> *Figure 5: Matching photomasks and insert arrays generated from shared dimensions for three literature-inspired layouts. (A, C, E) Single-device and wafer-scale photomask DXFs; (B, D, F) single and arrayed resin-insert renderings. The examples are an oligodendrocyte myelination layout inspired by Kerman et al. (2015) (A, B), an axon-guidance layout with gradients at right angles to each other inspired by Taylor et al. (2015) (C, D), and a three-compartment layout inspired by Coquinco et al. (2014) (E, F). Plate-frame outputs are omitted for clarity. Physical fabrication and assay testing of these generated designs remain future work.*

R53. Updated DMSO counts and Figure 4 text **Table row R53**

The later DMSO package has 72 rows and five counting-artifact exclusions, giving 22, 21 and 24 devices (67 total); both drafts still report three exclusions and 69 devices, so reconcile the flags with the exclusion rule, then update Methods, Protocol S1, Results, Figure 4 and its legend.

My suggestion: Integrate the recent DMSO plot and match the text

Related feedback: Feedback L51 Feedback L291 Feedback L551 Feedback L607

V7 P41

V7 P64

V7 P81

V7 P116 **Table row R53**

> *Figure 4: Directional fluid movement and neuronal compatibility. (A) Twenty-four two-compartment devices immediately after one-time loading (Hour 0) and after 72 h without volume adjustment. Alexa Fluor 568-conjugated secondary antibody (red; 25 µL) was*

> *loaded into the left compartment and Alexa Fluor 488-conjugated secondary antibody (green; 50 µL) into the right compartment, each at 1:1000 dilution in PBS; devices were kept in a cell-culture incubator between imaging time points. (B) DIV11 CTβ-647 plate overview with two representative device regions aligned and cropped to the same area. (C) Mean CTβ-positive cell-body count per analyzed device for each of three plates containing neurons from the same E18 rat embryo (technical replicates); error bars show the within-plate standard deviation, with 69 interior devices analyzed in total.*

V7 P245 **Table row R53**

> *Devices were excluded only when late-stage microbial contamination or an observable technical failure in culture or imaging prevented cell counting. Every usable count was included, whether high or low.*

V7 P247 **Table row R53**

> *In this experiment, 69 of the 72 interior devices had images suitable for cell counting. Three devices had late-stage contamination or a technical failure that prevented cell counting in the final image.*

V9 P48

V9 P79

V9 P80

V9 P95

V9 P106 **Table row R53**

> *Figure 4: Directional fluid movement and neuronal compatibility. (A) Twenty-four two-compartment devices immediately after one-time loading (Hour 0) and after 72 h without volume adjustment. Alexa Fluor 568-conjugated secondary antibody (red; 25 µL) was loaded into the left compartment and Alexa Fluor 488-conjugated secondary antibody (green; 50 µL) into the right compartment, each at 1:1000 dilution in PBS; devices were kept in a cell-culture incubator between imaging time points. (B) DIV11 CTβ-647 plate overview with two representative device regions aligned and cropped to the same area. (C) Mean CTβ-positive cell-body count per analyzed device for each of three plates containing neurons from the same E18 rat embryo (technical replicates); error bars show the within-plate standard deviation, with 69 interior devices analyzed in total.*

V9 P235 **Table row R53**

> *Devices were excluded only when late-stage microbial contamination or an observable technical failure in culture or imaging prevented cell counting. Every usable count was included, whether high or low.*

V9 P237 **Table row R53**

> *In this experiment, 69 of the 72 interior devices had images suitable for cell counting. Three devices had late-stage contamination or a technical failure that prevented cell counting in the final image.*

R54. Figure 4 scale bars **Table row R54**

The newer supplementary-biology views supply calibrated 500 µm CTβ bars and crop dimensions; integrate these views into Figure 4, while adding the acquisition scale and bar to the separate tracer panel.

My suggestion: Use the recent calibrated CTβ views; add tracer bars

Related feedback: Feedback L301

V7 P43

V7 P116

V9 P49

V9 P106

R55. Counting-method settings Table row R55

The newer analysis supplies software, count tables, overlays and saved settings; identify the configuration matching the chosen dataset and integrate it into Methods and Protocol S1, with the reconciled exclusions from R53.

My suggestion: Integrate the saved supplementary-biology counting method

Related feedback: Feedback L51

V7 P241 Table row R55

> *Identify and count CTβ-positive neuronal cell bodies using the same cell segmentation settings and criteria for CTβ labeling in every cropped image. Record one CTβ-positive cell-body count for each cropped device.*

V9 P231 Table row R55

> *Identify and count CTβ-positive neuronal cell bodies using the same cell segmentation settings and criteria for CTβ labeling in every cropped image. Record one CTβ-positive cell-body count for each cropped device.*

R56. Animal approval details Table row R56

Both protocols mention an approved procedure but omit the approving body, guidelines and approval number.

My suggestion: Add the approving body, guidelines and approval number

Related feedback: Feedback L576

V7 P226 Table row R56

> *Isolate cortical neurons from one embryonic day 18 (E18) Sprague Dawley rat embryo following the procedure described by Harris et al. (2007), under the laboratory's approved animal-use procedure.*

V9 P216 Table row R56

> *Isolate cortical neurons from one embryonic day 18 (E18) Sprague Dawley rat embryo following the procedure described by Harris et al. (2007), under the laboratory's approved animal-use procedure.*

R57. Supplementary table layout Table row R57

The data are unchanged, but V9 removes S1's explicit 10-point cell text and inherits a 12-point default; in its rendered page 21, 5.567 splits into 5.56 and 7, and the other heights wrap similarly. The earlier V7/V8 render kept these values on one line. V9 also splits long words in S2; restore readable numerical cells and widen the comparison columns.

My suggestion: Restore compact S1 numbers; widen the S2/S3 columns

Related feedback: Feedback L494 Feedback L507 Feedback L606

V7 P121 **Table row R57**

> *Each pin height was measured individually with a micrometer across one 8 × 12 resin-insert array (rows A-H, columns 1-12; n = 96). Values are reported in millimeters (mm) and as deviations from the mean in micrometers (µm).*

V7 P129 **Table row R57**

> *Representative strategies for combining micrometer-scale features with millimeter-scale wells or other tall structures are summarized below. The comparison emphasizes the operations and equipment each strategy adds to conventional PDMS soft lithography.*

V7 P131 **Table row R57**

> *The tested design used two-well device units spaced 18 × 9 mm apart and a 200 µm SUEX well layer. These values provide starting points for new layouts. Changes to gaps, margins, taper allowances, and adhesive contact areas require testing to determine which values work reliably.*

V9 P111 **Table row R57**

> *Each pin height was measured individually with a micrometer across one 8 × 12 resin-insert array (rows A-H, columns 1-12; n = 96). Values are reported in millimeters (mm) and as deviations from the mean in micrometers (µm).*

V9 P119 **Table row R57**

> *Representative strategies for combining micrometer-scale features with millimeter-scale wells or other tall structures are summarized below. The comparison emphasizes the operations and equipment each strategy adds to conventional PDMS soft lithography.*

V9 P121 **Table row R57**

> *The tested design used two-well device units spaced 18 × 9 mm apart and a 200 µm SUEX well layer. These values provide starting points for new layouts. Changes to gaps, margins, taper allowances, and adhesive contact areas require testing to determine which values work reliably.*

R58. Separate supplement and height CSV Table row R58

The 96-height CSV matches Table S1 and the existing separate supplement matches V7's supplementary text and ten tables; V9 retains those contents unchanged.

My suggestion: Use the existing separate supplement and height CSV

Related feedback: Feedback L511 Feedback L513 Feedback L609

V7 P119 **Table row R58**

> *Supplementary Information*

V7 P121

V9 P109 **Table row R58**

> *Supplementary Information*

V9 P111

R59. Later channel-width measurements Table row R59

The later check supplies nine brightfield ruler estimates from three devices, spanning 10.7–13.1 µm; moving each blurred-wall endpoint by one pixel changes a span by up to 2.37 µm, so report the pixel scale, endpoint rule and overlays with the estimates.

My suggestion: Report the width estimates with their measurement method

Related feedback: Feedback L136

V7 P53

V9 P65

V9 P89

R60. Individual mold-production records Table row R60

Both report more than 50 casts, more than eight months of use and both mold losses; existing per-mold records could add detail, but no new durability study is proposed.

My suggestion: Keep the reuse observations; add per-mold records if available

Related feedback: Feedback L138 Feedback L522

V7 P54

V9 P68

V9 P87

R61. Channel depth and between-cast comparisons Table row R61

The saved discussion proposes depth and between-cast measurements; integrate the available width estimates in R59 without adding those studies.

My suggestion: No new depth or cast-to-cast study proposed

Related feedback: Feedback L136 Feedback L346

V7 P53

V7 P72

V9 P65

V9 P89

R62. Figure layout and additional plots Table row R62

All five legend titles, device/alignment views and Figure 5's three application rows already exist; V9 removes the legends' explicit 11-point size and inherits 12-point text. Revisit image sizes and layout and integrate the recent device-dot plot and calibrated views; height and aggregate mold plots are possible uses of existing data, not proposed additions here.

My suggestion: Revisit image sizes and Figure 3/5 layout; integrate the supplied biology views

Related feedback: Feedback L214 Feedback L238 Feedback L239 Feedback L262 Feedback L291 Feedback L313 Feedback L327 Feedback L398

V7 P110 Table row R62

> *Figure 1: Shared device dimensions connect photomask and insert design to plate assembly. (A) Editable preset containing the shared device dimensions. (B, C) Single-device and wafer-scale photomask DXFs. (D) Single and arrayed 3D-printable resin-insert models. (E) Plate-frame model. (F) Assembled plate-format device.*

V7 P112 **Table row R62**

> *Figure 2: Insert transfer and wafer bonding for the hybrid mold. (A) Wafer and 3D-printed resin insert array before clamping. (B) Top view showing the relative positions of the printed insert, SUEX alignment hole, adhesive contact area, and gap around the alignment pin. (C) Clamped assembly used for acetone washing and room-temperature cure. (D) Side view of the clamping layers, including the compressible EPDM rubber sheet. (E) Bond between a resin insert and the wafer. (F) SUEX alignment-hole micrograph.*

V7 P114 **Table row R62**

> *Figure 3: Repeated PDMS device casting and plate assembly. PDMS is poured over the full hybrid mold within an aluminum-foil reservoir, degassed in a cylindrical vacuum chamber, and cured in a six-shelf wafer rack. The circular cast is demolded, placed with the molded channels facing upward, taped to paper along the cutting guides, and trimmed with a paper guillotine. The trimmed device is plasma-bonded to glass and autoclaved at 121°C for 15 min before frame assembly. The frame is inverted with its adhesive groove facing upward, and the device is seated with the glass facing upward. After a 3-day room-temperature cure, the assembled plate is turned upright and plasma-treated before coating and culture. One-time mold fabrication is detailed in Supplementary Protocol S2.*

V7 P116

V7 P118 **Table row R62**

> *Figure 5: Matching photomasks and insert arrays generated from shared dimensions for three literature-inspired layouts. (A, C, E) Single-device and wafer-scale photomask DXFs; (B, D, F) single and arrayed resin-insert renderings. The examples are an oligodendrocyte myelination layout inspired by Kerman et al. (2015) (A, B), an axon-guidance layout with gradients at right angles to each other inspired by Taylor et al. (2015) (C, D), and a three-compartment layout inspired by Coquinco et al. (2014) (E, F). Plate-frame outputs are omitted for clarity. Physical fabrication and assay testing of these generated designs remain future work.*

V9 P100 **Table row R62**

> *Figure 1: Shared device dimensions connect photomask and insert design to plate assembly. (A) Editable preset containing the shared device dimensions. (B, C) Single-device and wafer-scale photomask DXFs. (D) Single and arrayed 3D-printable resin-insert models. (E) Plate-frame model. (F) Assembled plate-format device.*

V9 P102 **Table row R62**

> *Figure 2: Insert transfer and wafer bonding for the hybrid mold. (A) Wafer and 3D-printed resin insert array before clamping. (B) Top view showing the relative positions of the printed insert, SUEX alignment hole, adhesive contact area, and gap around the alignment pin. (C) Clamped assembly used for acetone washing and room-temperature cure. (D) Side view of the clamping layers, including the compressible EPDM rubber sheet. (E) Bond between a resin insert and the wafer. (F) SUEX alignment-hole micrograph.*

V9 P104 **Table row R62**

> *Figure 3: Repeated PDMS device casting and plate assembly. PDMS is poured over the full hybrid mold within an aluminum-foil reservoir, degassed in a cylindrical vacuum chamber, and cured in a six-shelf wafer rack. The circular cast is demolded, placed with the molded channels facing upward, taped to paper along the cutting guides, and trimmed with a paper guillotine. The trimmed device is plasma-bonded to glass and autoclaved at 121°C for 15*

> *min before frame assembly. The frame is inverted with its adhesive groove facing upward, and the device is seated with the glass facing upward. After a 3-day room-temperature cure, the assembled plate is turned upright and plasma-treated before coating and culture. One-time mold fabrication is detailed in Supplementary Protocol S2.*

V9 P106

V9 P108

R63. Small supplement edits Table row R63

Add Figure S1/S2 links within the corresponding notes; Table S3 already separates tested and generated-only designs, and the protocol's open-channel result can become an inspection instruction; no raised-edge measurement experiment is proposed.

My suggestion: Add note citations and clarify the protocol inspection instruction

Related feedback: Feedback L450 Feedback L451 Feedback L475 Feedback L477 Feedback L541

V7 P152 Table row R63

> *During development of the 200 µm SUEX K200 layer, unfiltered exposure produced ridge-like artifacts around patterned features. These ridges interrupted the seal between the printed inserts and the mold, allowing epoxy to reach nearby SU-8 features and weakening the insert bond.*

V7 P153 Table row R63

> *Exposure through an LP360 long-pass filter substantially reduced the ridges and produced a flatter bonding surface.*

V7 P155 Table row R63

> *Supplementary Figure S1. SUEX K200 features after unfiltered exposure (left) and LP360-filtered exposure (right). Ridge-like edge artifacts are visible after unfiltered exposure and substantially reduced with the filter. The images provide a qualitative process-development comparison.*

V7 P157 Table row R63

> *SU-8 microchannels patterned directly on bare silicon at a target thickness of 5 µm delaminated during development or subsequent processing.*

V7 P158 Table row R63

> *A 5 µm flood-exposed SU-8 2005 base layer was therefore added before the patterned microchannel layer. The base layer is exposed without a mask and is not developed. It provides sufficient adhesion without a dehydration bake or HMDS treatment.*

V7 P159 Table row R63

> *The developed two-layer SU-8 stack also requires a 10 min hard bake at 150°C before SUEX lamination. No controlled ramp up or cooldown is used because the combined SU-8 stack is approximately 10 µm thick. Eight wafers were processed in parallel during one process-development batch in which this hard bake was inadvertently omitted. Stress during cooling after subsequent SUEX processing caused the SU-8 stack to warp and delaminate around well features on part of the batch, although some wafers remained intact. A subsequent two-wafer batch included the 150°C hard bake, and both wafers remained intact. These observations distinguish adhesion to bare silicon, addressed by the*

> *flood-exposed base layer, from stabilization of the completed SU-8 stack before SUEX processing.*

V7 P161 **Table row R63**

> *Supplementary Figure S2. Representative optical micrographs of SU-8 microchannel delamination when features were patterned directly on bare silicon without the flood-exposed base layer. Scale bars: 100 µm (left) and 200 µm (right).*

V7 P332 **Table row R63**

> *Microscopy confirmed open channels after coating and casting. Specified microchannel dimensions in the manuscript and design tables refer to the design dimensions before parylene coating.*

V9 P142 **Table row R63**

> *During development of the 200 µm SUEX K200 layer, unfiltered exposure produced ridge-like artifacts around patterned features. These ridges interrupted the seal between the printed inserts and the mold, allowing epoxy to reach nearby SU-8 features and weakening the insert bond.*

V9 P143 **Table row R63**

> *Exposure through an LP360 long-pass filter substantially reduced the ridges and produced a flatter bonding surface.*

V9 P145 **Table row R63**

> *Supplementary Figure S1. SUEX K200 features after unfiltered exposure (left) and LP360-filtered exposure (right). Ridge-like edge artifacts are visible after unfiltered exposure and substantially reduced with the filter. The images provide a qualitative process-development comparison.*

V9 P147 **Table row R63**

> *SU-8 microchannels patterned directly on bare silicon at a target thickness of 5 µm delaminated during development or subsequent processing.*

V9 P148 **Table row R63**

> *A 5 µm flood-exposed SU-8 2005 base layer was therefore added before the patterned microchannel layer. The base layer is exposed without a mask and is not developed. It provides sufficient adhesion without a dehydration bake or HMDS treatment.*

V9 P149 **Table row R63**

> *The developed two-layer SU-8 stack also requires a 10 min hard bake at 150°C before SUEX lamination. No controlled ramp up or cooldown is used because the combined SU-8 stack is approximately 10 µm thick. Eight wafers were processed in parallel during one process-development batch in which this hard bake was inadvertently omitted. Stress during cooling after subsequent SUEX processing caused the SU-8 stack to warp and delaminate around well features on part of the batch, although some wafers remained intact. A subsequent two-wafer batch included the 150°C hard bake, and both wafers remained intact. These observations distinguish adhesion to bare silicon, addressed by the flood-exposed base layer, from stabilization of the completed SU-8 stack before SUEX processing.*

V9 P151 **Table row R63**

> *Supplementary Figure S2. Representative optical micrographs of SU-8 microchannel delamination when features were patterned directly on bare silicon without the flood-exposed base layer. Scale bars: 100 µm (left) and 200 µm (right).*

V9 P322 **Table row R63**

> *Microscopy confirmed open channels after coating and casting. Specified microchannel dimensions in the manuscript and design tables refer to the design dimensions before parylene coating.*

V7 P162

> *Supplementary Protocol S1. Post-mold device assembly, primary cortical neuron culture, and CTβ retrograde axon tracing*

V7 P249

> *Supplementary Protocol S2. One-time hybrid SU-8/SUEX/resin mold fabrication*

V9 P152

> *Supplementary Protocol S1. Post-mold device assembly, primary cortical neuron culture, and CTβ retrograde axon tracing*

V9 P239

> *Supplementary Protocol S2. One-time hybrid SU-8/SUEX/resin mold fabrication*

Additional sources and images

Later brightfield width check

Widths were measured from existing brightfield images during this review.

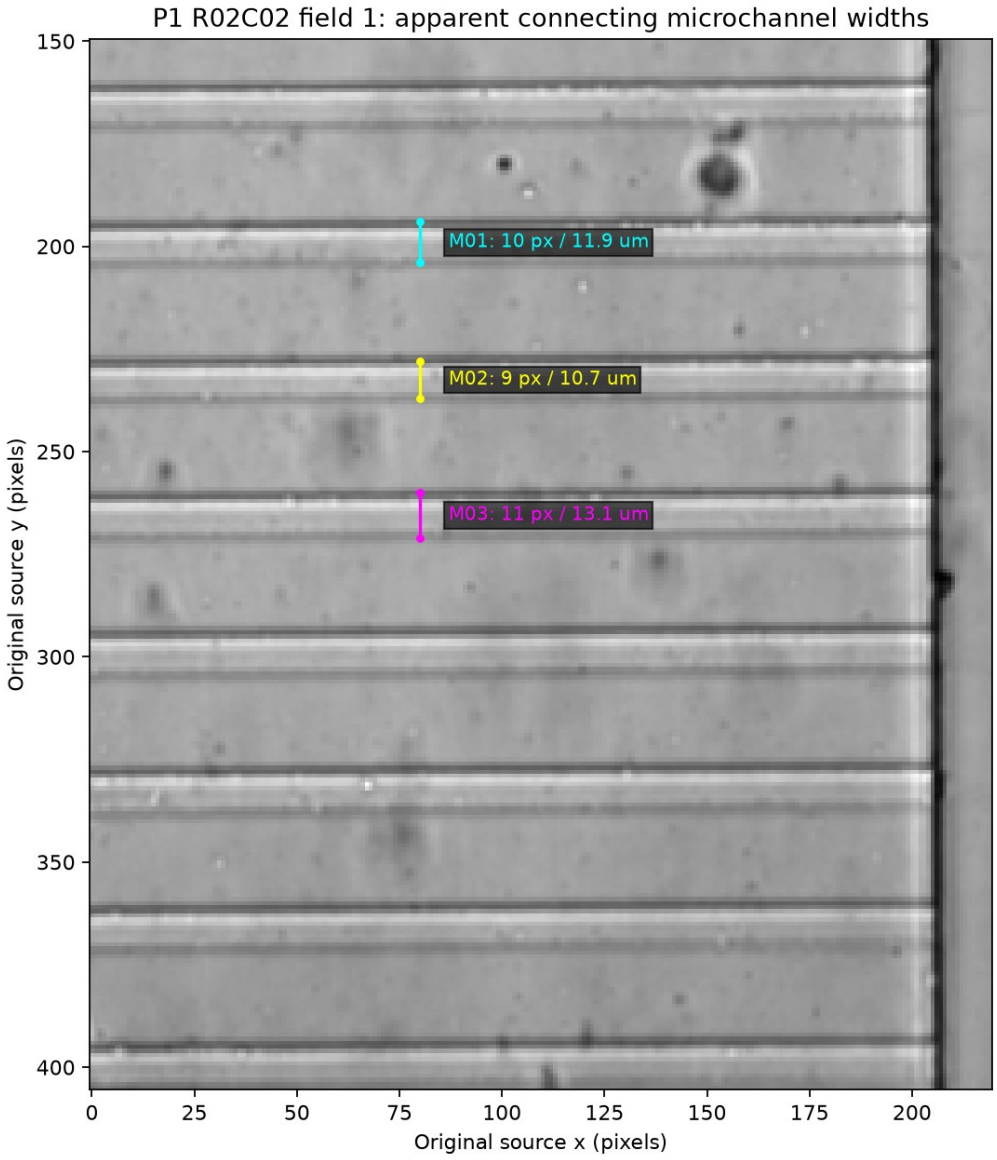
Nine rulers span 9–11 original pixels, or 10.7–13.1 µm, at 1.18675253 µm/pixel from Index.xml. Endpoints mark the centers of opposing dark wall bands. Blur affects edge placement.

Ruler	Plate / device	Endpoints (x, y), pixels	Pixels	µm
M01	P1 / R02C02	(80, 194) to (80, 204)	10	11.9
M02	P1 / R02C02	(80, 228) to (80, 237)	9	10.7
M03	P1 / R02C02	(80, 260) to (80, 271)	11	13.1
M04	P2 / R02C02	(600, 183) to (600, 192)	9	10.7
M05	P2 / R02C02	(600, 216) to (600, 225)	9	10.7
M06	P2 / R02C02	(600, 249) to (600, 258)	9	10.7
M07	P3 / R02C02	(500, 183) to	9	10.7

Ruler	Plate / device	Endpoints (x, y), pixels	Pixels	µm
		(500, 192)		
M08	P3 / R02C02	(500, 216) to (500, 225)	9	10.7
M09	P3 / R02C02	(500, 249) to (500, 259)	10	11.9

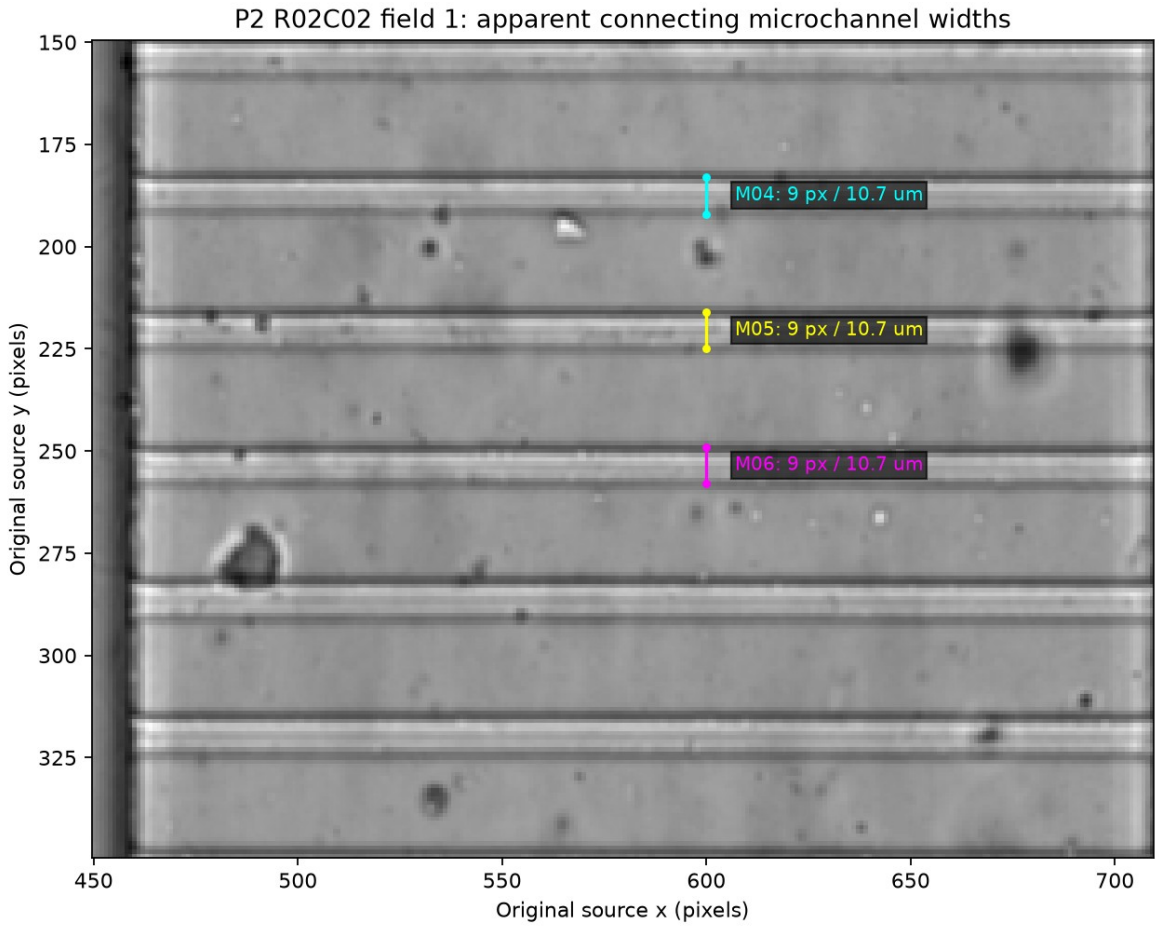
All rulers use field 1, channel 1 of r02c02f01p01-ch1sk1fk1fl1.tiff at original resolution. Moving each endpoint by one pixel changes the span by up to 2.37 µm.

P1: TS_MFD_PRE_P1_20260203__2026-02-03T19_23_43-Measurement 1



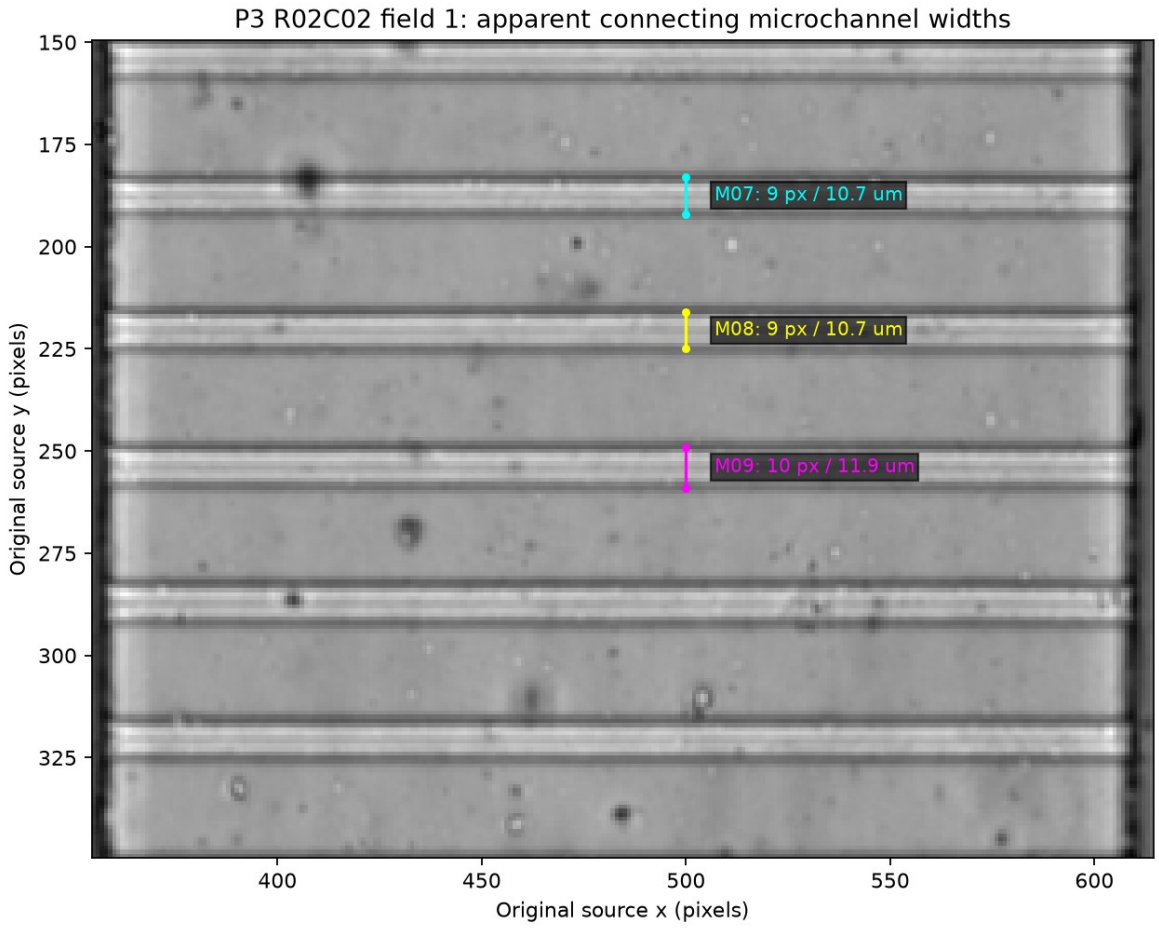
P1 raw-field rulers

P2: TS_MFD_PRE_P2_20260203__2026-02-03T19_50_17-Measurement 1



P2 raw-field rulers

P3: TS_MFD_PRE_P3_20260203__2026-02-03T20_12_32-Measurement 1



P3 raw-field rulers

R07. Introduction: ways to combine small channels and large wells

Table row R07

R31. Results: pin heights and clamping

Table row R31

Table S1: A3/A4/A6 = 5.599 mm; H12 = 5.483 mm; A1 = 5.567 mm.

Recalculated from all 96 S1 values: mean 5.550770833 mm, sample SD 26.67996653 um, range 116 um.

R33. Results: repeated casting and mold use

Table row R33

Related hybrid-fabrication paper: Ristola et al. (2019), Methods 2.2 and Results 3.1. They inspected devices from the fifth fabrication batch and estimated about 15 mold uses before significant degradation. They report reuse and wear observations, not a cast-by-cast durability curve. V7 already reports its own cast total and elapsed use; no new durability study is proposed here.

R49. Code, files and data statement

Table row R49

RSC article guide: Chip design masks guidance says "no compulsion" to supply masks.

R50. Conflict-of-interest statement

Table row R50

RSC article guide: Data availability section says "after the conflicts of interest statement and before any acknowledgements".

R52. Reference list removed

Table row R52

Verified records: Kerman 26015546, Taylor 26000554, Coquinco 24662423.

R53. Updated DMSO counts and Figure 4 text

Table row R53

The DMSO manifest identifies the checked count CSV: 72 rows, 24 per plate, with plate and device IDs, counts, exclusion flags and reasons.

Included CTβ counts: Plate 1 22, Plate 2 21, Plate 3 24. The exclusions, recalculated summaries and updated plot are included here.

The checked DMSO CSV (R53) names five exclusions: p1 R03C02 and p1 R06C02, "PRE segmentation artifact identified by plate-mosaic review"; p2 R02C02, p2 R05C03, p2 R06C03, "Extreme pre-cell-count artifact".

The legend says: "Five devices with counting artifacts were excluded from B and E".

DMSO figure code, lines 812-820, selects non-flagged devices and uses mean(values) and stdev(values).

Recalculated from R53: Plate 1 177.7273 ± 43.7244, n=22; Plate 2 269.3333 ± 62.2971, n=21; Plate 3 203.6250 ± 32.3914, n=24.

Recent supplementary-biology CTβ count plot

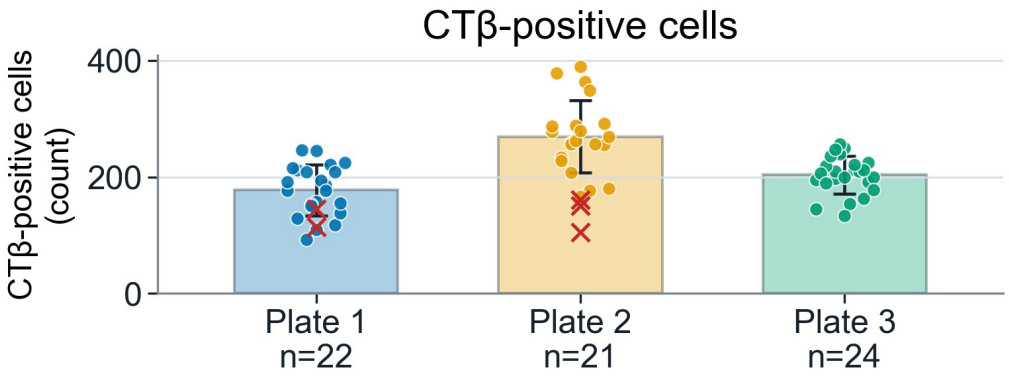
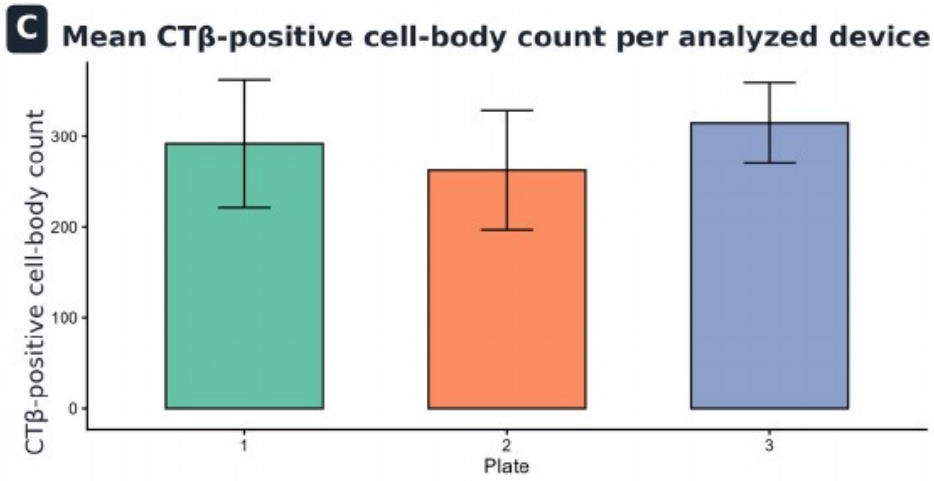


Figure 4C: original count bars (unchanged figure)



R54. Figure 4 scale bars

Table row R54

The old Figure 4 has no visible bars.
DMSO scale check: "The scale bars in Panel G are calibrated correctly." Calibration: 1.1867525298988041 µm/pixel; 500 µm = 421.318 source pixels. PRE crop: 2292 × 3700 pixels; POST width: 1000 pixels.
Figure code, lines 747-748, supplies pixel scale and 500 µm length to the image renderer.

Recent supplementary-biology calibrated CTβ views

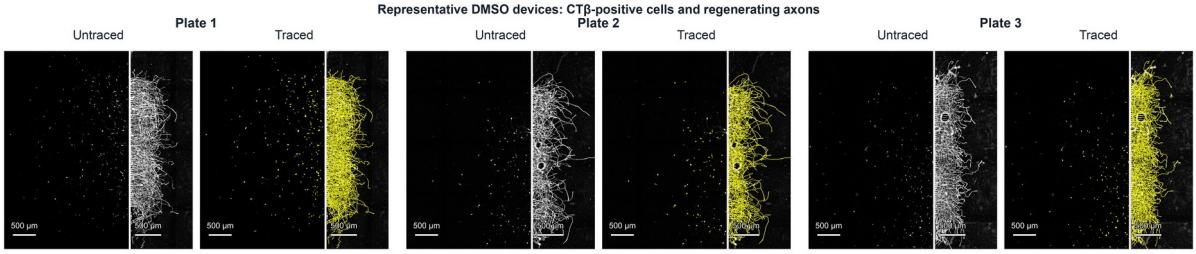
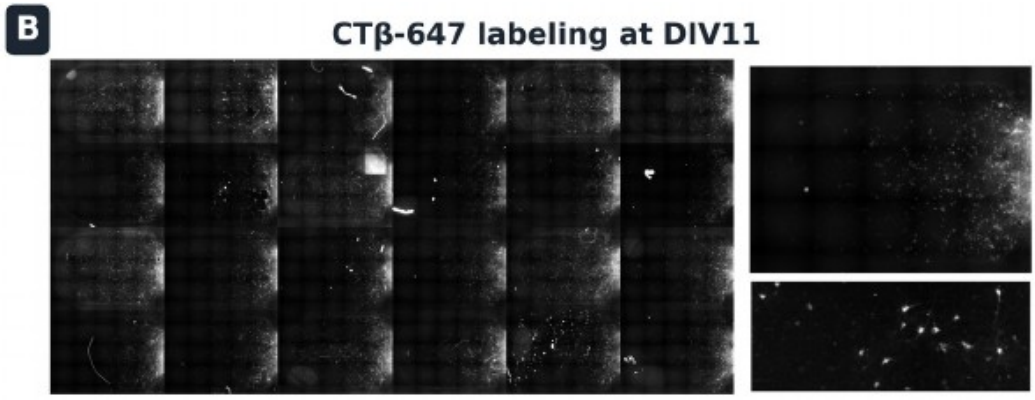


Figure 4B: original microscopy panels (unchanged figure)



R55. Counting-method settings

Table row R55

The recent supplementary-biology figure legend identifies OpenHCS 0.5.15. Its manifest identifies CTβ-647 channel 2 and traced/untraced views.

The archived Stitched_Plates/final_scripts/pre_analysis_v2.py specifies channel 2; normalization from 0.1 to 99.99 percentiles; white top-hat radius 35 pixels; count_cells_simple; maximum object area 2000 pixels; maximum eccentricity 0.9; watershed above 200 pixels; seed separation 10 pixels.

R56. Animal approval details

Table row R56

RSC welfare policy: "institutional/local ethics committee"; "where possible the approval or case number should be provided".

R57. Supplementary table layout

V9 Table S1, rendered page 21: heights split across lines

The cell value remains 5.567 in the DOCX; V9's layout displays 5.56 above 7. The earlier-render crops below show the same values without this wrapping.

S1A. Measured pin heights (mm)

Row	1	2	3	4	5	6	7	8	9	10	11	12
A	5.567	5.588	5.599	5.599	5.591	5.599	5.598	5.598	5.597	5.596	5.581	5.562
B	5.557	5.588	5.591	5.591	5.592	5.588	5.593	5.592	5.592	5.591	5.587	5.562
C	5.55	5.55	5.55	5.56	5.56	5.55	5.56	5.55	5.56	5.56	5.55	5.55

Table row R57

Table S1A: column headings and row A (earlier render)

S1A. Measured pin heights (mm)

Row	1	2	3	4	5	6	7	8	9	10	11	12
A	5.567	5.588	5.599	5.599	5.591	5.599	5.598	5.598	5.597	5.596	5.581	5.562

Table S1A: row H (earlier render)

H	5.499	5.516	5.525	5.525	5.521	5.520	5.529	5.517	5.517	5.503	5.495	5.483

Table S1B: row A deviations (earlier render)

Row	1	2	3	4	5	6	7	8	9	10	11	12
A	+16.2	+37.2	+48.2	+48.2	+40.2	+48.2	+47.2	+47.2	+46.2	+45.2	+30.2	+11.2

Table S1B: row H deviations (earlier render)

H	-51.8	-34.8	-25.8	-25.8	-29.8	-30.8	-21.8	-33.8	-33.8	-47.8	-55.8	-67.8

Table S2: headings and first row (earlier render)

Approach	Micrometer-scale features	Millimeter-scale wells or tall structures	Plate-format and automation compatibility	Added workflow and equipment	Representative references
Conventional PDMS soft lithography with manual punching	SU-8 photolithography for micrometer-scale channels	Openings made with a biopsy punch after casting	Each opening is made separately after casting	Biopsy punching and manual debris removal during every device-production cycle	Baseline approach; see Taylor et al., 2005

Table S3: separation of fabrication layers (earlier render)

Separation of fabrication layers	5 µm SU-8 microchannel layer plus 200 µm SUEX well and alignment-hole layer	The microchannels and taller well features are fabricated as separate aligned layers using matching solid and hollow alignment marks	Combining layers, changing SUEX thickness, or moving alignment holes into the microchannel layer

R58. Separate supplement and height CSV

Table row R58

Imported height CSV: row labels A-H and columns 1-12; A1 5.567 mm, A3 5.599 mm, H12 5.483 mm. All 96 values match V7 Table S1A and are included below.

The existing separate supplement (Word and PDF) contains the same 212 nonempty paragraphs and ten tables as V7's supplement. Its cited passages, table excerpts and images are included in this document.

RSC article guide: Supplementary information is published **"as is"**. R58 identifies the existing height CSV; R58 identifies the existing separate supplement. S2 p. 17 breaks long words.

Pin heights (mm), in the original 8 x 12 array order.

Row	1	2	3	4	5	6	7	8	9	10	11	12
A	5.567	5.588	5.599	5.599	5.591	5.599	5.598	5.598	5.597	5.596	5.581	5.562
B	5.557	5.588	5.591	5.591	5.592	5.588	5.593	5.592	5.592	5.591	5.587	5.562
C	5.555	5.556	5.555	5.563	5.563	5.556	5.560	5.557	5.560	5.560	5.558	5.555
D	5.549	5.552	5.552	5.555	5.552	5.552	5.549	5.552	5.549	5.552	5.550	5.545
E	5.543	5.546	5.549	5.550	5.547	5.547	5.548	5.544	5.544	5.543	5.536	5.532
F	5.522	5.531	5.548	5.544	5.540	5.540	5.543	5.542	5.536	5.531	5.528	5.522
G	5.519	5.528	5.542	5.543	5.536	5.535	5.538	5.538	5.532	5.529	5.521	5.501
H	5.499	5.516	5.525	5.525	5.521	5.520	5.529	5.517	5.517	5.503	5.495	5.483

R62. Figure layout and additional plots

Legend formatting: V7 explicitly sets 11-point text; V9 removes those size settings and inherits the 12-point default. The legend wording and image bytes are unchanged.

Table row R62

Calculation: (2.00 - 1.85) / 2 = 0.075 mm = 75 um per side.

Figure 1 legend title (unchanged figure)

Figure 1: Shared device dimensions connect photomask and insert design to plate assembly. (A) Editable preset containing the shared device dimensions. (B, C) Single-device and wafer-scale photomask DXFs. (D) Single and arrayed 3D-printable resin-insert models. (E) Plate-frame model. (F) Assembled plate-format device.

Figure 1 heading (unchanged figure)

OpenMFD turns one parameterized design into masks, printed parts, and a packaged device

Figure 3 heading (unchanged figure)

Post-mold device replication and assembly

Figure 2C: clamping fixture (unchanged figure)



Figure 1A: shared parameters and outputs (unchanged figure)

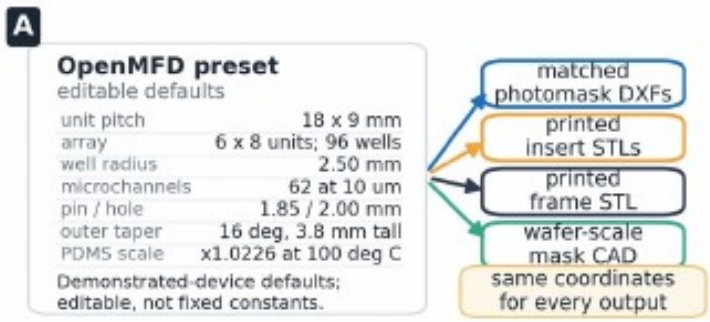


Figure 1B: connected two-well design (unchanged figure)

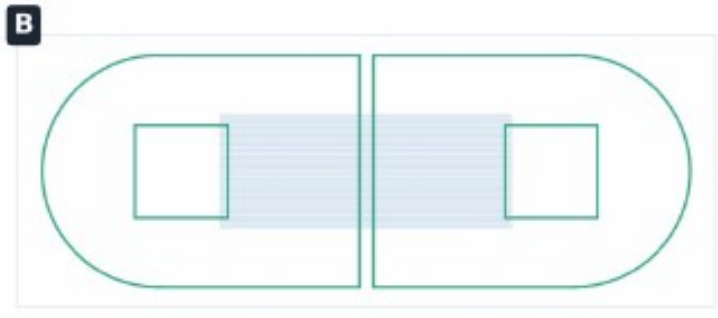


Figure 2B: pin and opening (unchanged figure)

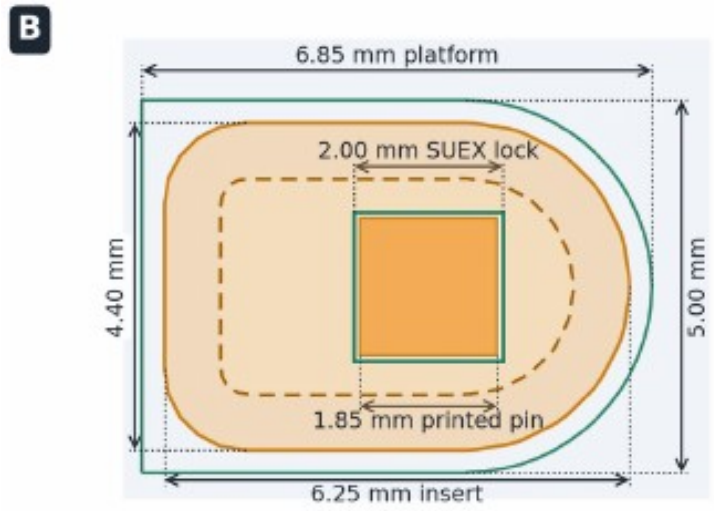


Figure 2D: backing and clamps (unchanged figure)

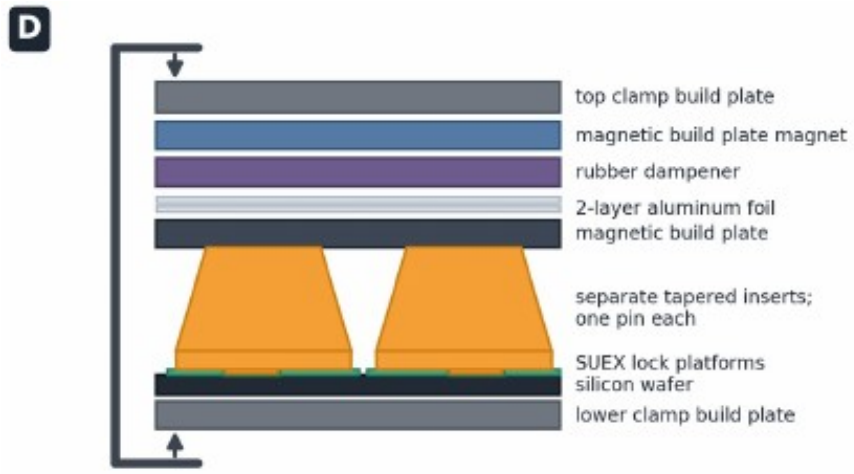


Figure 2A: components before bonding (unchanged figure)

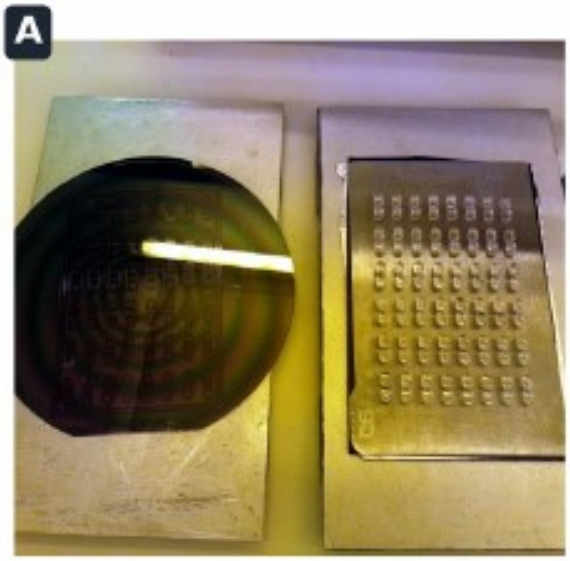


Figure 2E: bonded inserts, close-up (unchanged figure)

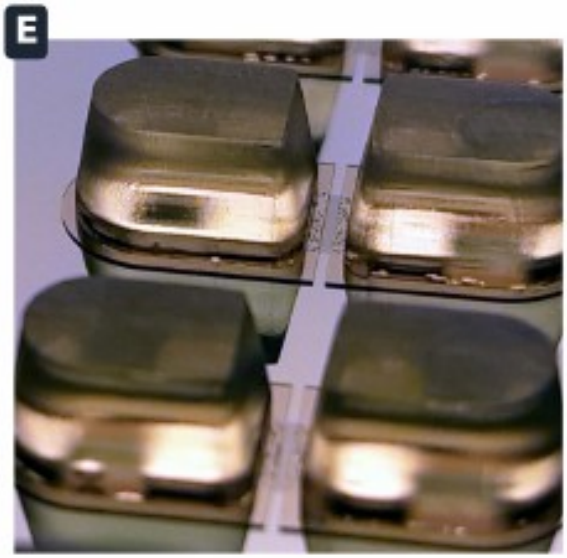


Figure 3: first three illustrated steps (unchanged figure)

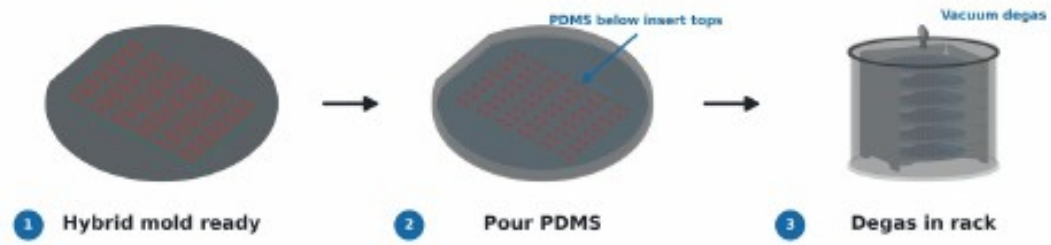


Figure 4A: tracer time points (unchanged figure)

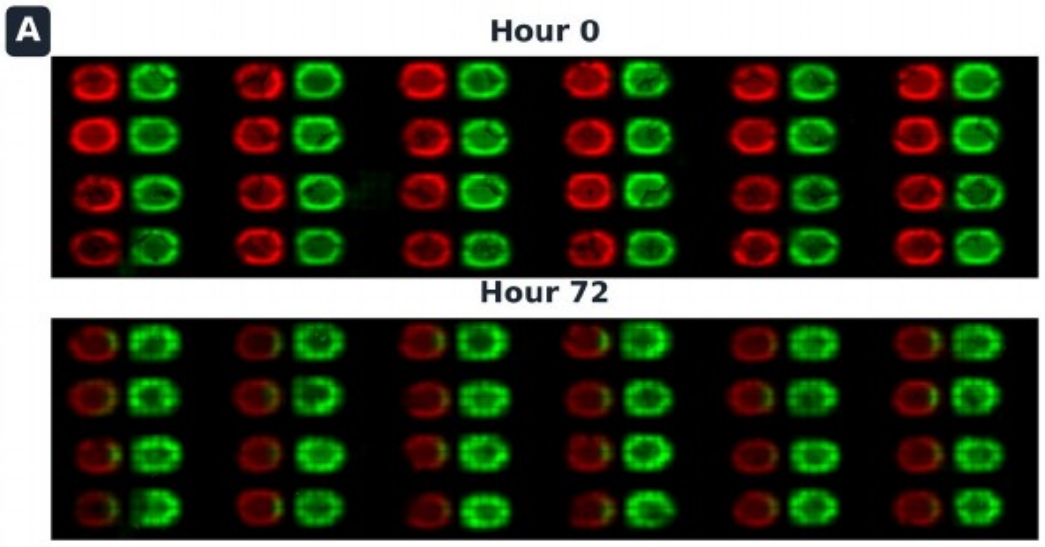


Figure 5: three existing application rows (unchanged figure)

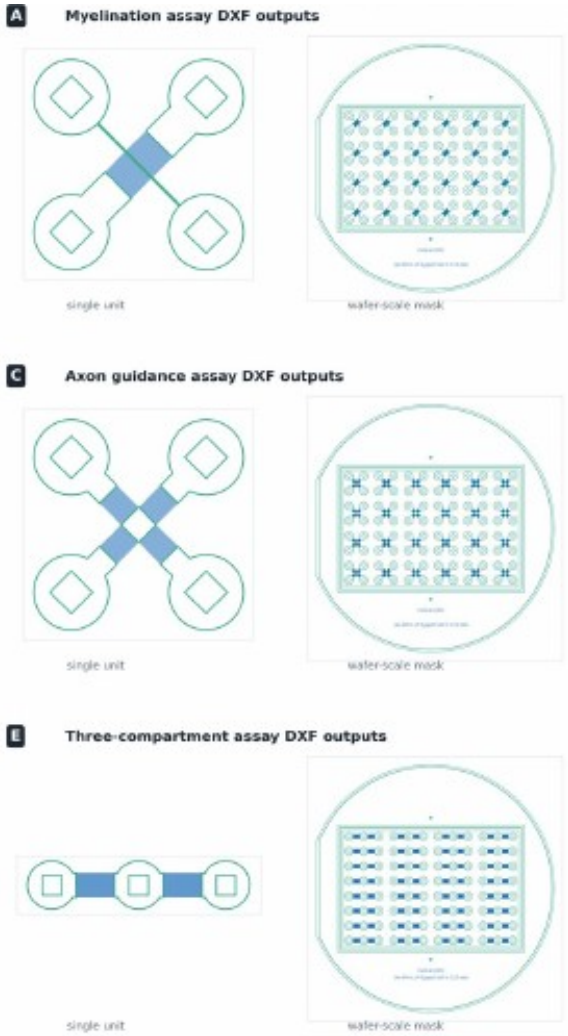


Figure 5: output views in the first row (unchanged figure)



R63. Small supplement edits and extra failure comparisons

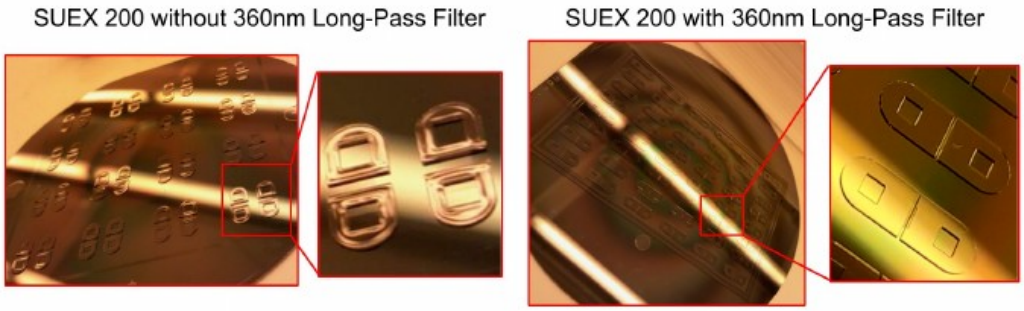
Table row R63

Note S1 beside Figure S1 (unchanged figure)

Supplementary Note S1. LP360 long-pass filter improves SUEX K200 top-surface flatness

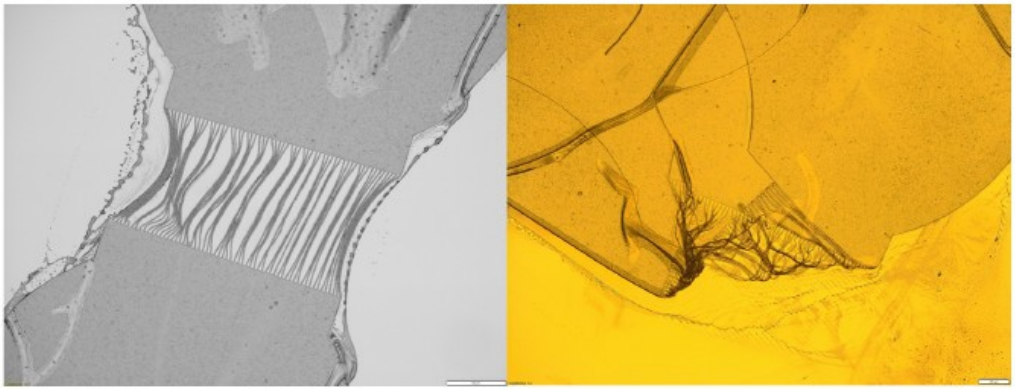
During development of the 200 µm SUEX K200 layer, unfiltered exposure produced ridge-like artifacts around patterned features. These ridges interrupted the seal between the printed inserts and the mold, allowing epoxy to reach nearby SU-8 features and weakening the insert bond.

Exposure through an LP360 long-pass filter substantially reduced the ridges and produced a flatter bonding surface.



Supplementary Figure S1. SUEX K200 features after unfiltered exposure (left) and LP360-filtered exposure (right). Ridge-like edge artifacts are visible after unfiltered exposure and substantially reduced with the filter. The images provide a qualitative process-development comparison.

Figure S2: failed-condition images and legend (unchanged figure)



Supplementary Figure S2. Representative optical micrographs of SU-8 microchannel delamination when features were patterned directly on bare silicon without the flood-exposed base layer. Scale bars: 100 µm (left) and 200 µm (right).

Published Lab on a Chip comparisons

The papers group fabrication, device use and testing in comparable ways. Lin has five Results groups, Compera three and V7 four.

Page locations identify the versions checked for this review. Public links open the article DOI or full text.

Paper 1

Paper

Lin, Do, Kwon and Lillehoj (2017). *3D printed metal molds for hot embossing plastic microfluidic devices*. DOI; Public article (DOI).

Published headings and location

Experimental, journal p. 242 / PDF p. 2: "Mold fabrication"; "Surface roughness and replication fidelity measurements"; "Microchannel characterization". Results, journal p. 245 / PDF p. 5: "Mold reproducibility for multiple embossing cycles".

What matches

Both papers separate mold fabrication, device production and assessment in Methods, then report reuse in Results.

What differs

Lin uses hot embossing of plastic; this paper casts PDMS.

Paper 2

Paper

Moon et al. (2024). *Reversible bonding in thermoplastic elastomer microfluidic platforms for harvestable 3D microvessel networks*. DOI; Public full text.

Published headings and location

Experimental, PDF pp. 11-13: "Cell culture"; "Microfluidic cell seeding and culture"; "Statistical analysis". Fabrication and bonding tests are separate sections before these biological methods.

What matches

Both combine device-fabrication methods with cell culture, imaging and statistical analysis.

What differs

Moon places Methods after Results and studies microvessel cultures.

Paper 3

Paper

Taylor, Menon and Gupton (2015). *Passive microfluidic chamber for long-term imaging of axon guidance in response to soluble gradients*. DOI; Public full text.

Published headings and location

Experimental, author-manuscript PDF pp. 7-9: "Device fabrication"; "Device assembly and preparation"; "Neuronal culture"; "Imaging and analysis". Results/discussion begins on PDF p. 4 with device design, followed by gradient and neuronal responses.

What matches

Both group device fabrication, assembly, neuronal culture and imaging in Experimental. Taylor reports device design and neuronal responses in Results.

What differs

Taylor places Experimental after combined Results and Discussion.

Paper 4

Paper

Compera et al. (2021). *Upscaling of pneumatic membrane valves for the integration of 3D cell cultures on chip*. DOI; Public full text.

Published headings and location

Experimental section, exact section-title elements: "Master mold fabrication"; "Chip fabrication"; "Surface measurements"; "Microscopy and image analysis". Results/discussion then covers valve development and biological use.

What matches

Both separate mold fabrication, chip fabrication, measurements and imaging, then report development and biological use in Results.

What differs

Compera's printed and SU-8 molds form separate layers; this paper bonds well-forming inserts onto a wafer.

Paper 5

Paper

Carnero et al. (2024). *Versatile hybrid technique for passive straight micromixer manufacturing by combining pulsed laser ablation, stereolithographic 3D printing and computational fluid dynamics*. DOI; Public full text.

Published headings and location

Journal pp. 2670-2672 / PDF pp. 2-4: "2.1 Manufacturing"; "2.2 Micromixer testing"; "2.3 Data collection"; "3.1 Laser parameters for manufacturing". The first three belong to Methods; the last belongs to Results/discussion.

What matches

Both place fabrication and testing procedures in Methods and observed process optimization in Results.

What differs

Carnero studies a laser-patterned micromixer.

Paper 6

Paper

Meynard et al. (2025). *Fabrication of a bioreactor combining soft lithography and vat photopolymerisation to study tissues and multicellular organisms under dynamic culture conditions*. DOI; Public full text.

Published headings and location

Journal pp. 4131-4135 / repository PDF pp. 14-18, Materials and methods: "Design and printing process"; "System fabrication process"; "Statistical analysis". Printed-structure measurements and culture methods also appear there.

What matches

Meynard retains design, construction and measurement methods in the main text.

What differs

Meynard uses resin in finished-device parts; this paper keeps the resin in the mold.

Paper 7

Paper

Broeren et al. (2023). *On-demand microfluidic mixing by actuating integrated magnetic microwalls*. DOI; Public full text.

Published headings and location

Materials and methods sections 2.1-2.2: "Fabrication of the microwall mixer and the magnetic actuation"; "Quantification of mixing". Section 2.1 points fabrication detail to ESI I; 2.2 points measurement implementation to ESI III. Main Figure 1 shows fabrication steps.

What matches

Broeren includes a main fabrication explanation, a process figure and a detailed supplementary procedure.

What differs

Broeren uses two broad Methods sections and has no cell culture.

Paper 8

Paper

Roy, Galas and Veres (2011). *Thermoplastic elastomers for microfluidics: Towards a high-throughput fabrication method of multilayered microfluidic devices*. DOI; Public article (DOI).

Published headings and location

Journal p. 3194 / PDF p. 2: "2. Materials and methods"; "3. Results and discussion". Results begins: "For the thin layer fabrication, with reflow of SU-8 photoresist being impossible, we developed a method"; further process details occur on pp. 3195-3196.

V7 P51: "Microchannel features patterned directly on bare silicon detached from the wafer."

What matches

Both report fabrication problems and process development in Results.

What differs

Roy is a short Technical Note with no Methods subsections and combined Results/Discussion. V7 has Methods subsections and separate Results and Discussion.

Paper 9. Bélisle et al. (2008): LAPAP

Bélisle, J. M., Correia, J. P., Wiseman, P. W., Kennedy, T. E., and Costantino, S. Patterning protein concentration using laser-assisted adsorption by photobleaching, LAPAP. *Lab on a Chip* **8**, 2164–2167. DOI: 10.1039/B813897D.

This Communication presents and characterizes a protein-patterning method, with axon guidance as biological proof of principle. That method-first framing is relevant to V7: fabrication is the contribution, and neuronal culture demonstrates its use.

Paper 10. Roy et al. (2013): engineered culture substrates

Roy, J., Kennedy, T. E., and Costantino, S. Engineered cell culture substrates for axon guidance studies: moving beyond proof of concept. *Lab on a Chip* **13**, 498–508. DOI: 10.1039/C2LC41002H.

This Critical Review examines biological insights from engineered substrates for axon guidance. It provides context for the neuronal application and for making engineered substrates useful in biological research.